ESPI
European Space Policy Institute

European Space Weather Services: Status and Prospects

Report 68
February 2019

Marco Aliberti
Leyton Wells

Short title: ESPI Report 68
ISSN: 2218-0931 (print), 2076-6688 (online)
Published in February 2019

Editor and publisher:
European Space Policy Institute, ESPI
Schwarzenbergplatz 6 • 1030 Vienna • Austria
http://www.espi.or.at
Tel. +43 1 7181118-0; Fax -99

Rights reserved – No part of this report may be reproduced or transmitted in any form or for any purpose without permission from ESPI. Citations and extracts to be published by other means are subject to mentioning "Source: ESPI Report 68; February 2019. All rights reserved" and sample transmission to ESPI before publishing.

ESPI is not responsible for any losses, injury or damage caused to any person or property (including under contract, by negligence, product liability or otherwise) whether they may be direct or indirect, special, incidental or consequential, resulting from the information contained in this publication.

Design: Panthera.cc

Table of Contents

1. Introduction **5**
1.1 Background and Rationales 5
1.2 Objectives and Scope 6
1.3 Research Methodology 7
1.4 Structure of the Study 7

2. Outlining Space Weather Services **8**
2.1 Defining Space Weather 8
2.1.1 SWE Causes 8
2.1.2 SWE Events 9
2.1.3 SWE Impact 10
2.2 Defining Space Weather Services 12
2.2.1 SWE Service Categories 13
2.2.2 Stakes for the delivery of operational services 14
2.3 SWE Service Enablers 15
2.3.1 Technological Enablers 15
2.3.2 Market Enablers 20
2.3.3 Organisational Enablers 24

3. European and International Efforts in SWE **25**
3.1 The European Architecture for SWE Services 25
3.1.1 Background: From National to Pan-European Efforts 25
3.1.2 European Space Agency 27
3.1.3 European Union 33
3.1.4 EUMETSAT 40
3.2 International Framework for SWE Services 42
3.2.1 International Space Environment Service (ISES) 42
3.2.2 United Nations 43
3.2.3 International Organisations 46
3.2.4 Coordination Group for Meteorological Satellites (CGMS) 50
3.2.5 International Space Weather Initiative (ISWI) 50
3.2.6 Research and Education: COSPAR and ILWS 51
3.2.7 Other International Service Providers 52
3.3 Summary: Status of Supply in the European and International Context 54
3.3.1 Europe 54
3.3.2 International Context 54

4. Towards Operational SWE Services in Europe **56**
4.1 Addressing the Technical Gaps 56
4.1.1 Filling Data Gaps 57
4.1.2. Improving Software Maturity 59
4.1.3 Advancing Product and Service Maturity 61
4.2 Addressing Demand/Market Requirements 63
4.2.1 Fortifying Relations with End-Users 63
4.2.2 Identifying Customers 65
4.3 Defining an Appropriate Organisational Setting 67
4.3.1 Scenarios for Operational SWE Services 69
4.3.2 Scenarios Assessment 74
4.4 The Bottom Line: Enhancing Awareness and Preparedness 75
4.5 Elements for a European Roadmap 76

5. Findings and Recommendations **78**

Annexes **81**
A.1 Explanation of Terms 81
A.2 NOAA Space Weather Scales 83
A.3 Space Weather Service Demand 85
A.4 Selected National SWE Weather Activities in Europe 98
A.5 Selected Worldwide Institutions Involved in SWE 101
A.6 ESA's Expert Groups Overview 103
A.7 SWE Projects in EU Framework Programmes (FP7 and H2020) 105
A.8 ESA and Operational Services 111
A.9 Long-Term Sustainability Guidelines of Relevance to SWE 112
A.10 List of External Contributors to the Research 114

List of Acronyms **115**

References **119**

About ESPI **130**

About the Authors **130**

1. Introduction

1.1 Background and Rationales

The shift in recent years towards the development of space-based services marks the convergence of space technologies, science and research in addressing user needs - be they societal or commercial. This shift in the overall practical output of space-related activity is of significance also because of its tangible interaction with wider society for functional purposes. Whilst there is no doubt about the value of the advancements made, and the necessary foundations that have been laid down, by space activities since the 1950s, it is only in recent times that the practical utility of the space sector has rapidly expanded. This applies not only to the scope of missions and research, but also to the technological infrastructures that are becoming increasingly available and accessible for exploitation. One needs only to glance at the current societal dependence on the satellite applications sector to grasp how substantial a service space sector can become in only a few decades, presenting a multitude of socio-economic benefits.

In this vein, the further development and potential role of space applications and services has been increasingly iterated in both Europe and internationally. In the development of a service-oriented space sector, however, challenges persist in the transition from demonstration to operations. In fact, services can be seen as step(s) beyond the technical feasibility or successful demonstration of a particular new technological system that fulfils a market demand. Essentially, a service is the application, or functional product, of research and innovation, and generally has a market or user base. Importantly a service *has* end-users. A *pre-operational service* is one that whose technological feasibility and capabilities have been proven, and a basis for estimating overall costs has been demonstrated, but whose organisational and institutional grounding has not yet been fully finalised. By contrast, an *operational service* is sustainable from a technological standpoint, has a strong institutional and organisational basis and, crucially, has a well-identified user-base. In this sense, it can be called self-sufficient. However, to reach this point, considerable research and financial investment is required for an innovation to become a sustainable service.

While value adding, sustainable services have been successfully established in the fields of telecommunications, meteorology and, more recently, navigation and Earth observation, challenges remain in areas outside these traditional domains. A clear case in point is Space Weather (SWE) services, a rapidly emerging issue-area that has been identified by all European stakeholders as needing more pro-active action, and the potential for European autonomy.

It should be first highlighted that substantial space weather-related observation has already been gathered in the preceding decades and a framework for transforming the subsequent data into functional, value-added services is already envisaged within the current European framework. In addition, at national level some European countries already provide operational SWE services for certain sectors (e.g. the commercial airlines, the satellite industry, power grid operators, etc).

However, from a pan-European perspective, space weather remains a rather novel area of action for the different European actors, with their envisaged transition to operational services necessitating an increasingly integrated and networked approach to ensure sustainability. To become fully operational, space weather services require a strong engagement with user communities to develop preparedness and responses to space weather risks. In addition, it calls for synergies between different stakeholders on the supply side (in both the space and non-space domain) as well as coordination/cooperation efforts at international level. Indeed, as also stressed by several studies, even though there currently are numerous national space-based and ground-based assets that could be used to improve space weather services, these assets are generally not effectively coordinated, or easily available beyond the community that operates them. Observations are not systematically in-

teroperable, shared in near-real time, or documented with metadata that would enable their most efficient use. In line with this, although each area of application for space weather services has its own specific requirements, the increased interconnectivity of all stakeholders involved, and coordination efforts at national and international policymaking levels, are essential in building a framework that would enable a flourishing and sustainable market base.

As a result, when considering the provision of value-adding space weather services, several questions remain:

- How will the space community move from the provision of space weather data to the provision of value-adding operational services?
- What are the gaps in the current European and international architecture for space weather services?
- Who should fill these gaps (private business/ public institutions/ international cooperation mechanisms)? And how can they be filled?
- What are the drivers for ensuring European autonomy on space weather services? How can it be enabled?

Considering this background, the overarching objective of this research project is to elaborate on the possible role and actions of various European stakeholders (ESA, the EU, EUMETSAT, private actors) towards the all-round development and delivery of operational space weather services to end-users. To this end, the report will first provide general observations on the importance of operational SWE services in Europe and on the issues associated with the establishment of sustainable services. Building on this, the demand and supply conditions will be explored with respect to both the European and international contexts, and a fit/gap analysis of the technological, business and policy dimensions will be provided. Subsequently, the study will reflect on different models for ensuring the smooth provision of value-adding services, including those centred on public institutions, private actors and international cooperation mechanisms. Finally, a cost-benefit analysis of the different models will be provided to identify the optimal way forward.

1.2 Objectives and Scope

The overarching objective of this study is to provide an in-depth investigation of the possible future functioning of space weather services in Europe. This ESPI study will more specifically:

- Assess and characterize the demand conditions for space weather services by elaborating on the various services domains and the user and customer base
- Assess and characterize the supply conditions of space weather services by elaborating on the current European and international architecture for space weather
- Investigate the required steps to move from the provision of space weather information to the provision of fully fledged and sustainable operational services
- Identify the technological, business and policy gaps in the current European and international architecture for space weather services
- Elaborate different scenarios for filling these gaps, including scenarios centred on the private sector, public institutions, and international cooperation mechanisms
- Assess the pros and cons of each scenario
- Identify key elements of a possible roadmap for SWE service delivery

In terms of scope, two layers of research have been identified.

- The first is related to the provision of space weather services at the European level. This part of the research will devote particular attention to ESA's, the EU's and EUMETSAT's SWE activities, including their interactions (or lack thereof) and their role in the international context. This layer of the research will also consider the relations between the different European stakeholders, as well as the interests of private firms.
- The second layer is related to international mechanisms. In this respect, the report will assess international cooperation formats already established in the field of space weather, such as those led by the International Space Environment Service (ISES), and the different UN specialised organisations, including UNCOPUOS, the WMO and ICAO.

1.3 Research Methodology

The study has been primarily prepared on the basis of an in-house analysis featuring a profound literature review of publicly available documents, external and internal databases, conference proceedings and other bibliographic sources, spanning both space-related and general contributions in the area of space-based services. In addition, the research has leveraged external contributions by relevant European and international stakeholders in the form of preliminary discussions and feedbacks on the study, as well as interviews and expert meetings. Finally, a peer review with experts was carried out to validate research findings before preparation of the final report.

The starting point has been to dissect what services are, and how this applies in the context of SWE services. In doing so, and in exploring the potential areas of applications, a characterisation of space weather has been conducted in regards to the technological systems that enable the provision of related data and subsequent services, market aspects, and the organisational framework. For the most part, the technological systems, observational capabilities, and data delivery aspects are already in place, and so the analysis has focused on the final stages of the process, from demonstration to operations, assessing the constituting organisational frameworks into which they are constructed and through which they are interlinked. Moving forward, the study has performed a mapping of the institutions involved in SWE activities at both European and international level and with respect to both space and non-space aspects. This enabled identification of stakeholders for the interviews, which in turn helped to perform an examination of the SWE service demand and supply environments with an inventory of existing solutions, the fit-gap analysis and the elaboration of different governance scenarios with comparative evaluations.

In line with the mission of ESPI to provide European stakeholders with informed analyses in the field of space policy and to facilitate the decision-making process, the study has an inherent normative dimension: the research more specifically aims to identify what is at stake for the various stakeholders in Europe and elaborate on possible policy actions for European decision-makers.

1.4 Structure of the Study

Following an overview of rationales and objectives in this chapter, Chapter 2 provides an introductory overview of SWE services. First, a definition of what SWE is – i.e. causes, events and impacts – leading to evaluation of the importance and stakes of SWE service delivery, before providing a basis for defining the provision of SWE services through characterisation of all their technological, market, and organisational dimensions as a foundation for sustainable operation.

Chapter 3 will focus on the ongoing efforts at European and international level towards operational SWE service provision. Beginning with an assessment of the current European organisational architecture before moving onto the international framework, this chapter will address the institutional and organisational elements of SWE service provision by identifying the various stakeholders, organisations and initiatives relevant to the field of SWE services.

Together, this assessment will culminate in a demand-supply fit/gap analysis that will be provided in Chapter 4 through an integrated approach involving all of its technical, economic and policy dimensions. Scientific and technological, market, and organisation gaps will be highlighted and included within the analysis of potential future governance scenarios that seek to strengthen SWE service provision and address the current gaps. In this penultimate segment, the report will also provide an overview of potential models of governance involved in the delivery of operational services. Specifically, public, private and public-private frameworks for the provision of SWE services will be evaluated in reference to one another; looking in detail at each model's merits as well as disadvantages for facilitating different types of services, and which models best suit the purpose of a particular service in various scenarios.

Finally, the overall findings and recommendations of the report will be presented in Chapter 5, with particular attention paid to what actions can be taken at the European level to support the transition phase of SWE services from demonstration to operation.

2. Outlining Space Weather Services

2.1 Defining Space Weather

The term Space Weather (SWE) designates both the physical environment and dynamic phenomenological state of the space environment, particularly the Sun and the planetary and interplanetary environments, as well the scientific discipline associated with its study.[1] The objective of SWE studies, also known as the meteorology of space, is to observe, understand and predict the "dynamic, highly variable conditions in the geospace environment including those on the Sun, in the interplanetary medium, and in the [Earth's] magnetosphere-ionosphere-thermosphere system"[2] which can affect "performance and reliability of space-borne and ground-based technological systems and endanger human life or health".[3]

2.1.1 SWE Causes

Most of space weather phenomena are directly linked to solar activities, which can vary dependent on the solar activities and cycle, as well as phenomena originating from sources external to the solar system. The main drivers of space weather phenomena interactions with the near-earth environment are listed below.

Solar Flares

A solar flare is an eruption of radiation – right across the electromagnetic spectrum from kilometric radio waves through the infrared, visible and UV ranges to X-rays and Gamma-rays[4] – emitted from the energy accumulated in the sun's magnetic fields as they become increasingly unstable, most frequently discharged from above sunspots[5]. These outbursts can last from minutes to hours and, travelling at the speed of light, this electromagnetic energy has an effect on the Sun-facing side of the Earth's atmosphere that is observable at the same time as these events occur[6]. Solar flares can disrupt the area of atmosphere in which radio waves travel by producing a brief (10-20 minute) atmospheric layer that absorbs high frequency radio waves[7], leading to degradation and in the worst scenarios, complete radio blackouts.[8]

Coronal Mass Ejections (CMEs)

CMEs are large expulsions of plasma and magnetic field originating from the Sun's coronal atmosphere. Ejecting vast amounts of coronal material as well an embedded magnetic field, CMEs travel at speeds from 250 km/s to nearly 3000 km/s, often reaching earth in 15-18 hours.[9] CMEs are sometimes associated with solar flares but can also occur separately. Both consist of large eruptions of energy, however they emit different properties, as well as appearing and travelling differently, having different impacts on affected planets[10]. Magnetic alterations around the Earth stimulated by CMEs can affect technological systems.[11]

Solar Energetic Particles (SEPs)

A form of cosmic ray but determined by solar activity, SEPs are high energy charged particles accelerated by the Sun (i.e. predominantly protons and electrons) that travel at close to the speed of light[12]. The acceleration of these charged particles is caused by dynamic processes in the magnetised coronal and interplanetary plasma and are commonly

[1] World Meteorological Organisation, 2016
[2] Baker, 1998
[3] World Meteorological Organisation, 2008:3
[4] Schwenn, 2006:21
[5] European Space Agency, 2018a
[6] National Oceanic and Atmospheric Administration, 2018a
[7] Hapgood, 2010:9
[8] Gleber, 2014
[9] National Oceanic and Atmospheric Administration, 2018b
[10] Glaber, 2014
[11] Glaber, 2014
[12] Gleber, 2014

associated with solar flares and coronal mass ejections (CMEs).[13][14]

Galactic Cosmic Rays (GCRs)

Similar to SEPs, galactic cosmic rays are also energetic charged particles, produced instead by the occurrence of supernovae explosions external to our solar system. These particles are trapped and directed by magnetic fields through interstellar space and can enter into our solar system, potentially damaging spacecraft.[15]

2.1.2 SWE Events

When interacting with the near-Earth environment (the ionosphere, magnetosphere and thermosphere – see Appendix A.1 for the terms of reference), the SWE drivers detailed above can cause one or more SWE events, also known as SWE storms, those being: geomagnetic storms, solar radiation storms, radio blackouts and ionospheric disturbances.[16] A brief description is provided in Table 1. These events additionally lead to increased geomagnetic activity, energy particle radiation and x-rays, which in turn have various impacts on human health and technologies, as further outlined in the following sections.

SWE event	Description
Geomagnetic Storm	Geomagnetic storms are strong disturbances in the Earth's magnetic field that occur when CMEs or solar wind streams interact with the geomagnetic field. As a result of this interaction, the Earth's magnetic field adjusts to this jolt of energy and is altered. **Frequency**: Most common during the solar maximum and during the declining phase, but can occur anytime during the solar cycle. **Duration**: From a few hours to a few days
Solar radiation storm	Solar radiation storm events occur when the near-Earth environment is immersed in large quantities of charged particles, primarily protons, which are accelerated by solar activity (solar flares). **Frequency:** Most common during the solar maximum years, but can occur at any stage of the solar cycle. **Duration:** Proportional to the magnitude of the solar eruption and received spectrum – from hours to a week.
Radio black out (Solar Flares)	Radio blackouts are the consequence of solar flares causing enhanced electron densities that ionise the sun-side of the Earth- disrupting radio waves as they pass through this region. **Frequency:** Very common – minor events occurring on average 2,000 times each solar cycle, most frequent during the peak years of the solar cycle, almost absent during solar minimum. **Duration:** Minutes to hours.

Table 1: Outline of SWE drivers (sources: adapted from Nation Oceanic and Atmospheric Administration, 2015 and International Civil Aviation Organisation, 2010:15-20)

Each of the above-described SWE phenomena (solar flares, CMEs, SEPs, GCRs) can have different effects on the interplanetary medium (from the Sun to the Earth), the near-Earth environment (magnetosphere-ionosphere-thermosphere), to impacts on the Earth itself. Accordingly, there are a number of additional factors, phenomenological and physical state, that require observation in order to understand and model the space weather status. Monitoring of solar activity and the solar drivers discussed above is of course essential, however beyond this, further monitoring and observational data are necessary on a number of other SWE components from space-based and ground-based sensors – in some instances from both, although in certain cases these observations can only be made from space. Key areas of observation from space-borne and ground-based instruments include: solar activity, solar wind, space radiation, the geomagnetic environment, upper atmosphere monitoring (ionosphere and thermosphere); while space-based sensors can potentially extract a wider array of, or more in-depth, observations within these areas, or on additional phenomena such as microparticles, which are

[13] Zheng and Evans, 2014
[14] National Aeronautics and Space Administration, 2012
[15] Hapgood, 2010:8
[16] National Oceanic and Atmospheric Administration, 2014

difficult to observe from within Earth's atmosphere.[17] Within these domains, specific measurements can be taken on SWE effects including: geomagnetic storms, ionospheric scintillation, total electron content (TEC), aurora etc.[18] The monitoring of and resultant data from these SWE effects is not only important for SWE science and modelling, but the real-time or near real-time provision of such data is crucial for forecasting and nowcasting SWE impacts– with each industry sector impacted having its own specific observational requirements in terms of data products and services made available by space- and ground-based sensors.

2.1.3 SWE Impact

It is important to understand that, as a discipline, SW aims not only to scientifically study phenomena involving ambient plasma, magnetic fields, electromagnetic radiation and energetic charged particles in space, but also to predict how all these phenomena can "influence the functioning and reliability of space-borne and ground-based systems and services, thereby potentially endangering human health and wellbeing through impact on this infrastructure."[19]

In recent history, there have been several well-documented major space weather events with tangible impacts. Whilst there has not been an episode on the same scale of the "Carrington event" of 1859, since the emergence of the space age, the largest geomagnetic storm of the modern era occurred in March 1989 and caused the collapse of Canada's Hydro-Quebec power grid[20], leaving millions without an electricity supply for up to 9 hours.[21] In addition to the outage in Quebec, this storm damaged transformers in the UK and other countries, and caused "the loss of positional knowledge for over 1,000 space objects for almost a week.[22] To further clarify the frequency and "not if but when" nature of such space weather events, within the same year of 1989, additional solar flare events were experienced in August, March and October, as well as a large-scale SEP event occurring in October 1989[23].

A more recent major space weather event, and as such well observed and documented, occurred at the end of October 2003 – often referred to as the 'Halloween event'. Although weaker than the March 1989 event, the wealth of data available provided "clear evidence that large geomagnetic storms can disrupt space-based navigation systems by inducing rapid and large changes in the morphology of the ionosphere and plasmasphere".[24] Only a few days later, the same region of the Sun that caused the Halloween event produced the largest observed X-ray solar flare since space measurements began, however due to the Sun's rotation in those few days Earth fortunately experience a near-miss of a "Carrington-class CME as well as intense particle fluxes".[25]

The possibilities, frequencies and intensities of such events should not be underestimated - modern technological dependencies would have been catastrophically affected if such an occurrence had hit Earth. Indeed, in the current state, modern society and its technological reliance's can be characterised as "a complex interweave of dependencies and interdependencies among its critical infrastructures" and as such the socioeconomic analysis of extreme space weather impacts must be inclusive of both "direct, industry specific effects (such as power outages and spacecraft anomalies)" as well as "the collateral effects of space-weather-driven technology failures on dependent infrastructures and services".[26] A graphic overview of these transversal and plethoric impacts of space weather is presented in Figure 1.

[17] European Space Agency, 2013:18-21
[18] National Oceanic and Atmospheric Administration, 2018c
[19] European Space Agency, 2017a
[20] Pinheiro et al, 2016
[21] National Research Council, 2008:3
[22] Royal Academy of Engingeering, 2013
[23] Guhathakurta, 2011
[24] Royal Academy of Engingeering, 2013:18
[25] Royal Academy of Engingeering, 2013:18
[26] National Research Council, 2008:12

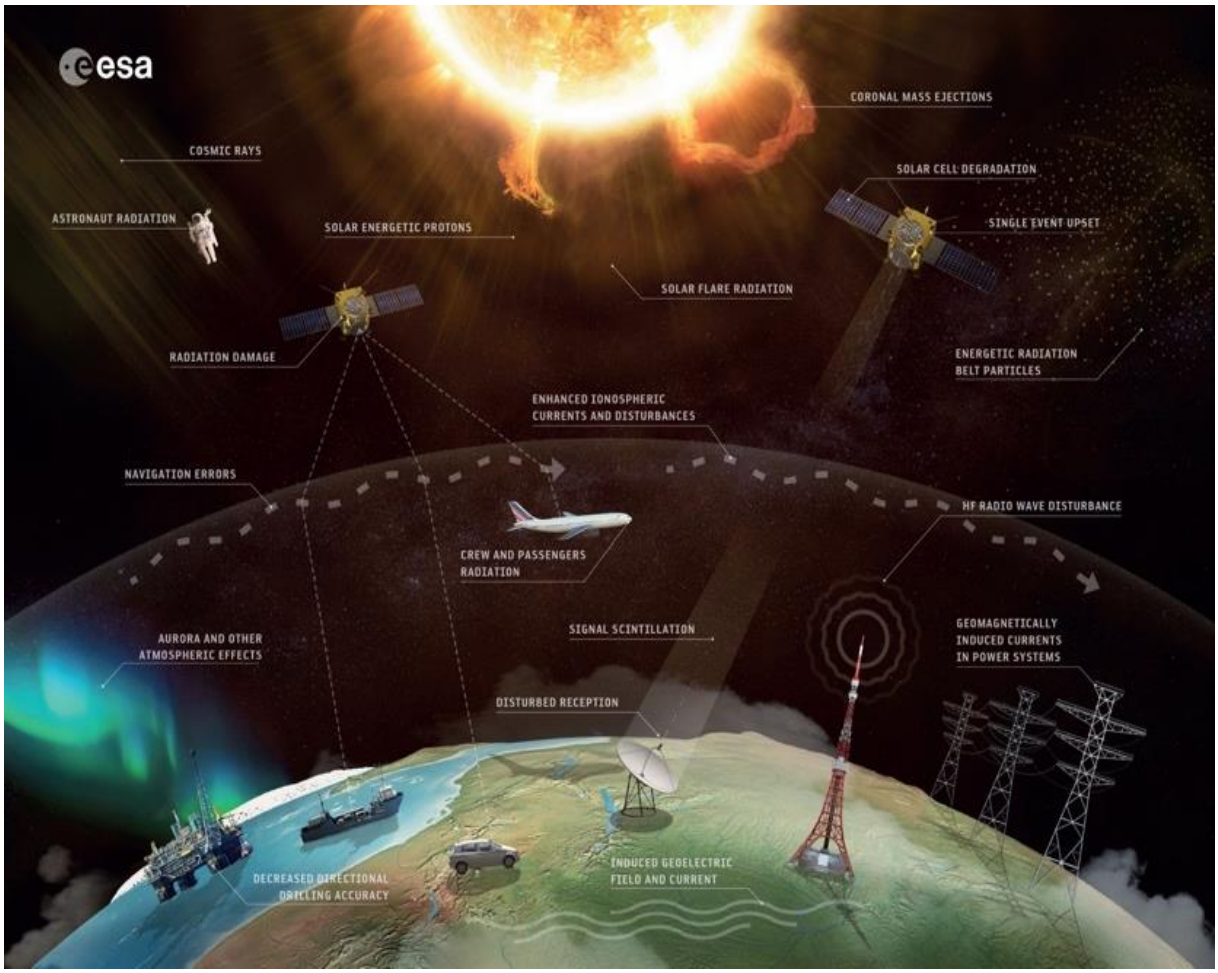


Figure 1: The Impact of Space Weather: An Overview (credit: European Space Agency, 2018b[27])

As evident from Figure 1, in today's economy, a large number of sectors could potentially be affected by SWE storms. These range from space-based telecommunications, broadcasting, navigation and weather services, through to aviation, power distribution, and terrestrial communications, especially at northern latitudes.

In the space environment, SWE can have detrimental impacts on space technologies such as satellites and spacecraft, damaging them directly but also having a knock-on effect to the services they provide for Earth-based users. Satellites, in particular, are interweaved into the network of critical infrastructure, particularly in the areas of communication and navigation systems, on which society is increasingly dependent. As such, disruptions to these satellite services, for example the Galileo system, could have a serious impact on a magnitude of sectors on Earth, affecting aviation, road transport, shipping, and other sectors reliant on precision positioning to function[28]. The safety of astronauts could also be directly affected by space weather events, with additional risks to health indirectly posed by disruptions that might occur in human-space flight operations on Earth.

The effects of space weather could also have an impact on technological systems and human health on Earth; potentially affecting the electricity grid, pipelines, ground and ocean transportation systems (e.g. road, rail and maritime), aviation communication and control, air passenger and crew health, mobile telephones, and high frequency radio communication systems.[29] In this regard, extreme space weather events could disrupt a vast number of modern technological systems across a variety of industry sectors, the socio-economic cost of which would be tremendous if not effectively predicted and mitigated, placing great significance on space weather service provision *before* a major space weather event takes place.

[27] European Space Agency, 2018b
[28] European Space Agency, 2017a
[29] Royal Academy of Engineering, 2013:5; Hapgood, 2010

In the European context, it is estimated that a single extreme space weather event (encompassing radio blackout, geomagnetic storm and solar radiation storm), could cost €14,971 M across a number of sectors (in 2016 economic conditions). See Table 2.

Domain	2016 (year 1)	2024 (year 9)	2032 (year 17)
Spacecraft design and operations	- €912.9 M	- €1,123.2 M	- €1,389.4 M
Launch operations	- €0.008 M	- €0.037 M	- €0.051 M
Aviation	- €6,635.6 M	- €11,139.8 M	- €18,701.5 M
Resource exploitation	- €197.5 M	- €234.9 M	- €279.5 M
Power system operators	- €5,630.5 M	- €6,364 M	- €7,195.2 M
Road & Transportation	- €1,595.4 M	- €1,783 M	- €1,992.8 M
TOTAL	- €14,971.9 M	- €20,644.9 M	- €29,558.4 M

Table 2: Statistics reported as NPVs over the period of 2016-2032 (source: PricewaterhouseCoopers, 2016:2)

The exact socio-economic impact of SWE events on different engineered and biological systems depends on the intensity and duration of an SWE event. NOAA has developed a scale for categorising geomagnetic storms, solar radiation storms, and radio blackouts, which include five levels of strength: extreme, severe, strong, moderate, and minor. Each of these levels is associated with a specific type of impact. An overview of these impacts is provided in Annex 1.

The important point to highlight here is that as dependency on space- and ground-based technological systems increases, "the sensitivity of our society and our economy to space weather effects is expected to increase" and, as such, within the overall European economy, each individual sector "has a need for specific space weather data and services, together with a further requirement for those services to be tailored to their particular applications and uses".[30]

2.2 Defining Space Weather Services

Given the ever-growing dependence of contemporary societies on technologies that could be impacted by space weather, there is a growing demand for operational space weather services to safeguard vulnerable sectors: e.g. air navigation on polar routes exposed to space weather events; fleets of operational satellites used for telecommunications, broadcasting, observation or positioning; use of satellite-based navigation and timing signals that are affected by ionospheric disturbances; electric power grids that are exposed to geo-magnetically induced currents with potentially disastrous cascading effects".[31] As a result, societal interest in SWE is developing, stakeholders across the globe – ranging from scientists, emergency management agencies, commercial airlines, satellite operators, pipeline designers, power grid operators, rail operators etc. – are becoming increasingly aware of SWE services as they mature into their operational phase. As science and society increasingly recognise "the impacts of space weather on the infrastructure of the global economy, interest in, and dependence on, space weather information and services grows rapidly".[32]

SWE services can be defined as the final output of the transformation of space weather data and products into practical applications for specific customers to defend against the potentially harmful impacts of SWE. As the definition makes plain, it is important to make a distinction between SWE *data* and *products* on the one hand, and SWE *services* on the other. Both draw on data (i.e. raw or processed measurements of any space weather parameter), but whilst the former simply aim to describe a certain condition in the space en-

[30] European Space Agency, 2017a
[31] World Meteorological Organisation, 2016:3
[32] Schrijver et al., 2015:2747

vironment, SWE services are intended to enable specific users to take actions when adverse SWE might be occurring in order to minimize the impacts on their systems and/or customers. For example, an SWE product may provide information on the radiation risks associated with a solar storm, while an SWE service may be intended to enable airlines to modify polar flights routes during the forecasted period of SWE activity to protect their passengers and crew.

2.2.1 SWE Service Categories

To add more depth to the definition, operational SWE services aim to "monitor, specify, and forecast the space environment in order to provide timely, accurate, and reliable space weather services for domestic and international customers" that can be presented in the form of forecasts, current SWE status reviews, as well as SWE event alerts as they happen.[33]

In this sense, reviews and forecasts "focus on presenting analyses of current conditions and developing trends of space weather activities, such as solar X-ray flares, geomagnetic activity, solar proton events (SPEs), and the relativistic electrons in the radiation belts"; whereas "event alerts are triggered and issued automatically whenever the activity level of solar, geomagnetic, or high-energy particles reached threshold, which aims at notifying customers of space weather disruptions immediately".[34]

As such, and similar to terrestrial weather, SWE services can be categorised according to the service timeframe, i.e. forecasts, nowcasts and postcasts or hindcasts, as outlined in Figure 2.

To better illustrate the distinction between SWE products and services in relation to different timeframes involved, some examples are provided in Table 3.

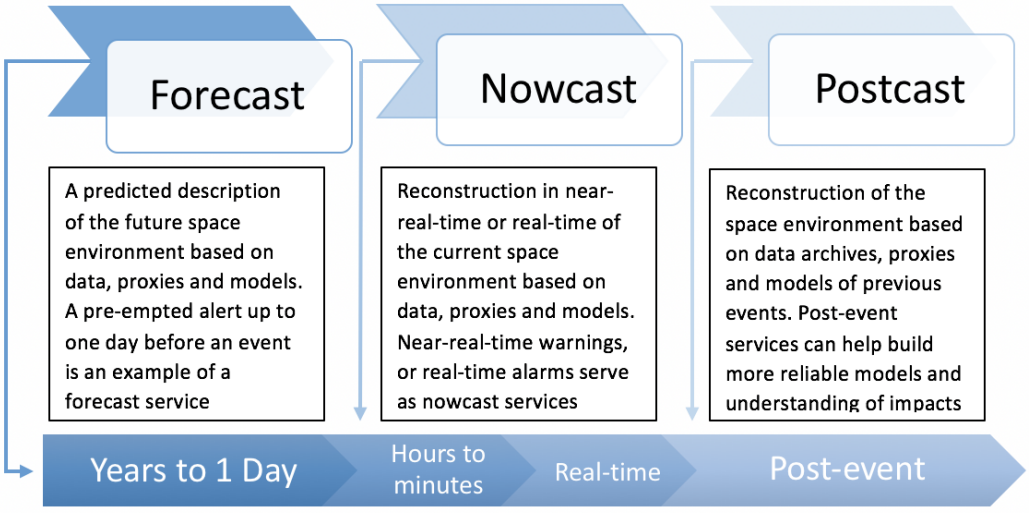


Figure 2: Services in relation to their timeframes (source: PricewaterhouseCoopers, 2016)

Time Frame	SWE Product	SWE Service
11 years	Solar activity cycle	Forecast for satellite orbit planning
27 days	Solar radio emissions	Forecast for optimal radio-communication planning
1-3 days	Solar x-rays	Alert to power grid operations
30-60 minutes	Geo-magnetic storms	Warning to airline company on polar routes
Real-time	Solar energetic protons	Alarm to astronauts inside ISS

Table 3: Examples of services in relation to timeframes

[33] Lui and Gong, 2015:599

[34] Lui and Gong, 2015:599

Beyond this categorisation, SWE services can be differentiated according to the two general components of significance to a service: the source of funding and the operating body or organisation. Generally, both of these components fall into the categories of public, private, or public-private models of services which, in themselves, can highlight the general purpose of the service (e.g. a societal need or a market demand). Variance in funding sources, and providing groups of a service can have different implications for the user, but the user, or more broadly the demand side, can also dictate which type of service model might be best for them. To expand on this, a public model might suit a public demand or societal issue, a private model will certainly fixate on areas in which demand will return profit, and joint public-private models will most likely be utilised for a service that is desirable for society whilst also having the ability to create revenue. While the majority of SWE services worldwide are managed under public models, being considered non-excludable and non-rivalrous goods, there are also instances of SWE services as private goods.

In terms of service provision, one further distinction can be made between national and internationally provided SWE services, both in terms of providers and end-users. In this sense, services can be initiated, funded, and ultimately provided by international organisations or frameworks, or national actors, or a combination of the two.

2.2.2 Stakes for the delivery of operational services

Whereas SWE services are already used today in several countries in various sectors (e.g. the commercial airlines, the satellite industry, drilling and surveying operations, power grid operators, pipeline designers and users of satellite-based navigation systems, etc.), it is anticipated that "this demand will expand with broader awareness of the impact of space weather events, increasing exposure of the society, and greater maturity of space weather products and services" (WMO, 2016). Indeed, as dependency on space- and ground-based technological systems increases along with the sensitivity of our society and our economy to SWE, the stakes associated with the delivery of SWE services will become higher.

The majority of these stakes have been framed in terms of direct and indirect socioeconomic effects, primarily on technological infrastructures and human health, caused by SWE events. In the European context, the study conducted by ESA and PwC estimated that ESA's SWE programme services could provide a net saving of €2,635 million in comparison to a "do nothing scenario", which could cost -€13,135 million – from just a single major space weather event – across several sectors (see Table 4).[35]

Cost/Benefit	Do nothing scenario	Do ESA scenario	Value added of ESA Services
User domain benefits			
Satellite operations	- €283 M	- €267 M	€26 M
Launch operations	- €0.3 M	- €0.1 M	€0.2 M
Resource exploitation	- €327 M	- €135 M	€192 M
Power grid operations	- €5,771 M	- €4,546 M	€1,225 M
Aviation	- €3,312 M	- €3,066 M	€246 M
Logistics/Road transport	- €3,432 M	- €2,888 M	€544 M
Investment benefits			
GDP impact	None	€952 M	€952 M
Total Benefits (b)	- €13,135 M	- €9,950 M	**€3,185 M**
Programme Costs (c)	None	- €529 M	**- €529 M**
Total Net Benefits	- €13,135 M	- €10,479 M	**€2,656 M**
		Benefit / Cost ratio (b/c)	6

Table 4: Statistics reported as NPVs over the period of 2016-2032 (source: PricewaterhouseCoopers, 2016:2)

Whilst the critical infrastructures and societal ventures that space weather can potentially influence are vast, in many cases intertwined, and the complexity ought not be underestimated – there are also wider stakes or incentives for the delivery of operational SWE services. These qualitative benefits can be broken down into four distinct "macro categories" - strategic, economic, societal, and environmental domains - on which SWE services can have positive effect.[36] These are summarised in Figure 3.

[35] PricewaterhouseCoopers, 2016

[36] PricewaterhouseCoopers, 2016:15-17

Strategic	Economic	Societal	Environmental
• Stimulate advances in basic and applied research in critical areas •Development of end-to-end capability •Increased autonomy with independent SWE data •Equal partner in data exchange agreements internationally •Acheive greater European cooperation •Access sensitive data	•Improve reliability of utilities •Improve design and operations of spacecraft •Positive impact on European Economy •Support commercial gain forvarious industries and businesses •Enable opportuities for commercial spin-offs	•Improved safety of European infrastructure and services (space systems, human space flight, aviation, transport, power systems...) •Improved safety of human life (navigation, radiation environment) •Reduction of morbidity and mortality due to prolonged electrical blackouts •Reduced loss of time in road, rail and aviation transport	•Reduced risk and faster recovery from spill-over from underwater oil wells •Reduced greenhouse gas emissions through improved logistics and transport (e.g. alternate flight routes and heights)

Figure 3: Stakes in SWE Service Delivery (source: adapted from Horne, 2001 and PricewaterhouseCoopers, 2016)

2.3 SWE Service Enablers

The successful provision of any service requires the fulfilment of three requirements that ensure its continuity, availability and sustainability, namely: (i) structured demand; (ii) reliable, affordable technology that is adapted to user needs; and (iii) structures that ensure both appropriate funding allocation and generation as well as adequate programmatic overview. In essence, this implies three distinct categories of operational service enablers – market, technological, and organisational enablers.

With specific respect to SWE services, the operationalisation of these services requires continuity, availability and sustainability in:

- The technological enablers for SWE observation and scientific understanding, data management, and communication;
- An active market/user base requiring the identification of users and associated needs.
- An organisational framework in which SWE stakeholders operate through the value chain in an interconnected and functional manner

2.3.1 Technological Enablers

To provide timely and accurate space weather services and meet user requirements, there is a need for the continuous availability of SWE data from both ground- and space-based observation systems as well as tools that process raw data and turn them into timely and accurate space weather information, nowcasts and forecasts.

a. Ground-Based Observations

Many of the overall measurement parameters for SWE observation can be taken from the ground - a study of the European SWE resource network found that out of 222 observation sources, 99 were from ground segments.[37] Whilst there are certainly limitations, ground-based segments for SWE observation and data provision provide several advantages: namely, that in comparison to space-based (in-situ) systems they are relatively simple and inexpensive (avoiding associated costs with qualifying space instruments and launch operations)[38]; they are flexible in terms of upgrading and maintaining; they come with reduced telemetry constraints and delays, increasing cadence of observation and analysis capabilities, as well as being available in real-time; and limitations in observing time can be overcome by networks of observatories.[39] Thus, for these reasons, ground-based

[37] Hapgood, 2001
[38] Hapgood, 2001
[39] Veronig and Pötzi, 2016:4

segments serve as a preferential choice over space-based, however certain SWE observation capabilities can only be conducted by, or have significant advantages from, in-situ space-based segments.

The range of instruments that monitor SWE is large and varied. The most important are summarised in Table 5.

Instrument	Description
Riometers	Riometer instruments – Relative Ionospheric Opacity Meters – are ionospheric monitors that passively measure the strength of galactic radio noise, and through inference, the absorption of HF radio waves propagating above the riometer in the ionosphere.
Ionosondes	Ionosondes are also ionospheric monitors that measure the plasma constituting the ionosphere. To do this, Ionosondes "sound" the plasma by sending a spectrum of radio pulses from the ground and measure the time taking from the pulse to return after reflecting from a particular ionospheric layer.
Magnetometers	Magnetometers measure the strength and direction of the local magnetic field. They can be static or mobile, ground-based or located above on satellite platforms.
Neutron Monitors	Neutron monitors are based on the ground and count the number of neutrons passing through the instrument. They measure the highest energy charged particles in the atmosphere, allowing for monitoring of the near-Earth Radiation environment.
Radio Telescopes	Radio telescopes are used to monitor the solar radio emissions passing through the ionosphere. Solar activity, specifically solar flares, can emit bursts of radio frequency, which can be disruptive to certain GPS/GNSS systems.
Optical Telescopes	Optical telescopes are used to monitor the sun in order to view solar activity – i.e. eruptions magnetic field activity, sunspots and coronal holes – in various wavelengths respective to the activity that is to be observed.

Table 5: Instruments used for SWE measurements (source: adapted from International Civil Aviation Organization, 2010)

Examples of European and international observatories and instrument networks are shown in Table 8. Within the European context, there is already a wealth of activity in regards to ground-based SWE observation, conducting a variety of types of SWE measurements from several European observatories (see Table 6 below). These observatories and observational instruments tend to work within networks, both pan-European (EISCAT and THEMIS) and internationally (INTERMAGNET and SuperDARN), to produce collective and interoperable observational data products.

Observatory/Network	Details	Type of Observation
Global High Resolution H-alpha Network	A network of observatories comprising the Big Bear Solar Observatory (USA), the Kanzelhöhe Solar Observatory (Austria), and the Yunnan Astronomical Observatory (China)[40]	Solar images in various wavelengths
Global Oscillation Network Group (GONG)	A network of six solar observatories: Teide Observatory (Canary Islands), the Learmonth Solar Observatory (Western Australia), the Big Bear Solar Observatory (California), the Mauna Loa Solar Observatory (Hawaii), the Udaipur Solar Observatory (India) and the Cerro Tololo Inter-American Observatory (Chile)	Conducting detailed studies on the solar internal structure and dynamics using helioseismology[41]
Dominion Radio Astrophysical Observatory (DRAO)[42]	Operating, designing and development of telescopes	Solar radio wave spectrum of varying wavelengths
Compound Astronomical Low-cost Low-frequency Instrument for Spectroscopy and Transportable Observatory (e-Callisto)	A worldwide radio-spectrograph network with 24/7 monitoring. A list of the Callisto instruments /observatories and their respective countries is on their website.[43]	Solar radio bursts
Real-time Neutron Monitor Database (NMDB)[44]	A database collected from 18 neutron monitors across the globe	Cosmic ray neutron flux on the Earth's surface
International Real-time Magnetic Observatory Network (INTERMAGNET)[45]	A global network of observatories monitoring the Earth's magnetic field. A list of participating observatories and their respective countries is provided on the INTERMAGNET website[46]	Vector magnetic field (Magnetograms at Earth's surface)
Super Dual Auroral Radar Network (SuperDARN)	A global network of ground-based coherent-scatter radars operating in high-frequency[47]	Primarily measuring plasma convection in the ionosphere, also has wider uses for studying other magnetospheric and ionospheric phenomena
European Incoherent SCATer Scientific Association (EISCAT)[48]	Scientific association with member institutions from Finland, Norway and Sweden, using radars to conduct observations.	Atmospheric and ionospheric observations, e.g. the effects of the aurora borealis
European Digital Upper Atmosphere Server	A pan-European digital data collection for forecasting and nowcasting purposes[49]	Upper atmospheric/ionospheric conditions
Global Muon Detector Network (GMDN)	An international muon space weather telescope network: Germany, Japan, Australia, Brazil, Kuwait, USA	CME detection through cosmic ray anisotropy precursors

Table 6: European and International ground-based observatories and instrument networks (source: European Space Agency, 2018b

[40] Steinegger et al., 2000
[41] Armet, 2017
[42] Government of Canada National Research Council, 2017
[43] Russu et al., 2015
[44] Mavromichalaki et al., 2011
[45] INTERMAGNET, 2018
[46] INTERMAGNET, 2017
[47] Chisham et al., 2007
[48] European Incoherent SCATer Scientific Association, 2018
[49] Belehaki et al., 2005

b. Space-Based Observations

To obtain all the necessary, accurate, real-time data needed for future SWE warning services, data from space-based instruments are essential. While the baseline working approach for many SWE sensors operators is to collect as much of the required measurement data using ground-based instruments as possible, ground-based systems, as mentioned, can have observational limitations. As such, space-based measurements are necessary to sufficiently meet the whole range of observational requirements. The main reason for this the effects of the magnetosphere and atmosphere surrounding the Earth, respectively deflecting solar wind and charged particles as well as filtering out electromagnetic radiation[50]. As an example, space observations have increased performance through in-situ measurements over those taken from the ground in:

- Practicability, e.g. ground-based solar UV and X ray images are impossible because of atmospheric absorption. Such observations must be space-based.
- High quality, e.g. coronagraph images of coronal mass ejections are much clearer when taken in space because of the absence of stray light from atmospheric scattering (indeed CMEs were not recognised prior to their observation with the Skylab coronagraph).[51]

The most typical and widely used instruments for space-based SWE observations are outlined in Table 7.

Instrument	Description
Magnetometers	Space-based magnetometers function to measure the magnetic fields of planetary bodies, Interplanetary Magnetic Field, and for navigational uses.
X-ray Instruments	Used in SWE applications to mainly observe the Sun and its active phenomena, covering wavelengths from 10-0.01 nm
Plasma Instruments	The purpose of plasma instruments is to observe a variety of metrics regarding space-born plasma, including electron and ion density, temperature, and velocity.
Electric Field Measurement Devices	These devices are used to study the plasma environment – i.e. take measurements of electric field and density.
Mass Spectrometers	Mass spectrometers are used to determine the mass of atoms or molecules in any given sample – current devices generally include an ion source, a mass analyser and a detector.
Electrostatic Analysers	Electrostatic analysers are devices used to measure charged particle energy distributions – e.g. electron or ions.

Table 7: Space-based SWE observation instruments (source: Peitso, 2013:20-24)

Space-based observations can be categorised into two distinct types of SWE measurements - in-situ observation systems and remote sensing. In-situ technologies observe the resultant effects of solar activity on the near-Earth environment from within the magnetosphere; these include "hosted payload instruments", or "hitchhikers[52]" flown on spacecraft (increasing economic efficiency), small-sats, and cubesats[53]. Remote sensing observation, on the other hand, aims to boost forecasting and nowcasting capabilities by directly monitoring solar activity outside of the magnetosphere, i.e. the solar corona and free solar wind.[54] Some of the prominent European and international SWE missions and instruments are shown, as examples, in Table 8.

[50] European Space Agency, 2017b
[51] Hapgood, 2001
[52] Hapgood, 2001
[53] European Space Agency, 2017b
[54] European Space Agency, 2017b

Mission/Satellite	Details	Type of Observation
Solar and Hemispheric Observatory (SOHO)[55]	• NASA and ESA • Launched in 1995	Studying the internal structure of the Sun, the Sun's outer atmosphere, and the origin of solar wind
Advanced Composition Explorer (ACE)[56]	• NASA • Launched in 1997	Observing particles of solar, interplanetary, interstellar, and galactic origins, spanning the energy range from solar wind ions to galactic cosmic ray nuclei[57]
Cluster	• ESA • Launched in 2000 (four satellites)	To observe small-scale structures of the magnetosphere and its environment in three dimensions[58]
Double Star	• CNSA and ESA • Launched in 2003 and 2004 (two satellites)	Investigating the effects of the Sun on the Earth's magnetosphere
Solar Terrestrial Relations Observatory (STEREO)[59]	• NASA • Launched in 2006	Stereoscopic measurements of the sun and SWE phenomena (e.g. CMEs)
Deep Space Climate Observatory (DSCVR)	• NOAA • Launched in 2015	Monitoring CMEs and solar wind

Table 8: European and International space-based observatories and instrument networks

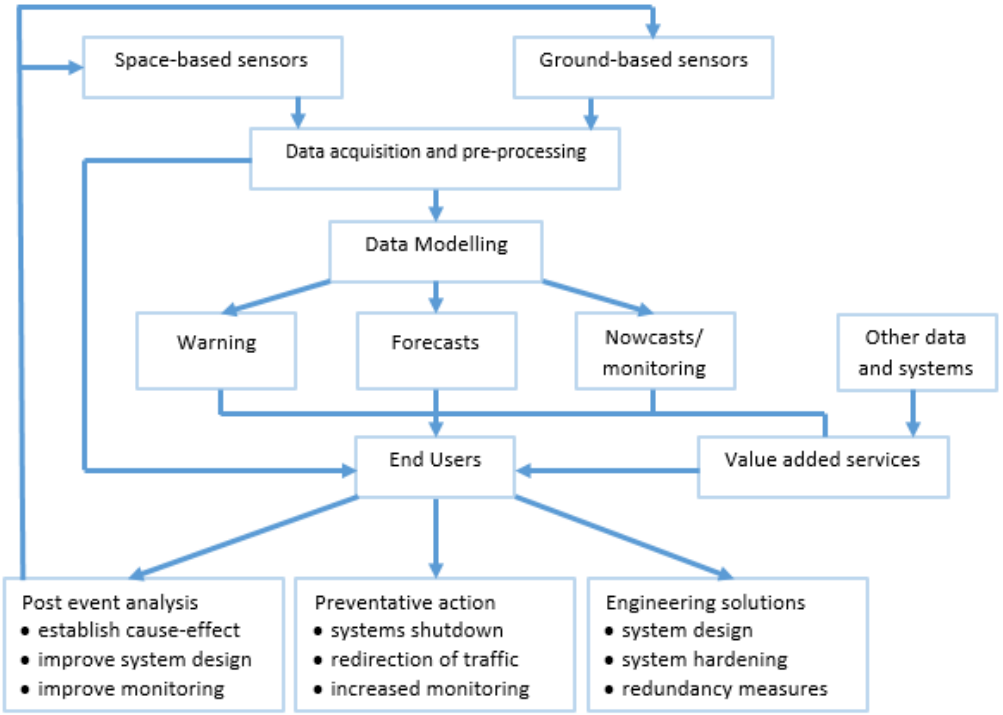


Figure 4: Importance of SWE modelling in the supply chain (source: Shaw, 2001)

[55] Garner, 2017
[56] Christian and Davis, 2017
[57] National Aeronautics and Space Administration, 2016a
[58] European Space Agency, 2005
[59] National Aeronautics and Space Administration, 2017

c. Software Enablers

An important component for the provision of SWE services is the software that makes them possible. As made visible through the technological enablers, there is a multitude of different missions and observational technologies (ground- and space-based), which record an even larger variety of types of data. This requires compatibility of these technologies, adequate data management, storage, and processing techniques. In addition, techniques of turning the available SWE data and information into reliable models rely heavily on effective software. Data assimilation into models is indeed a central activity for the successful provision of SWE services, as outlined in Figure 4.

Accordingly, the provision of predictive and forecasting services depends on accurate modelling systems that need to take into account a plethora of factors, and be continuously updated as novel data and insights emerge. From the perspective of users, who expect forecasts, alarms, warnings etc., there is a need for appropriate systems and interfaces with which to communicate such information, often through visual means of graphical presentation.

In the European context, SWE data from ESA missions is processed and stored at the SWE Data Centre at ESA Redu, Belgium. From here, the "calibrated and verified measurement data will then be disseminated in near-real-time to the teams in the Space Weather Service Network for further processing, validation and utilisation in in customer/end-user applications and services"[60] Beyond this, the provision of SWE services to users requires appropriate software that makes such SWE data products accessible, presentable and intelligible, and often in real-time scenarios, in the form of user interface platforms – e.g. the ESA SSA Space Weather Service Network, with individual Expert Service Centres designed for specific customers.[61][62]

2.3.2 Market Enablers

For sustainable and successful SWE services to be available, it is essential that there be a stable user base across multiple sectors. This requires sufficiently designed processes for user identification, sectorial analysis of user requirements, and essentially a functional service product that fulfils user needs, which also demands adequate processes for user feedback.

The first step is to promote the use of SWE services among potential user communities. Actions need to reach, inform, and educate new user communities to increase their understanding of how SWE services can address their needs. Given the transversal impact of SWE events, and the growing demand for SWE data products, there are several well-defined service domains in which SWE services can be of use. The identification of these domains can be directly derived from the general areas of impact of SWE presented in Section 2.1. A graphic illustration is presented in Figure 5.

As evident from Figure 5, two macro-categories of domains can be identified: those relevant to space operations: spacecraft design, spacecraft operation, human space flight, launch operation – and those concerning ground operations: aviation, rail, resource exploitation, power grid operation, pipeline operation, and auroral tourism. Several space and non-space operations (e.g. satellite using satcom links, transport and finance), can be impacted either directly or indirectly, as a result of their reliance on space assets.

Appendix 3 will detail SWE impacts on various service domains, addressing the aspects of user identification, the potential benefits for each sector, and the technological systems needed to fulfil them.

Once users are identified, the second objective is to establish an effective platform for dialogue between the service providers and users in order to gauge their needs. This task has become central in the actions of many space agencies worldwide, as well as in the work of several international organisations (e.g. ICAO and the WMO). Also in the European context there are already a number of publications on the relation between user identification and relative requirements, and the technological system requirements needed to meet demand[63]. A succinct overview of SWE services' end users and their relative requirements is provided in Table 9.

[60] European Space Agency, 2017b
[61] European Space Agency, 2018c
[62] European Space Agency, 2018d

[63] Horne, 2001; European Space Agency, 2011; European Space Agency, 2013

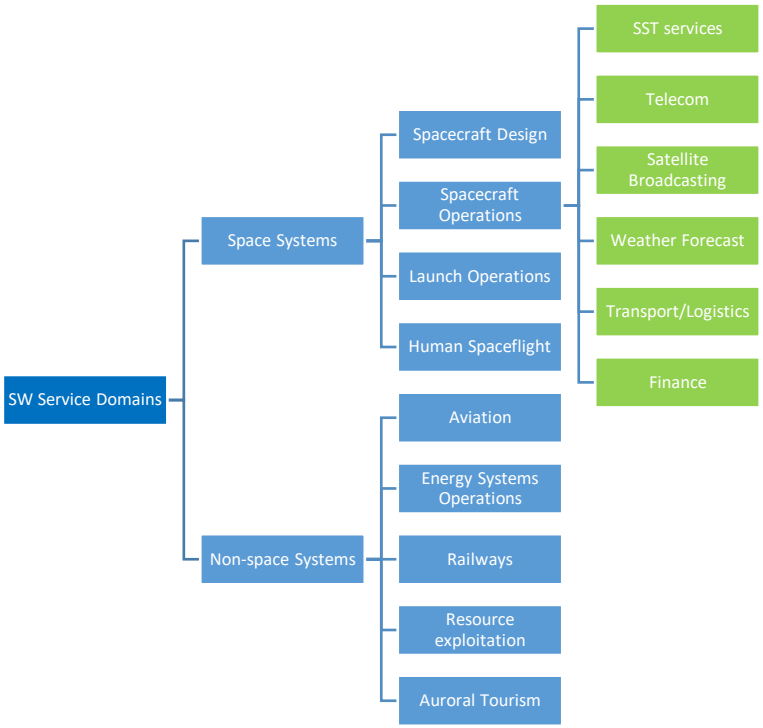


Figure 5: SWE Services Domains (sectors impacted indirectly in green)

Domain	***End Users***	***Requirements***
Spacecraft Design	• Personnel involved in generating space environment specifications for the design of spacecraft	• Prevent over-design • Establish common design standards • More reliable satellite operations • Achieve longer design life • Identify risks due to new technology
Spacecraft Operations	• Flight Control Teams and operations support teams of European and national space agencies. • Public and private spacecraft operators.	• Reduce risk of anomalies and failures • More reliable service provision • Reduction in lost revenue • Extended satellite lifetime • More competitive service • Better planning for orbit manoeuvres • Reduced risk of uncontrolled re-entry • Better station keeping • Conservation of fuel • Reduce risk of collision damage with debris • Continuity and reliability of service
Human space flight	• Operations teams for human spaceflight including during launch, activities inside and outside of the ISS. • Future space tourism flight operators and future human missions in outer space.	• Optimisation of launch procedures • Reduced radiation dose to astronauts • Cost savings
Launch operation	• Personnel involved in launch operation of space agencies and European entities operating launchers	• Optimisation of launch procedures • Cost savings

Space Surveillance and Tracking	• Personnel involved in the Space Surveillance and Tracking segment of the SSA system.	• Adequate modelling of the geospace environment • Improved precision of SST methods • Prevention of damage to space systems • Increased awareness and understanding of re-entry events.
Electrical Power Grids	• Power grid operators	• Better service continuity • Minimise lost revenue due to down time • Reduced risk of transformer damage • Better planning of generating capacity, identify system behaviour, and identify total risk • Identify links to space weather
Aviation	• Commercial airline companies • Aircraft manufacturers	• Optimisation of flight-paths • Reduced radiation exposure to aircrew and passengers • Compliance with legislation • Reduced cost of having to re-schedule aircrew • Increased flight safety
Railways	Railway Operators	• Increased safety of signalling and train control • Reduced cost of delays
Pipeline Operators	Pipeline Operators	• Improving the management and lifetime of pipelines
Resource Exploitation	Geophysical Resource surveyors and extractors	• Better planning of aerial surveying • Reduced loss of revenue due to corrupt data • Reduced interruptions to drilling operations Reduced loss of revenue due to errors in navigating drill heads.
Auroral Tourism	Tourism companies and commercial airlines	• Enables predictions of the aurora • Development of tourist market

Table 9: Overview of SWE service domains, users, and requirements

Whereas identifying customers and customers' needs is a crucial step towards creating effective SWE services, it is also clear that different customers have different needs. Therefore, a third enabling step is to address the segmentation of market demand, and the likely presence of common requirements, by federating demand. This step consists of aggregating fragmented user communities and leading them to expressing common requirements. It is a difficult exercise, especially because user communities making use of SWE services are not homogenous, comprising both civil and military users, ranging from satellite operators to airline companies, and power grid operators.

Within the European context, three broad categories of users can be identified for SWE services:

- Commercial entities
- Public civil authorities
- Military authorities

On the basis of these types of users, and in relation to the domains described above, several high-level requirements for user needs have been outlined, namely:[64]

- Provision of comprehensive knowledge, understanding and maintained awareness of the natural space environment and space weather;
- The detection and forecasting of space weather and its effects;
- The detection and understanding of interferences due to space weather;
- The prediction and/or detection of permanent or temporary disruption of mission

[64] European Space Agency, 2011:14; European Space Agency, 2013:24

and/or service capabilities due to space weather;

- The monitoring of the Sun, the solar wind, the radiation belts, the magnetosphere and ionosphere to the extent that it supports services related to effects that include radiation and spacecraft charging hazards, spacecraft drag, ionospheric perturbations, aircraft radiation hazards, geomagnetic disturbances, and current induced in large conductive networks such as power lines and pipelines;
- The provision of all required predicted local spacecraft and launcher radiation, plasma and electromagnetic environment data.

From these high-level user requirements, a set of system requirements then needs to be derived by the stakeholders involved in the provision of SWE services. In the ESA Space Weather Study led by the Rutherford Appleton Laboratory (RAL), for instance, a set of general requirements that describe what a space weather service must deliver has been created, together with the basic functionality that should be associated with each user requirement (see Table 10).

In addition to all these steps, market factors dependent on the typology of service (i.e. how the user or customer receives or pays for a service) need to be reflected within the organisational elements of service development and provision, allowing for continued reinvestment in the development of new services and the procurement of new technologies. Irrespective of whether the source of funding is private or public, the capacity to monetise such services requires investigation, especially within the European context where clear cases for public good SWE services have been made.

User Requirement	Service Function Element
Timely and reliable data from multiple sources	Networked and reliable data access
	Retrieval scheduler
Good documentation of system and data	On-line help
	Comprehensive metadata
	Human support
Consistent interface with multiple datasets	Generic, comprehensive and accessible data output format
Easy to identify relevant datasets	Data dictionary
	Yellow pages system
Access to enhanced products	Data aggregation
	Models and forecasts
Access to past data	A local archive of relevant data
Personalised regular data retrieval	User accounts with personal profiles
Access to informed advice and scientific technical supports	Technically and scientifically competent personnel
Background information on science and impart of space weather	On-line introduction to space weather
	Outreach materials
Graphical presentation	Graphics engine
Continuous service development	Regular service monitoring
	User feedback facilities
	Medium to long-term strategy

Table 10: User requirements and service functions (source: Hapgood, 2001)

2.3.3 Organisational Enablers

To enable the successful delivery of operational services, the availability and continuity of SWE data and services should be secured, and sustainable governance and funding of the services should be set up.

Towards this, well-defined relations, and even synergies, between different stakeholders are critical. As outlined in Section 2, there are different categories of stakeholders across the supply-demand value chain of any operational service delivery; in the context of SWE services these have been summarised in Figure 6.

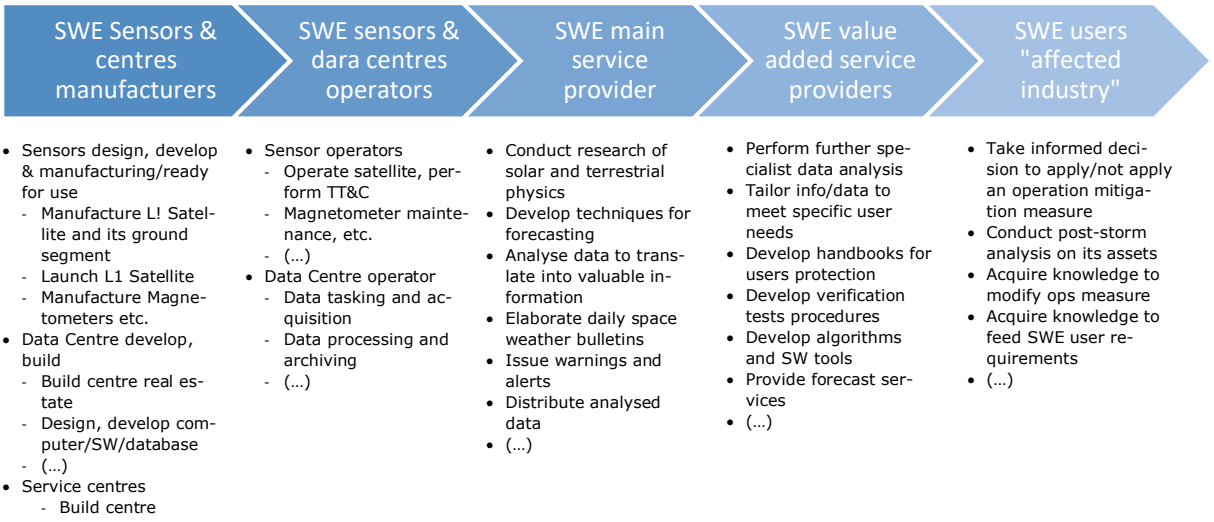


Figure 6: SWE Service Value Chain (source: adapted from PricewaterhouseCoopers, 2016)

The relations between the different stakeholders can be based on varying organisational architectures characterised by different degrees of government or commercial ownership, and different degrees of international cooperation.

The need for international cooperation on SWE service provision is most apparent because of its indiscriminate global-reaching impacts, exacerbated by the dual effects felt on both ground- and space-based infrastructures, and even more so as global reliance on these intertwined infrastructures increases; "even temporary loss of services from global navigation satellite systems (GNSS) would have an impact on numerous transportation sectors and potentially the global financial system, which relies on accurate timing".[65] Accordingly, UN COPUOS has expressed that "effective progress in advancing space weather services requires coordinated global actions that will serve to focus efforts on the needed forecasting, monitoring and awareness raising with the goal of protecting life, property and critical infrastructure".[66] This, of course, requires not only political will but also mechanisms for cooperation from the international community through international entities, with increased international coordination in "both resilience analysis and scientific research with a view to improving future space weather services and impact mitigation".[67]

However, different mechanisms for action on SWE issues can take varying forms depending on the typology of service that is being developed – i.e. national or international, public or commercial services. The most prominent example of contrasting models for such services is clear when looking at the U.S. SWE service setting, where there exist both public typologies of services (functioning under an open/free data policy for users) and private ones (provided by private companies on a subscription basis to specific customers). Additionally, it is still the case in the U.S. that public entities such as the NOAA Space Weather Prediction Centre (NOAA SWPC) provide both national typologies of services (e.g. alerts and warnings communicated to U.S. government agencies and infrastructure operators in case of SWE events)[68] and international services – e.g. the global SWE services to be provided under the ICAO framework to the aviation sector (see Chapter 3 for a more detailed overview).

[65] United Nations Committee on the Peaceful Uses of Outer Space, 2017a:2
[66] United Nations Committee on the Peaceful Uses of Outer Space, 2017a:3-5
[67] United Nations Committee on the Peaceful Uses of Outer Space, 2017a:3
[68] Krausmann et al., 2016:10

3. European and International Efforts in SWE

3.1 The European Architecture for SWE Services

3.1.1 Background: From National to Pan-European Efforts

Over the past decades there have been increasing efforts to establish SWE as a discipline in Europe. While these have been primarily led by national institutions of most European member states (an overview of which is provided in Annex A.4), pan-European approaches have progressively been devised.

A first European round table on Space Weather was organised by ESA in 1996 with the goal of investigating options for a European counterpart to the US National Space Weather Programme. Two years later ESTEC launched the first Space Weather workshop for furthering perspectives on a coordinated effort in the field of Space Weather. Since then, ESTEC has maintained a key role in structuring this scientific field in Europe, primarily by organising annual workshops from 1998 to 2003. During the same period, it financed a feasibility study on a European Space Weather Programme. To conduct the study, two international consortia were appointed and a Space Weather Working Team (SWWT) was created to coordinate their work.

The two studies, published in 2001, clearly determined that Europe had very strong assets in the physics and effects of Space Weather that could potentially be exploited in European or International SWE services. While they proposed several strategies to progress SWE activities in Europe, the baseline assessment was that "the coordination and development of Space Weather services would enhance the efficiency of these activities and provide new opportunities for the use of resources across domains that are currently separated from each other". [69]

Following the conclusion of the two feasibility studies, the SWWT continued to play an active role in advising ESA on the space weather strategy, acting as an open forum for discussion amongst the European space weather community, and promoting coordinated European space weather activities at both national and industry levels. One of the recommendations made by the SWWT was to apply for an action targeting Space Weather science under the umbrella of the European Cooperation in Science and Technology (COST), an intergovernmental framework for supporting the coordination of nationally funded research at a European level.

Under COST, two actions were started to develop an interdisciplinary network among European scientists and researchers in the field of SWE:

- COST 724 (running from November 2003 to November 2007)[70]
- COST ES0803 (running from June 2008 to November 2012)[71]

The main objective of *COST 724* was to "develop further within a European framework the science underpinning space weather applications, as well as exploring methods for providing a comprehensive range of space weather services to a variety of users, based on modelling and monitoring of the sun-Earth system"[72]. The main motivation and benefit of establishing a European Space Weather programme was highlighted as providing organisations at risk of SWE impacts with a resource

[69] Lilensten, 2008
[70] European Cooperation in Science and Technology, 2002
[71] European Cooperation in Science and Technology, 2008
[72] European Cooperation in Science and Technology, 2002

to allow them to manage such risks through a system with traceable quality standards[73].

At the time, there was a distinct user reliance on SWE information provided by the U.S. Space Environment Centre (SEC), and so to provide a counterpart to the activity within the USA, the Action set out and agreed upon a European definition of SWE, and a number of general aims:

- To coordinate European research into modelling and prediction of space weather;
- To promote where necessary the deployment of new instrumentation to satisfy data requirements, and the development of new models;
- To educate potential users of space weather data;
- To gather feedback from users which may be used to improve services;
- To create a forum for exchanging "best practice" among users and providers of space weather services;
- To set standards on data exchange.

This action included four Working groups to achieve its aims. Working Group 4 (WG4) was explicitly on Space Weather Observations and Services and was tasked with implementing the basis of the European Space Weather Network; coordinating the network of data, models, prediction, public outreach, developing methods and standards for coupling varying SWE models, and disseminating information to users.[74]

COST Action ES0803 on Developing Space Weather Products and Services in Europe had the primary objective of fostering "an interdisciplinary network between European scientists dealing with different issues of Geospace, as well as warning system developers and operators"[75] in order to:

- Foster the ties between European Geospace research and space technology establishments,
- Assess the European potential in advanced Space Weather observational and modelling techniques and in reliable products and services,
- Define the needs of a broad range of users and,
- Determine and recommend the specifications for new products and services that best meet the user's requirements.[76]

In this context, Europe had a very strong basis of scientific research on the physics and effects of SWE, however its optimal use was limited by a lack of coordination between national research programmes[77]. Accordingly, the primary goal of this Action sought to address this issue, and was accompanied by more specified secondary objectives:[78]

- Cross-disciplinary collaboration of Geospace researchers with biologists, physicians, engineers and economists, to form an interdisciplinary network for the more efficient study and modelling of the space weather effects on technological and biological systems and to explore how space weather effects can interact with the economic behaviour of key infrastructures such as power grids
- Stimulation of development and delivery of reliable computer codes for predicting key Space Weather parameters
- Further development of partially existing links between the space weather research community, space weather service providers and space weather service users to their mutual benefits
- Efforts to launch a new European Journal on Space Weather in collaboration with the European Geosciences Union
- Training, through organisation of dedicated courses, of young researchers and post graduate students
- Raise public awareness through targeted outreach and education activities

Action 803 had a number of intended societal, scientific and technological benefits in developing collaboration, coordination and strengthening the SWE research community within Europe – i.e. establishing a network for coordination of research and applications amongst European researchers, and creating closer links between the SWE research community and SWE impacted industries within Europe – as well as recommending and improving on validation of SWE models, ultimately the quality of SWE services, and increasing public awareness on SWE issues[79].

Importantly, these two notable Actions under COST also induced the EU and ESA to take a more integrated and pro-active role in the field

[73] European Cooperation in Science and Technology, 2002
[74] European Cooperation in Science and Technology, 2002a:11
[75] Belehaki, 2010
[76] Ibid.
[77] Ibid.
[78] Crosby, 2010
[79] European Cooperation in Science and Technology, 2008

of SWE, which has essentially increased European capacity in SWE research and applications since 2003.

In addition to the SWWT and COST, it is also important to acknowledge the role played by the recently established "European Space Weather Assessment and Consolidation Working Group" within the European Space Science Committee (ESSC).[80] Consistent with ESSC's mandate to deliver independent scientific advice to ESA, the European Commission, European national space agencies, and other decision-makers on space matters, this new Working Group (chaired by Hermann Opgenoorth, IRF, Sweden) aims to prepare detailed recommendations for a consolidated and strategic European approach to SWE, within which identifying the appropriate efforts and investments that need to occur in all parts of the so-called SWE "progress iteration loop", which is "defined by:

- new science understanding
- the improved potential to deliver SWE products (based on the most recent science findings)
- evolving requirements of European end-users and infrastructure providers".[81]

3.1.2 European Space Agency

Building on the recognition that Europe as a whole had matured "a wealth of expertise and assets providing high-quality scientific data and, in some cases, space weather 'products' to a wide variety of customers", but that these capabilities remained "largely fragmented across national and institutional boundaries"[82], the European Space Agency (ESA) was mandated by its Member States to initiate efforts in the field with the aim of avoiding duplication of efforts, federating the existing assets, and ensuring the timely dissemination of reliable services for customers.

Since 2008, ESA's SWE activities have been undertaken as part of the Space Situational Awareness (SSA) Programme (see Box 1 for an overview of this programme).

Box 1: ESA SSA Programme

ESA's SSA programme was first approved in 2008 and implemented as an ESA optional programme with financial contributions from 19 Member States.[83] The programme was funded at approx. €240 million for the period 2009-2020 and has been executed in periods, namely:

- Period 1 decided at MC in November 2008 and running from 2009 to 2012
- Period 2 decided at MC in November 2012 and running from 2012 to 2016
- Period 3 decided at MC in November 2016 and running from 2017 to 2019.

The overarching objective is "to support Europe's independent utilisation of, and access to, space through the provision of timely and accurate information and data regarding the space environment, and particularly regarding hazards to infrastructure in orbit and on the ground".[84] Ultimately, the SSA programme intends to "enable Europe to autonomously detect, predict and assess the risk to life and property due to man-made space debris objects, re-entries, in-orbit explosions, in-orbit collisions, disruption of missions and satellite-based service capabilities, potential impacts of Near-Earth Objects (NEOs), and the effects of space weather phenomena on space- and ground-based infrastructure"[85]

The programme comprises three major segments:

- Space Surveillance and Tracking (SST) for detecting active and inactive satellites, discarded launch stages and fragmentation debris orbiting Earth
- Space Weather (SWE), for watching and predicting the state of the Sun and the interplanetary and planetary environments, including Earth's magnetosphere, ionosphere and thermosphere, which can affect space-borne and ground-based infrastructure thereby endangering human safety
- Near-Earth Objects (NEO), for watching natural objects that could potentially impact Earth and assessing their impact risk and potential mitigation measures.[86]

[80] European Space Science Committee, 2018
[81] http://sites.nationalacademies.org/cs/groups/ssbsite/documents/webpage/ssb_182864.pdf
[82] European Space Agency, 2018e
[83] European Space Agency, 2018f
[84] European Space Agency, 2018e
[85] Ibid.
[86] European Space Agency, 2018f

Within ESA's SSA programme, the major focus is on SWE-related activities, as evident when looking at the breakdown of the financial envelope agreed upon at the ESA Ministerial Council of 2016 (Figure 7) and at the participation of ESA Member States in the various SSA segments (Table 11).

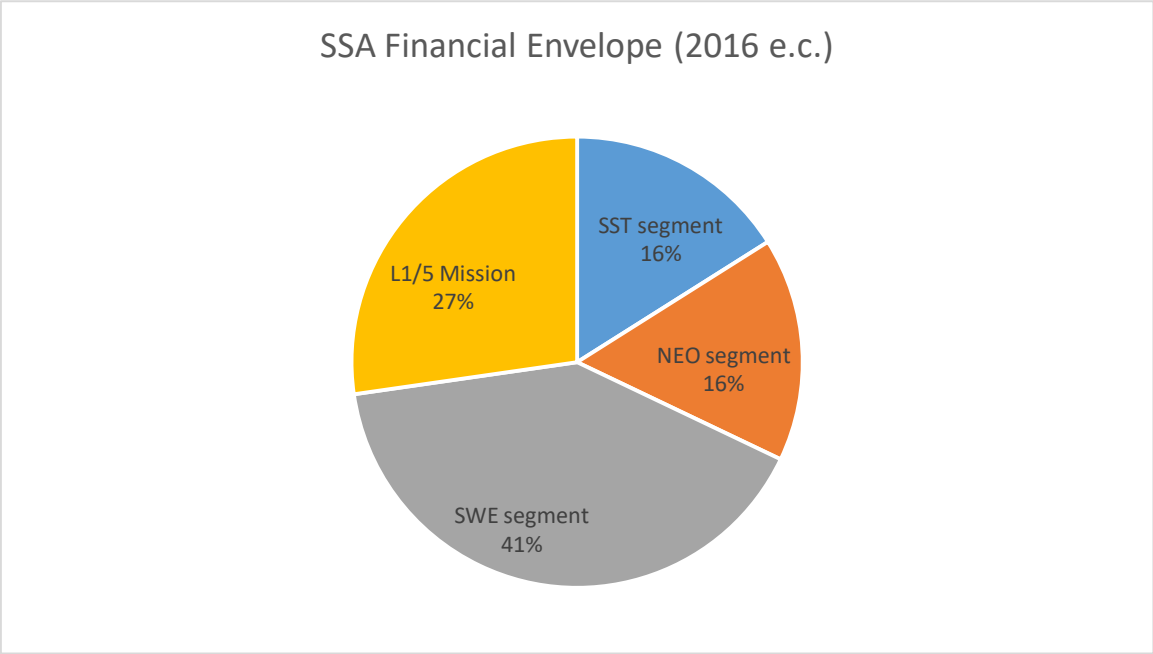


Figure 7: SSA Budget Breakdown (source: Space Weather Working Team, 2016)

Of the €187 M allocated at the M/C 2016, 68% (€ 127 M) was devoted to SWE-related activities (€76 M for the SWE segment and + €51 M for the preparation of the Lagrange mission).

Currently, the large majority of ESA Member States (19 out of 22) participate in the SSA programme, with the UK, Germany and Italy being the top three contributors. Of these 19 participating states, 17 have invested in the SWE segments, 11 in the SST segment, 10 in the NEO segment, and 7 in the L1/L5 mission preparation.

	SWE	NEO	SST	L1/L5
Austria	X	X	X	X
Belgium	X	-	-	-
Czech Republic	X	X	X	X
Denmark	X			
Finland	X	X	X	-
France	X	-	-	-
Germany	X	X	X	X
Greece				
Italy	X	X	X	-
Luxembourg	X	X		
Netherlands	X			
Norway	X	-	X	-
Poland	X	X	X	
Portugal				
Romania	X	X	X	X
Spain	X	X		
Sweden	X	-	X	X
Switzerland	X	X	X	X
United Kingdom	X	X	X	X

Table 11: SSA Programme Status Priorities of Member States (source: Bobrinsky, 2017)

Within the SSA programme, the stated objective of the Space Weather Segment is "to provide owners and operators of critical space-borne and ground-based infrastructure timely and accurate information to enable mitigation of the adverse impacts of space weather". [87] In order to achieve this objective, ESA has been federating existing assets and capabilities into a unified space weather network for the delivery of space-weather applications tailored to European user needs. The ultimate objective of the network is to "enable end-users in a wide range of affected sectors to mitigate the effects of space weather on their systems, reducing costs and improving reliability"[88]

SWE Segment Infrastructure and Organisation

ESA has not set up its own SWE centre but coordinates a virtual network of SWE service centres in various ESA member states. At present, ESA's Space Weather Service Network is organised around three major elements, namely:

- The SSA Space Weather Coordination Centre (SSCC)
- The Expert Service Centres (ESCs)
- The SWE Data Centre (SDC)

The SSCC, located at the Space Pole in Belgium, coordinates the provision of SWE data and services that are available at the SWE Data Centre or at federated sites. The SSCC has also set up a "European Space Weather Helpdesk", where operators provide first-level user support and answer questions about space weather conditions in general or the SWE precursor service network.[89]

ESA SWE data and products are gathered under the themes of Solar Weather, Space Radiation, Ionospheric Weather, Geomagnetic Conditions and Heliospheric Weather, and provided through the work of the five Expert Service Centres (ESCs). Each ESC consists of a distributed group of experts from organisations across Europe who collaborate to provide tailored data, products and/or expertise and services for Space Weather Network customers.[90] The five ESCs are organised according to the above-mentioned domains, each having a coordinating institute (see Table 12):

ECS Thematic Priority	Coordinating Institute	Function
Solar Weather	Royal Observatory of Belgium Brussels, Belgium	Monitor and forecast solar activity from beneath the solar surface into the corona, and events and processes that drive space weather in our solar system
Space Radiation	Royal Belgian Institute for Space Aeronomy (BIRA-IASB) Brussels, Belgium	Monitors and forecasts space particle radiation (ambient plasma, solar energetic particles, radiation belts, galactic cosmic rays), micron-size particulates from meteoroids and space debris as well as all types of resulting effects on technologies and biological systems
Geomagnetic Conditions	Tromsø Geophysical Observatory (TGO), Tromsø, Norway	Monitor and forecast varying conditions in the Earth's magnetosphere, on various timescales, which may lead to induced currents generated in power distribution systems or long pipelines, disrupt magnetic surveying, and influence resource exploitation
Ionospheric Weather	DLR Ionosphere Monitoring and Prediction Centre (IMPC), Neustrelitz, Germany	Monitor and forecast ionospheric and upper atmospheric conditions, in particular the disturbances resulting from solar and geomagnetic activity that may impact radio signal propagation or lead to increased satellite drag
Heliospheric Weather	STFC RAL Space, Harwell, UK	Monitor and forecast changing conditions in interplanetary space that may lead to disturbances in

[87] European Space Agency, 2018g
[88] European Space Agency, 2018h
[89] European Space Agency, 2018h
[90] European Space Agency, 2018i

		space weather conditions at Earth and at other locations in the heliosphere

Table 12: ESCs and their respective coordinating institutes and functions (source: European Space Agency, 2018i)

As shown in the SSA Space Weather Network Service Product Catalogue Summary of 2018, SWE products are classified according to the ESCs and Expert Group. The different ESCs and contributing number of Expert Groups currently providing SWE data products are summarised in Table 13.

	Number of SWE data products	**Number of Expert Groups**
Solar Weather	26	5
Space Radiation	32	9
Heliospheric Weather	15	5
Ionospheric Weather	60	8
Geomagnetic conditions	31	6

Table 13: Quantity of SWE data products and expert groups for each ESC (source: European Space Agency, 2018j)

A more detailed account of the expert groups affiliated with the five ESCs, and of the related data products, is provided in Annex 2.

The third major element supporting ESA's Space Weather Network is the SWE data centre hosted at ESA's European Space Security and Education Centre in Redu, Belgium. The centre functions as a large data repository collecting data from the five ESCs, federated data archives, and collaborating sensors systems. It also hosts and provides access to the ESA Space Weather Service Network portal.[91]

As a whole, ESA's Space Weather service network has been organised to provide a variety of services distributed over 8 service domains targeting its specific groups of end users. The service domains and relative services are illustrated in Figure 8.[92]

[91] European Space Agency, 2018d

[92] European Space Agency, 2018k

Spacecraft Design (SCD)	Environment specification: data archive
	Environment specification: in orbit verification
	Post event analysis
	Space Weather in the Solar System
Spacecraft Operation (SCO)	In Orbit Environment and Effects Monitoring
	Post-event analysis - UNDER DEVELOPMENT
	In-orbit environment and effects forecast - UNDER DEVELOPMENT
	Mission risk analysis - UNDER DEVELOPMENT
	Space Weather in the Solar System
Human Space Flight (SCH)	In-flight Crew Radiation Exposure
	Cumulative Crew Radiation Exposure
	Increased Crew Radiation Exposure Risk - UNDER DEVELOPMENT
Launch Operation (LAU)	In-flight monitoring of radiation effects in sensitive electronics - UNDER DEVELOPMENT
	Estimate of radiation effects in sensitive electronics - UNDER DEVELOPMENT
	Forecast of radiation storms - UNDER DEVELOPMENT
	Atmospheric density forecast - UNDER DEVELOPMENT
	Risk estimate of service disruption caused by ionospheric scintillations - UNDER DEVELOPMENT
	Risk estimate of micro-particle impacts - UNDER DEVELOPMENT
Trans-ionospheric Radio Link (TIO)	Near real-time TEC maps
	Forecast TEC maps
	Quality assessment of ionospheric correction
	Near real-time ionospheric scintillation maps
	Monitoring and forecast of ionospheric disturbances
Space Surveillance & Tracking (SST)	Atmospheric estimates for drag calculations - UNDER DEVELOPMENT
	Archive of geomagnetic and solar indices for drag calculation
	Forecast of geomagnetic and solar indices for drag calculation - UNDER DEVELOPMENT
	Nowcast of ionospheric group delay - UNDER DEVELOPMENT
Non-space System Operation (NSO)	Service to power systems operators
	Service to pipeline operators
	Service to airlines
	Service to resource exploitation system operators
	Service to auroral tourism sector
General Data Services (GEN)	Space weather data archive
	Latest data guaranteed service
	Space weather nowcast and forecast products (daily, weekly)
	Event based alarms
	Virtual space weather modelling system - UNDER DEVELOPMENT
	Guaranteed data service for third-party/added-value service providers - UNDER DEVELOPMENT
	Space Weather Support Material - UNDER DEVELOPMENT

Figure 8: ESA SWE Services Overview (source: adapted from European Space Agency, 2018k)

Current Status and Planned Developments

As of mid-2018, ESA's network has been receiving raw data from a large number of ground- and space-based sensors enabling over 140 separate products providing scientific and pre-operational applications for 24 (out of the 39 services) to be provided to users (see Chapter 6.1). Most these services are currently in a pre-operational phase and, as reported by ESA, the broader Space Weather Network is "in an intensive development phase targeted at developing both customer-tailored interfaces and key models as well as other building blocks that will contribute to improving the accuracy of the information that can be provided to end-users. The Network is now being managed in a 'pre-operational' framework, with live support available only during normal working hours. In the future, steps will be taken to mature the service provision system

and prepare the network for transition to a fully operational framework".[93]

During the current phase of the programme (2017-2020), activities are taking place in the following areas:

- Federating additional data sources and expanding the Space Weather Service Network with new European capabilities, including development and integration of new state-of-the-art models and tools enabling enhanced nowcasting and forecasting of space weather conditions
- Maturing and enhancing space-weather products and services
- Supporting the development of new applications and enhancement of user interfaces
- Supporting the development and deployment of new cutting-edge sensors
- Placing space weather sensors on-board hosting missions from ESA or other partners
- Completing studies leading to future dedicated space-weather missions

As evident from this list of actions, together with advancements in ground-based measurements and computational capabilities, ESA has initiated efforts to complement ground observations with "in-situ" measurements from space. These measurements, which aim to ensure constant monitoring of the Sun and the broader space environment from a range of vantage points, will consist of:

- Hosted payload instruments that will be flown on spacecraft operated by ESA or by other organisations and will typically comprise monitoring of particles and fields within the magnetosphere and Auroral images
- Dedicated 'SmallSat' or cubesat missions "to complement hosted payload instruments and cover all the needed measurements. The hosted payload instruments and potential SmallSat missions will form the SSA 'Distributed SWE Sensor System' (D3S). Observations gathered by D3S will particularly benefit satellite operations, spacecraft engineering, anomalous-event analysis and space-environment impact studies".
- Dedicated missions outside the Earth's magnetosphere providing remote sensing of the Sun, the solar corona and the free solar wind (see Box 2 for an overview of ESA's Lagrange point space weather missions).

Box 2: ESA's Future SWE Missions

Together with the launch of dedicated SmallSats missions, ESA has initiated the assessment of two possible future SWE missions that will ensure a robust capability to nowcast and forecast potentially dangerous solar events. This assessment currently envisages positioning two spacecraft at the Lagrangian points so as to have a stable location from which to make observations. In particular, "of the five Lagrangian points of the Earth-Sun system, L1 and L5 have been deemed very good locations from which spacecraft can monitor interplanetary space and solar activity". More specifically, this is because:

- The L1 point is "located in the solar wind "upstream" from Earth, so measurements at L1 provide information about the space weather coming toward Earth".
- The L5 point, "located 60 degrees behind Earth, close to its orbit, will provide a way to monitor Earth-oriented coronal mass ejections (CMEs) from the 'side' so as to give more precise estimates of the speed and direction of the CME".

These measurements will be used to provide space weather warnings, alerts and status information to a variety of customers here on Earth. More specifically, the objectives of the L1 and L5 mission have been defined as:

- The primary objective of the L1 mission is "to provide in-situ observations of the interplanetary medium, including solar wind speed, density, temperature and dynamic pressure, as well as characteristics of the charged particle environment and the direction and strength of the Interplanetary Magnetic Field (IMF). The L1 mission will also monitor the solar disc and solar corona and measure solar energetic particles that may be associated with solar flares and the onset of coronal mass ejections".
- The L5 mission objective is "to complement measurements made from L1 by providing a view of the Sun away from the direct Sun-Earth line. This gives visibility of the propagation of plasma clouds emitted by the Sun toward Earth, as well as views of the solar disk before it rotates into view from Earth. The L5 mission will carry out heliospheric imaging of the space between the

[93] European Space Agency, 2018g

Sun and Earth, monitoring of the solar disc and corona and carry out measurements of the interplanetary medium".

To meet these objectives, "the satellites at the L1 and L5 positions have to carry different types of remote-sensing and in-situ instruments".[94] More specifically:

- Remote sensing instruments shall include:
  - Coronagraph
  - Heliospheric Imager
  - Magnetograph
  - EUV Imager
  - X ray flux monitor
- In situ instruments shall include:
  - Magnetometer
  - Plasma analyser
  - Medium energy particle spectrometer
  - Radiation monitor

The L1/L5 missions and the next SWE segment of the ESA SSA programme are to be approved at the ESA Ministerial Council of 2019. Overall, ESA's roadmap for the operationalisation of SWE services provision is summarised in Figure 9.

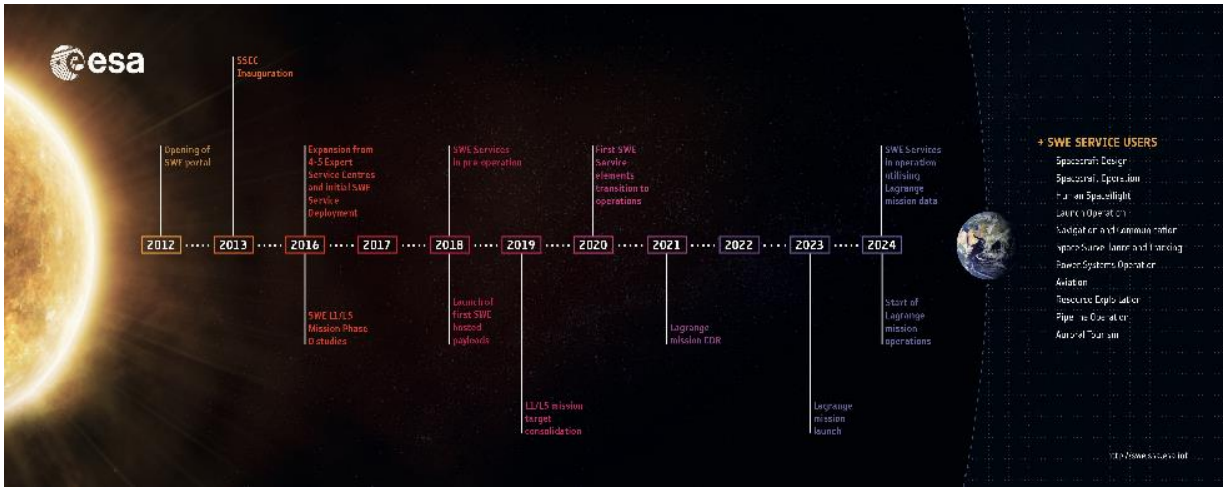


Figure 9: ESA SWE Services Timeline (credit: European Space Agency, 2018g)

3.1.3 European Union

In the context of SWE, the European Union has financed SWE-related research & innovation projects and carried out awareness raising activities. In June 2018, the European Commission proposed a Space Programme[95] that included SWE activities to provide operational services at EU level.

From a policy perspective, Council Directive 2008/114/EC on the "Identification and designation of European Critical Infrastructures and the assessment of the need to improve their protection", adopted in December 2008, was one of the first initiatives to provide an all-hazards approach to critical infrastructure protection. One such potential hazard is SWE. The EU Disaster Risk Management (DRM) policy covers prevention, preparedness, and response for all types of disasters, with risk assessment being seen as the very basis of DRM. The risk-assessment policy context is anchored in the Union Civil Protection Mechanism, which requires EU Member States to prepare a National Risk Assessment (NRA) and list the priority risks the EU is facing. As of 2018, six countries (Finland, Hungary, Netherlands, Sweden, UK and Norway) had included SWE as a priority risk in their NRAs.[96] In addition, 20 NRAs contained critical-infrastructure

[94] "The instruments on board the L1 and L5 missions will utilise technologies developed for, and tested on, earlier ESA and joint ESA/NASA solar science missions such as SOHO, STEREO and Solar Orbiter. For ESA's SSA missions, the instruments will be optimised for reliability and robustness to provide space weather monitoring data for operational applications, that is, for use in real-time systems that depend on a regular flow of data".

[95] European Commission, 2018

[96] European Commission, 2017a

loss or power outage scenarios as priority hazards. Clearly, SWE can be considered as a trigger of these scenarios.[97]

The EU's recognition of the need to protect space- and-ground-based infrastructure from SWE events has served as the backbone and instigator for the enactment of two broad types of actions, namely:

- Awareness raising and research activities undertaken by the European Commission's Joint Research Centre.
- Research and innovation activities under the EU's Framework Programmes for Research and Innovation

EU Framework Programmes

Research projects related to SWE have been funded through both the 7th Framework Programme (FP7) for the period 2007-2013 and Horizon 2020 (H2020) for the period 2014-2020 (see Box 3).´

Box 3: SWE in EU Framework Programmes

SWE-related research has been strongly supported by the EU's Framework Programmes for Research and Innovation. Since 2007, a total of €63.57 million of space weather-related activities has been provided by the EU through more than 30 projects involving a plethora of organisations (including universities, research institutes and companies)

Under the Space Theme of the FP7, the EU has more specifically supported the implementation of 23 projects related to SWE with a total funding of €43.9 million. These projects have been funded in various calls under the topics of:

- Space Sciences (2007)
- Security of space assets from space weather events (2009)
- Exploitation of space science and exploration data (2010)
- Key technologies enabling observations in and from space (2011)
- Space Weather events (2012)

Under the Space Theme of H2020, the EU has financed around €10.5 million for 7 projects during the period 2014-2018 and allocated €9 million for a call in 2019. The projects have been funded in four calls under the topics of:

- Protection of European assets in and from space (2014)
- Other Actions – GNSS Evolution, Mission and Services related R&D activities (2015)
- Competitiveness of European space technology (2017)
- Secure and safe space environment (2019)

A more detailed overview of the various calls, the funded projects and their relative budgets is provided in Annex 3.

The funded projects have encompassed research on a wide range of SWE physical phenomena, SWE effects on space and ground systems, as well as the development of models and applications. In order to provide a holistic account of this plethora of projects, they have been grouped according to their main focus, as shown in Figure 10.

[97] The Union Civil Protection Mechanism also requires Member States to submit a risk management capability assessment, the purpose of which is to understand the ability of Member States to address the identified priority risks. The risk-management capability should include administrative, technical, and financial factors (Krausmann et al., 2016).

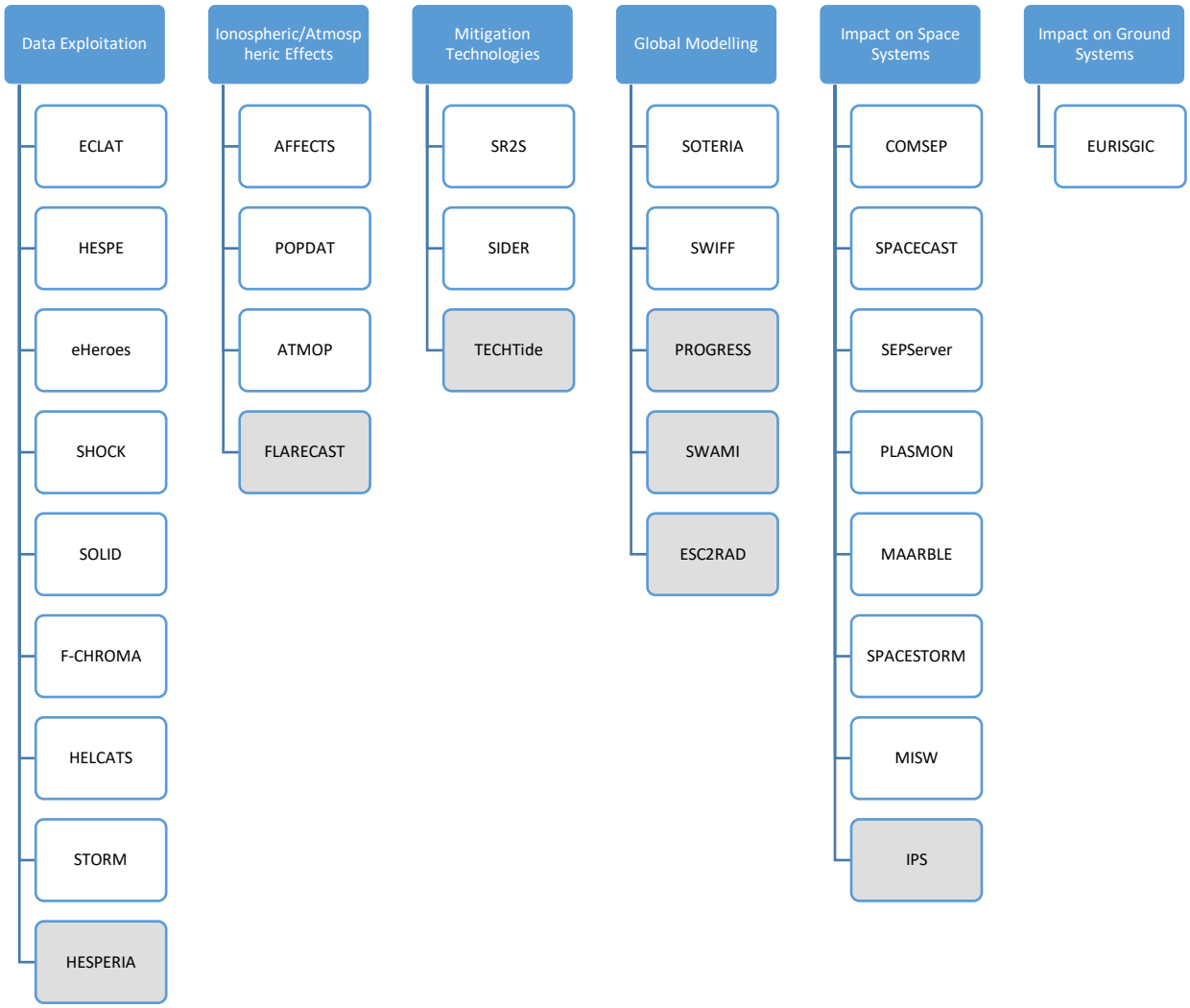


Figure 10: Projects funded by the FP7 and Horizon 2020 by category

As evident from Figure 10, the majority of research activities have focused on the exploitation of scientific data and the impact of SWE on space systems. This is mainly because the overarching themes of the calls under which SWE projects were funded were related to the security of space assets and to the exploitation of scientific data. A non-negligible focus has been also placed on the development of theoretical and simulation models for interpreting the SWE data. It is worth highlighting that the call, SU-SPACE-22-SEC-2019 - Space Weather, to be opened in 2019, will focus on addressing the "development of modelling capabilities and/or the delivery of prototype services able to interpret a broad range of observations of the Sun's corona and magnetic field, of the Sun-Earth interplanetary space, and of the Earth's magnetosphere/ionosphere coupling reliance on existing observation capacities."[98]

While providing a detailed account of the results of each of these projects exceeds the scope of the study, some examples can be provided to show that "the resources associated with these programmes have led to milestone results in space weather research, services development and international collaboration".[99] More specifically, it can be stated that the funded projects have greatly contributed to the goal of delivering operational SWE services at European level by:

- Improving understanding of the impact on space systems and terrestrial infrastructure of SWE phenomena. For instance, the project EURISGIC (European Risk from

[98] Proposals will address application domains that may include space as well as terrestrial infrastructure. Proposals will include architectural concepts of possible European space weather services in relation to the application domains addressed and they will demonstrate complementarity to and, if relevant, utilize, precursor Space Weather services already available through the Space Situational Awareness programme of ESA, and take into account global space weather service developments by the WMO" (European Commission, 2017a).

[99] Krausmann et al., 2016

Geomagnetically Induced Currents) identified the impact of Geomagnetically Induced Currents (GIC) on the electrical power networks in Europe based on in-situ solar wind observations and comprehensive simulations of the Earth's magnetosphere. Through utilisation of geomagnetic recordings and a developed prototype grid model, EURISGIC derived a geographic map that indicates the statistical occurrence of rapid geomagnetic variations and large geoelectric fields throughout Europe.

- Delivering "new insights into the detailed processes that generate space weather, with a view to enhancing the performance of space weather prediction."[100] The H2020 project HESPERIA (High Energy Solar Particle Events forecasting and Analysis), for instance, contributed to advance the knowledge of "physical mechanisms that result into high-energy solar particle events (SEPs) by exploiting novel datasets (FERMI/LAT/GBM; PAMELA; AMS)".[101]
- Supporting development of physics-based and empirical models. For instance, the FP7 project SWIFF (Space Weather Integrated Forecasting Framework) has developed some of the "most promising techniques to handle multi-physics and multiscale problems and has demonstrated the possibility of these methods in selected processes relevant to space weather modelling",[102] while the recently launched SWAMI (Space Weather Atmosphere Model and Indices), has been developing improved neutral atmosphere and thermosphere models with the aim of making a major leap forward by combining these physics-based and empirical models, and improving the forecast of the activity indices".[103]
- Adding value to Earth-based observations and space missions by making better use of existing data and developed databases. The studies conducted by SOTERIA (SOlar-TERrestrial Investigations and Archives), for instance, "involved the analysis and processing of the relevant data from 18 satellites, including several ESA and other European satellites, complemented by a large set of data from European ground-based observatories".[104]
- Improving forecasts and predictions of disruptive space weather events. The SPACECAST project developed the first European system providing a near-real-time (up to 3 hours ahead) forecast of the high-energy electron radiation belt, while FLARECAST (Flare Likelihood And Region Eruption Forecasting), for instance, has developed a flare prediction system aimed at providing more "accurate and reliable space-weather monitoring and forecasting capabilities".[105]
- Identifying best practices to limit the impacts of SWE on space- and ground-based infrastructures. TECHTide (Warning and Mitigation Technologies for Travelling Ionospheric Disturbances Effects), for instance, has designed new viable Travelling Ionospheric Disturbances (TID) impact mitigation strategies for technologies affected by TIDs and, plans to validate the added value of these techniques in close collaboration with operators of these technologies.[106]

EU-led Initiatives

In parallel to these activities on the research and innovation side, the European Commission has undertaken other actions related to SWE through the Joint Research Centre (JRC). The JRC is the European Commission's science and knowledge service. It aims to provide evidence-based scientific support for the European policy-making process. In this context, the JRC has provided support to EU policymakers in the area of space weather by:

- Raising awareness
- Performing risk assessment for critical infrastructures
- Conducting scientific research towards GNSS resilience[107]

Raising Awareness

The JRC has co-organized several international high-level meetings involving multiple stakeholders with the stated objective being to: [108]

[100] Community Research and Development Information Service, 2015
[101] HESPERIA also produced two novel forecasting tools based upon proven concepts Community Research and Development Information Service, 2017c
[102] Chiarini, 2013
[103] Community Research and Development Information Service,2018
[104] SOTERIA also put considerable effort into utilising and developing theoretical and simulation models for interpreting the space weather data, covering all aspects of the complex Sun-Earth connection. Chiarini, 2013
[105] Community Research and Development Information Service, 2017d
[106] Community Research and Development Information Service, 2018e
[107] Krausmann et al., 2016
[108] Krausmann et al., 2016

- Raise awareness of the potential impact of extreme space weather on technological infrastructures in space and on the ground,
- Identify related scientific, operational and policy challenges for disaster prevention, preparedness and response, and
- Develop proposals to go from awareness to action at the EU policy level.

Particular mention should be made of the Space Weather Awareness Dialogue (SWAD), which was organized by the JRC together with the Directorate-General Enterprise and Industry in Brussels, Belgium, on 25-26 October 2011. A summary of this high-level dialogue is in Box 4.[109]

Box 4: The Space Weather Awareness Dialogue (SWAD)

In view of the risk of catastrophic technological failure under the solar maximum expected in early 2013, the European Commission organised the Space Weather Awareness Dialogue (SWAD) Conference in Brussels on 25-26 October 2011. The aim of the dialogue was "to raise awareness of the potential impact of space weather on critical infrastructures in space and on the ground, and to recommend concrete actions to better protect them". The SWAD conference brought together about 70 high-level representatives from national organisations and authorities, international organisations with assets possibly affected by space weather, operators of critical infrastructures, academia, and European Union institutions. During the discussions, consensus was reached on the following points:

- Space weather is a threat to our critical infrastructures that needs to be addressed.
- An analysis of the space-weather threat to ground-based critical infrastructure (power grid, aviation, telecommunications, etc.) is of equal importance to the study of space-based infrastructures.
- There is no central entity that is taking the lead in the space-weather community.
- The assessment of space-weather impact on critical infrastructures requires a multidisciplinary effort from all stakeholders (scientists, engineers, infrastructure operators, policy makers).
- Ageing satellites that monitor space weather need to be replaced.
- A framework for better-structured communication between the stakeholders is required.
- Open space-weather data sharing is necessary for improving early warning and impact models.
- While there is some preparedness for normal space weather in some infrastructure sectors, nobody is fully prepared for extreme events.
- The topic of space-weather impacts would benefit from cross-sectoral discussion.
- Emergency exercises could help raise awareness of space-weather impact.
- International cooperation is required to cope with the problem as response capabilities may be beyond the capacity of individual countries. [110]

Risk Assessment

In terms of risk assessment, the JRC objective has been to "understand vulnerability of critical infrastructures and services to space weather and possible consequences for society", particularly in terms of risk to infrastructure, risk to services provided, and the risk of cascading effects. To achieve this purpose, the JRC has been conducting in-house analysis and organizing gatherings with representatives of European infrastructure operators, regulators, crisis response experts, national space agencies, academia, and others. The outcomes of these summits have been published in a series of relevant studies including:

- Space Weather and Power Grids: Findings and Outlook, 2013
- Space Weather and Financial Systems: Findings and Outlook, 2014
- Space Weather and Rail: Findings and Outlook, 2015
- Space Weather & Critical Infrastructures: Findings and Outlook, 2016

In addition, the JRC has carried out a preliminary assessment of the vulnerability of the North European power transmission grid to extreme space weather.[111]

Scientific Research

[109] European Commission, 2011
[110] Krausmann, 2011
[111] Piccinelli and Krausmann, 2018

Finally, with regard to SWE impact assessment, the JRC has been performing GNSS-based studies of the ionosphere, and more specifically, a quantitative assessment of the impact of space weather on GNSS navigation and timing receivers. As presented in Fig 18, this scientific research is a transversal effort involving multiple issue-areas.

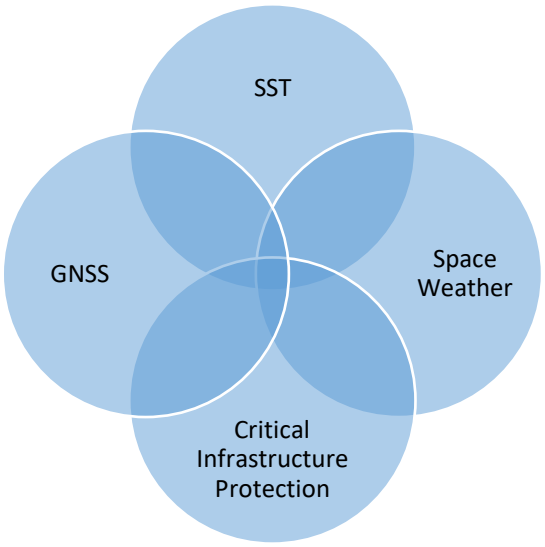


Figure 12: Relevant JRC scientific research areas

In order "to support the development of more resilient receivers, the JRC has deployed ionospheric scintillation monitoring stations in Peru, Norway and Vietnam, as well as two stations in Antarctica in collaboration with a European research consortium. Scintillation events are recorded at the monitoring stations and played back at the JRC to test standard receivers used, e.g., by aviation. This supports the development of GNSS receivers that are less vulnerable to space-weather impact, as well as also the preparation of standards for enhanced receiver reliability. The JRC hosts a database of scintillation events (intermediate frequency data) sourced from the monitoring stations during periods of high ionospheric activity. This data is made available to the research community for free. Future work will investigate the potential of using Formosat-3/COSMIC data, as well as information from the International Ground Station (IGS) network, to monitor ionospheric scintillation". [112]

Future Actions

Drawing on the various actions being taken directly or indirectly, the European Commission considers that reinforcing its role in SWE was, inter alia, affirmed in the Space Strategy for Europe of 2016. The document more specifically states that:

> The Commission will reinforce the SST support framework to improve the performance and geographical coverage of sensors. It will consider extending its scope to address other threats and vulnerabilities, for example cyber threats or the impact of space weather on satellites and on ground infrastructure such as transport, energy grids and telecommunication networks.
> In the long term, this SST model could evolve into a more comprehensive space situational awareness service, building on existing activities in the Member States and ESA, and taking into account international cooperation frameworks, particularly with the US.
> The Commission will engage with the user sectors concerned to develop responses to space weather risks and alerts. It will work with ESA and EUMETSAT to support research and promote international efforts in this domain.[113]

Building on this, in September 2017, in its Resolution on the Space Strategy, the European Parliament called on the Commission to support extending the scope of SST to allow space-based weather forecasts, and proposed a focus on near-earth objects to counter the potentially catastrophic risk of any such object colliding with Earth.

More recently, in June 2018, the European Commission published a 'Proposal for a Regulation of the European Parliament and of the Council – establishing the space policy programme of the European Union, relating to the European Agency for Space and repealing Regulations (EU) No 1285/2013, No 377/2014 and No 912/2010 and Decision 541/2014/EU'.[114] This programme proposal details its objectives, and suitable measures for achieving them, which also cover SWE:[115]

- Provide, or contribute to the provision of, high-quality and up-to-date and, where appropriate, secure space-related data, information and services without interruption and wherever possible at global level, meeting existing and future needs and able to meet the Union's political priorities, including as regards climate change and security and defence;
- Maximise the socio-economic benefits, including by promoting the widest possible use of the data, information and services provided by the Programme's components;
- Enhance the security of the Union and its Member States, as well as its freedom of

[112] Krausmann et al., 2016
[113] European Commission, 2016b:9
[114] European Commission, 2018
[115] European Commission, 2018

action and its strategic autonomy, in particular in technological and evidence-based decision-making terms;

- Promote the role of the Union in the international arena as a leading actor in the space sector and strengthening its role in tackling global challenges and supporting global initiatives, including as regards climate change and sustainable development.

Specifically concerning the future outlook of European SWE activity, this programme proposal highlights the risks of extreme SWE weather events to citizens and ground- and space-based infrastructures; states the need for further assessment of SWE risks and user needs, raises awareness of risks, and ensures the delivery of SWE services.[116] Furthermore, iterated within this paragraph is the necessity for Union-level service delivery according to user needs, requiring "targeted, coordinated and continued research and development activities to support space weather service development," building on present national and Union capabilities and enabling both Member State and private sector participation.[117]

Under Title VIII (Other Components of the Programme), Chapter I: SSA, Section II: Space Weather and NEO of the programme proposal, the Commission identifies a framework for the future delivery of SWE services to public and private European users. Article 59 states:[118]

1. "The space weather function may support the following activities:

   (a) the assessment and identification of the needs of the users in the sectors identified in paragraph 2(b) with the aim of setting out the space weather services to be provided;

   (b) the provision of space weather services to the space weather users, according to the identified users' needs and technical requirements.

2. Space weather services shall be available at any time without interruption and may be selected according to the following rules:

   (a) the Commission shall prioritise the space weather services to be delivered at Union level according to the needs of users, the technological readiness of the services and the result of a risk assessment;

   (b) the space weather services may contribute to the protection of the following sectors: spacecraft, aviation, GNSSs, electric power grids and communications.

3. The selection of entities to provide space weather services shall be performed through a call for tenders".

Additionally, with specific reference to Galileo, the EU is considering providing its own ionosphere prediction service for GNSS users through the GNSS Service centre in Madrid (GSC). Specifically, Article 44.2 of the proposed regulation states that Galileo shall also contribute to: "(c) space weather information and early warning services provided via the Galileo ground- based infrastructure, intended mainly to reduce the potential risks to users of the services provided by Galileo and other GNSSs related to space weather events". Towards this, in the H2020 Work Programme 2014-2015, one of the procurement topics in the area "GNSS Evolution, Mission and Services related R&D activities" was the development of an Ionosphere Prediction Service. (see Box 5).

Further comments on the possible implementation of these actions for SWE service delivery at European level will be provided in Chapter 4.3, but it is here important to note that this proposal is now in co-decision (Parliament and Council) and hence is subject to amendments, before its adoption and implementation.

Finally, it is worth mentioning that alongside these proposed activities on the intra-European front, the EU is sponsoring actions at the international level, with a prominent role assigned to the UN and its specialised agencies (WMO, ICAO). At the 55th session of the Scientific and Technical Subcommittee of the UN COPUOS, the EU delegation stressed its support for "cooperation among national/regional space weather services, including by free exchange of space weather data and forecasts, with the aim to achieve continuous, regular, global and consistent space weather forecasts, awareness and notification to users" as well as its support to "the WMO and ICAO initiatives aimed at the provision of space weather products to aviation". [119]

[116] European Commission, 2018: paragraph 70
[117] European Commission, 2018: paragraph 70
[118] European Commission, 2018
[119] Delegation of the European Union. 2018

Box 5: Ionospheric Prediction Service (IPS)

The shifting state of the Ionosphere, exposed to a number of SWE related effects and disturbances, is a major source of GNSS signal disruption. GNSS signals, products and services are crucial to a plethora of modern domains, public and private, and so the stability of GNSS systems is fundamental to a wide array of users. Within this line of thought, in 2015 the European Commission launched an open call for tenders to award a service contract with the objective of "translating the prediction and forecasting of the ionosphere into tangible results and user-devoted metrics".[120] As a result, the Ionosphere Prediction Service (IPS) project is under procurement by the European Commission through the framework of the Galileo programme, Galileo itself of course being highly dependent on GNSS signal stability.[121]

The overall aim of the IPS project is to "design, develop and operate a service prototype platform to monitor and predict the ionospheric behaviour and the potential effects on the performances of GNSS based applications."[122] It is in this sense that the IPS project is geared as a service towards stable performance for user bases that are vulnerable to GNSS disruptions from ionospheric disturbances. As such, the service aims to cater for the specific requirements of its wide potential user-base – key feature of the IPS is the generation and dissemination of customised, timely warnings as to allow them to prepare for approaching events which may pose issues to GNSS and dependent operations within the specific users' domain of application.[123]

As a system, the IPS "monitors and forecasts solar and ionospheric activity and predicts its effect on GNSS signals and on the final performance of user applications."[124] This predictive information is then translated into early warning issues that are targeted towards users, allowing them to anticipate potential degradation of performance as well as implement any pre-emptive mitigation measures.[125] In this sense the IPS service revolves around three core pillars: [126]

- Sensors which gather space weather and ionospheric measurements through a network of ground- and space-based instruments and GNSS reference stations
- A processing facility to generate the forecasting and nowcasting capabilities
- A user-interface – a web-based too to provide a platform of interaction with users, providing warnings and generated data products in various formats.

As mentioned, the potential user base for this service is considerable given the number of domains reliant on stable GNSS services. The aviation industry in particular is a key target user for a number of benefits can be gained regarding pre-planning of flight paths, real-time changes to flight-paths, and ultimately ensuring the safety of crew and passengers. Other industries that heavily rely on the precision positioning services provided by GNSS, e.g. energy extractors, electrical grid operators, construction, and civil engineering companies, are also likely beneficiaries. Beyond this, the IPS will also allow for other actors within, or interacting with, the field of operation space weather to reuse the predictions of the IPS, compare them to their own, or input the data into their own models to improve upon them.[127]

In addition, the delegation advocated that "UNOOSA should support the development of an international operational system with respect to space weather services, building on respective regional capacities and specificities", and that the EU "would be in favour of developing a coordination committee within the UN to support a collaborative research approach to the issue of space weather".[128]

3.1.4 EUMETSAT

Set up in 1986 to provide national meteorological agencies of member states and wider users with continuous weather and climate related satellite data through exploiting European weather and climate observation systems**,** EUMETSAT's involvement in SWE activities has been traditionally considered outside its core mandate and therefore very limited.

[120] European Commission, 2015
[121] Telespazio, 2018
[122] Telespazio, 2018
[123] Telespazio, 2018
[124] European Global Navigation Satellite Systems Agency, 2018
[125] European Global Navigation Satellite Systems Agency, 2018
[126] European Global Navigation Satellite Systems Agency, 2018; Telespazio, 2018
[127] European Global Navigation Satellite Systems Agency, 2018
[128] Ibid.

Over the past decade, however, the Darmstadt-based organization has expressed an interest in playing a more active role in the field of SWE in synergy with delivery of meteorological services. The rationale for EUMETSAT's involvement in SWE can be compared to that of the interests stated by WMO in the past decade (See Section 5.3). Since forecasting SWE has many similarities with forecasting the tropospheric weather, there are relevant synergies to be exploited with weather and climate data, science, and services to users, such as sharing observing platforms and issuing multi-hazard warnings.

In addition, as a meteorological satellite operating entity, EUMETSAT itself has an interest in the delivery of tailored SWE services. The reason for this is twofold: first, SWE poses risks to the functioning of meteorological satellites – i.e. direct damage to systems as well as interference to operations; second, EUMETSAT satellites' exposure to such SWE events places them in a very good position to contribute to SWE observations and hence become a source of SWE data.

EUMETSAT's interest in SWE can also be seen as an indirect by-product of the developments taking place in many of its Member States, the national meteorological services of which have, over the last decade, extended their involvement in SWE activities, thus compelling EUMETSAT to reflect this trend in its mandate. Equally important, all EUMETSAT's major partners, including the WMO and hydro-meteorological agencies such as NOAA, ROSHYDROMET and CMA, have included SWE activities within their portfolio, thus potentially placing EUMETSAT in an optimal position to benefit from its long-standing cooperative relations with these agencies. Finally, it should be highlighted that EUMETSAT's core mission is to provide and distribute forecasts 24/7 and in real-time to users.

EUMETSAT's involvement in SWE formally in SWE started in 2006, when the organisation started to host SWE instruments aboard its meteorological satellites. More specifically, the Metop-A and Metop-B polar orbit satellites host two instruments relevant to SWE:

- A GNSS Receiver Atmospheric Sounder (GRAS), which is primarily used for deriving atmospheric sounding and, by extension, relevant SWE-related information.[129]
- The Space Environmental Monitor (SEM-2), which is a non-meteorological instrument provided by NOAA to provide information on solar terrestrial phenomena and solar wind occurrences that might impair the satellites and instruments. Through SEM-2, NOAA and EUMETSAT guarantee "a continuity in the determination of auroral activity — intensities of charged particle radiation within the Earth's atmosphere that can degrade radio communications (occasionally making short wave radio communication impossible in the polar regions); occasionally disrupt the proper operation of satellite systems; and increase the radiation dose to astronauts in space (when intensities are high)". [130]

The data provided by SEM-2 are processed through NOAA SWPC to provide information on the state of the Earth's near space environment and possible warnings to customers (including EUMETSAT itself) whose systems are affected.

Besides GRAS and SEM-2, EUMETSAT plans to host dedicated SWE sensors aboard the next generation meteorological satellites. More specifically, the six Meteosat Third Generation (MTG), the six Metop Second Generation (Metop-SG) and the next Copernicus satellites (namely Sentinel 6) will all feature dedicated instruments for monitoring solar activity and SWE phenomena. Through these instruments, EUMETSAT will become an important source of SWE data, with the subsequent purpose of becoming involved in the exchange of SWE data with other meteorological agencies worldwide, thus complementing its current role of user of SWE data with the role of provider. From this standpoint, it is important to note that EUMETSAT is involved in the Space Weather Coordination Group of the Coordination Group on Meteorological Satellites (CGMS), which has been tasked with international coordination of SWE satellite operations and their derived products and services (see Section 5.2.4 for an overview of CGMS' role in SWE).

At the narrower European level, the SWE instruments flying on board EUMETSAT's MTG, Metop-SG and Sentinel-6 satellites are expected to contribute to ESA's planned D3S. Along the same line, the Darmstadt-based organisation is currently also involved in discussions with ESA concerning ESA's L5 mission preparations and, more specifically, with respect to the operations of this mission, the management of which might be eventually be entrusted to EUMETSAT. Besides these specific contributions, broader discussions are currently taking place between EUMETSAT and its member states to assess the opportunities for EUMETSAT to take a more defined role in the delivery of operational SWE services. Whereas

[129] European Organisation for the Exploitation of Meteorological Satellites, 2018a

[130] European Organisation for the Exploitation of Meteorological Satellites, 2018b

this development has not been detailed in EUMETSAT's 10-year strategy, adopted in June 2016 and named "Challenge 2025"[131], this is because EUMETSAT expects a consolidation of its Member States' views and a decision in this respect within the next few years.

3.2 International Framework for SWE Services

The development of European SWE services should also take into proper account existing collaborations and partnerships on the international level because, due to the nature and global reaching impacts of SWE events, global coverage from ground- and space-based observation systems is critical for ensuring the delivery of operational services.

Indeed, as expressed by UN COPUOS "effective progress in advancing space weather services requires coordinated global actions that will serve to focus efforts on the needed forecasting, monitoring and awareness raising with the goal of protecting life, property and critical infrastructure".[132] Thus, improved coordination in the field of SWE "is especially relevant for filling key measurement gaps, securing the long-term continuity of critical measurements, advancing global forecasting and modelling capabilities, identifying potential risks, and developing practices and guidelines to mitigate the impact of space weather phenomena, including on long-term observation of climate change and risk events".[133]

International collaboration is a well-established tradition in the area of SWE research, having longstanding roots. One of the first initiatives for "global coordination was taken in 1928 with the initiation of regular forecasts of radio conditions by the International Union of Radio Science (URSI). Cooperation was enhanced during the International Geophysical Year in 1957-1958 with the establishment of a calendar of "world days" for coordinated observations, and with the setting up of a series of Regional Warning Centres (RWC) and a World Warning Agency. These initiatives were combined in 1962 and, in 1996, were renamed the "International Space Environment Service (ISES)". [134]

Today there is a plethora of international initiatives in the field of SWE focusing on policy matters (e.g. UN bodies), on operational matters (e.g. ISES), and on research and education, e.g. the Committee on Space Research (COSPAR). An overview of these initiatives is provided in the following paragraphs.

3.2.1 International Space Environment Service (ISES)

Since its creation in 1962, ISES has been the primary multilateral body engaged in the international coordination of space weather services. ISES is "a collaborative network of space weather service-providing organizations around the globe, organized and operated for the benefit of the international space weather user community". [135] Its mission is to "improve, coordinate and deliver operational space weather services" and more specifically to:

- Provide real-time forecasting and monitoring of space weather to reduce and mitigate the risk of space weather impacts on technology, critical infrastructure, and human activities.
- Facilitate international communication and service coordination regarding space weather, particularly during periods of enhanced activity, and in the event of extreme space weather.
- Improve space weather services and promote the understanding of space weather and its effects for users, researchers, the media, and the general public (ISES).

These objectives are met through the work conducted by of ISES and its members. ISES currently comprises:

- 16 Regional Warning Centres (RWC) from member states (Australia, Belgium, Brazil, Canada, China, Czech Republic, India, Indonesia, Japan, Mexico, Poland, Republic of Korea, Russian Federation, South Africa, Sweden and United States)
- Four Associate Warning Centres from associate entities (three in China and one in France).
- One collaborative expert centre for data and product exchange in Europe (ESA)

The RWC "share data and services among the various centres and provide space weather

[131] European Organisation for the Exploitation of Meteorological Satellites, 2016
[132] United Nations Committee on the Peaceful Uses of Outer Space, 2017a
[133] United Nations Committee on the Peaceful Uses of Outer Space, 2017b
[134] World Meteorological Organisation and Coordination Group for Meteorological Satellites, 2008
[135] The International Space Environment Service, 2018

services to customers in their regions. The centres provide a broad range of services, including forecasts, alerts and warnings of solar, magnetospheric and ionospheric conditions, extensive space environment data, customer-focused event analyses and long-range predictions of the solar cycle". …The RWC hosted by the U.S. Space Weather Prediction Center in Boulder, Colorado, "plays a special role as "World Warning Agency", acting as a hub for data exchange and forecasts".[136] Whereas each centre focuses on its own region, ISES serves as a platform to "share data, exchange and compare forecasts, discuss customer needs and identify the highest priorities for improving space weather services"[137]. ISES also maintains the international geophysical calendar, which coordinates and recommends dates for solar and geophysical observations that cannot be performed continuously.

ISES is a Network Member of the International Council for Science World Data System (ICSU-WDS) and collaborates with the World Meteorological Organization (WMO) and other international organizations, including the Committee on Space Research (COSPAR), the International Union of Radio Science (URSI), and the International Union of Geodesy and Geophysics (IUGG) (see below).

3.2.2 United Nations

Within the UN system, SWE-related activities have been mainly carried out by the Committee on the Peaceful Uses of Outer Space (UNCOPUOS) and the UN Office for Outer Space Affairs (UNOOSA).

UNCOPUOS

UNCOPUOS serves as an international policy coordination body concerned with all aspects of space weather. The Committee started to specifically address this domain in 2003, when it included an agenda item on Solar-Terrestrial physics, as a single item for discussions, in the work of the Scientific and Technical Sub-Committee (STSC). Although being intended as a single item for discussion, since 2004 the Committee's involvement in space weather matters has become constant, thanks to the progressive enactment of a variety of initiatives. These are described below.

International Heliophysical Year 2007

In 2004, the STSC started to plan the organisation of an International Heliophysical Year (IHY) in conjunction with the fiftieth adversary of the International Geophysical Year (IGY) in 2007. The term "heliophysical" was coined as a broadening of the concept "geophysical," with the aim of referring specifically to the activity of studying the interconnectedness of the entire solar-heliospheric-planetary system.

The IHY served as an international programme of scientific exchange aiming to "focus worldwide attention on the importance of international cooperation in research activities in the field of solar-terrestrial physics".[138] The specific objectives of this UN-sponsored endeavour were:[139]

- To provide benchmark measurements of the response of the magnetosphere, the ionosphere, the lower atmosphere and Earth's surface to heliospheric phenomena, in order to identify global processes and drivers that affected the terrestrial environment and climate;
- To further the global study of the Sun-heliosphere system outwards to the heliopause, in order to understand the external and historical drivers of geophysical change;
- To foster international scientific cooperation in the study of heliophysical phenomena;
- To communicate the unique scientific results of the International Heliophysical Year to interested members of the scientific community and the general public

The IHY provided a successful model for the deployment of arrays of small scientific instruments in new and scientifically interesting geographic locations, and outreach, involving more than 70 countries during a two-year period from February 2007 to February 2009. The IHY concluded in February 2009, but its mission was largely continued via the International Space Weather Initiative (ISWI).

International Space Weather Initiative (ISWI)

Building upon the success of the IHY, in 2009 COPUOS endorsed the inclusion of "a new agenda item entitled the "International Space Weather Initiative" (ISWI) under a three-year work plan with specific focus on the effects of space weather on the Earth and its impact, inter alia, on communications and transport" Under the three-year work plan, from 2010 to 2012, the STSC:

[136] World Meteorological Organisation and Coordination Group for Meteorological Satellites, 2008
[137] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:15
[138] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:5
[139] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:5

- Finalised a report on regional and international plans,
- Identified gaps and synergies in ongoing activities,
- Encouraged continued operation of existing instrument arrays and new instrument deployments

Again, the agenda item on ISWI was concluded in 2012, but its activities were continued under the aegis of the international cooperation programme created out of this agenda item (see further).

Expert Group on Space Weather of the Long-Term Sustainability of Outer Space Activities (LTSSA)

In 2009 the STSC established a Working Group (WG) on the Long-Term Sustainability of Outer Space Activities (LTSSA). As part of its terms of reference, the WG established four expert groups, including one for space weather (expert group C). In 2012, this group submitted to the Working Group a working paper, which was then used as a basis for the development of two guidelines for the LTSSA pertaining to space weather, namely:

- Guideline 16: Share operational space weather data and forecasts; and
- Guideline 17: Develop space weather models and tools and collect established practices on the mitigation of space weather effects

The guidelines aim to promote "the collection, archiving, sharing, intercalibration, long-term continuity and dissemination of critical space weather data, model outputs and forecasts, the establishment of dissemination networks and the identification and filling of critical gaps in measurements, research and operational models and forecasting tools. They also recommend that satellite designs and mission plans incorporate features enabling them to withstand space weather effects".[140] The two guidelines, the texts of which are included in Annex A.7, have been considered to provide a basis for further action. Thus, COPUOS reiterates that progress in the implementation of the two guidelines needs to be encouraged in states.

Expert Group on Space Weather

Following the successful conclusion of the work of expert group C and a proposal from the STSC to introduce a regular agenda item on space weather, in 2014 COPUOS endorsed the creation of an Expert Group on Space Weather within the STSC. The Expert Group on Space Weather has the responsibility to "promote awareness, provide guidance, and enable communication and cooperation in space weather-related activities among member states of the Committee and related national and international organisations".[141]

In 2015, the expert group presented its multi-year work plan, which was subsequently endorsed by the STSC.[142] Under this work plan, the Expert Group has to "examine reports and other information related to space weather; complete an inventory of stakeholders, review their role in the global space weather effort and develop cooperation; and promote involvement by member states in providing space weather services and monitoring".[143]

The work plan was reviewed in February 2016 at the second meeting of the Expert Group, which agreed to continue meeting annually on the margins of the sessions of the STSC as well as inter-sessionally. At its third meeting, held on the margins of the February 2017 session of the STSC, the Expert Group "began to develop a road map for international coordination and information exchange regarding space weather events and the mitigation of its adverse impacts through risk analysis and assessment of user needs".[144]

Over the coming year, the Expert Group's members will "seek to actively engage with national critical infrastructure protection agencies and national and international electrical power distribution organizations to be able to better understand, characterize and ultimately examine steps to mitigate space weather damage to that critical infrastructure" [145]

Here Europe actions

Space Weather and UNISPACE+50

Marking the fiftieth anniversary of the UN Conference on the Exploration and Peaceful Uses of Outer space (1968) in June of 2018, UNISPACE+50 is an ambitious undertaking of COPUOS aiming to build a shared vision with all stakeholders for a "comprehensive Space2030 Agenda" in contribution to the UN's Sustainable Development Goals (SDGs), as part of the United Nations 2030 Agenda for Sustainable Development.[146]

[140] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:6
[141] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:6
[142] United Nations Committee on the Peaceful Uses of Outer Space, 2015a: paragraphs 163-169
[143] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:6-7
[144] United Nations Committee on the Peaceful Uses of Outer Space,2017b:7
[145] United Nations Committee on the Peaceful Uses of Outer Space, 2016: paragraph 171
[146] United Nations Office for Outer Space Affairs, 2018a

In preparation for UNISPACE+50, in 2016 COPUOS endorsed seven thematic priorities, one of which was been specifically dedicated to the "International framework for space weather services". The stated objectives of Thematic Priority 4 are to:

- Strengthen the reliability of space systems and their ability to respond to the impact of adverse space weather;
- Develop a space weather road map for international coordination and information exchange on space weather events and their mitigation, through risk analysis and assessment of user needs;
- Recognize space weather as a global challenge and the need to address the vulnerability of society as a whole;
- Increase awareness through developed communication, capacity-building and outreach; and identify governance and cooperation mechanisms to support this objective".

The implementing body for the thematic priority 4 is the Expert Group on Space Weather, which at its third meeting in February 2017, underlined that "important synergies existed between the tasks set out in its existing work plan and the objectives of the thematic priority".[147] The Expert Group also highlighted two main goals through which the Committee could make significant and actionable future contributions towards the mitigation of the adverse impacts of space weather:

- There was a need to develop an improved basis for international monitoring, forecasting and warning procedures, especially in the form of more coordinated international communication and coordination of warnings of extreme space weather events. The Expert Group noted that individual member states had some existing capabilities in that regard upon which to build;
- There was a need to define a set of best practices, operating procedures and actions to mitigate the adverse impacts of extreme space weather, which required a prior assessment in each member state of its exposure to risks from space weather and related socioeconomic impacts, as well as defined operating procedures, developed in partnership with administrations responsible for critical infrastructure and civil protection (UNCOPUOS, 2017).

UNOOSA

UNOOSA, the secretariat for the General Assembly's COPUOS, has been involved in SWE matters through the organisation and conduct of workshops, training courses, pilot projects, reports and publications as part of the UN Programme on Space Applications, and through the International Committee on Global Navigation Satellite Systems (ICG).[148]

UN Programme on Space Applications

Established in 1971, and implemented through UNOOSA, the United Nations Programme on Space Applications (PSA)[149] was set up with the objective of assisting member states in capacity building with space science, space technology and space applications, in support of the wider UN agenda of sustainable development and promoting international cooperation in space activities.[150] Since its establishment there have been several hundred training courses, workshops, meetings and seminars on a breadth of applications (including space weather). In regard to recent developments in space weather, there have been 10 workshops between 2005 and 2015 on the issue of observational limitations in key geographical locations restricting the understanding of the global ionosphere and its links to the near-Earth space environment – as recognised by the IHY early planning in 2007.[151] A major success resulting from these workshops was the instrument deployment programme, addressing the limitations in data observations, placing 16 instrument arrays (e.g. magnetometers to measure the Earth's magnetic field, radio antennas to measure solar CMEs etc.) in over 112 countries, providing measurements on heliosphere phenomena to fill the global gaps. To complement this, a number of science teams were additionally created to implement "coordinated investigation programmes", developing the instruments in the array. They participated in instrument operations, data collection, analysis, and publication of their findings.[152] Through the International Space Weather Initiative (ISWI – detailed below) the PSA has continued international research on understanding and predicting SWE impacts beyond the framework set out by the International Heliophysical Year (IHY) in 2007.

[147] United Nations Committee on the Peaceful Uses of Outer Space,2017b
[148] United Nations Office for Outer Space Affairs, 2018b
[149] United Nations Office for Outer Space Affairs, 2018c
[150] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:8
[151] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:8
[152] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:9

International Committee on Global Navigation Satellite Systems (ICG)

As an umbrella body of the UN, the ICG was established in 2005 to promote coordination on GNSS-related areas such as civil satellite-based positioning, navigation, timing, and value-added services.[153] Under the objective of seeking greater compatibility, interoperability and transparency in this field, the ICG consolidates coordination between GNSS providers, regional systems, and augmentations, with a particular focus on the needs of developing nations.[154] In the context of SWE, the ICG Working Group on Information and Dissemination and Capacity building has discussed GNSS applications for researching SWE related atmospheric phenomena – and with UNCOPOUS as a lead member of this Working Group, engagement and education is facilitated with the public and policy-makers on the impacts of SWE phenomena, in addition to training and seminar sessions for students and professionals on SWE data analysis and prediction.[155]

Other Activities

In relation to SWE, and beyond the SWE undertakings conducted through the PSA and ICG, UNOOSA additionally engages in a number of other activities within this field to, in general, promote awareness and collaboration in SWE research and applications. To note a few developments of late:[156]

- In accordance with the Dubai Declaration – the first High-level Forum on space as a driver for socioeconomic development, held in Dubai in November 2016, UNOOSA provides technical legal assistance and capacity building for states developing national level space policies and regulatory frameworks on areas including SWE.
- Leading up to UNISPACE+50, the UN/USA workshop on "Space Weather: the Decades after the International Heliophysical Year" was held in July/August 2017. This workshop focused on recent advancements in scientific research through ISWI data, held an international forum on extreme SWE socioeconomic and societal impacts, and also made recommendations for a future SWE activity roadmap
- The third International Civil Aviation Organisation and UNOOSA Aerospace symposium was held in August 2017 on the topic of "Emerging space activities and civil aviation: challenges and opportunities" – featuring a dedicated session on challenges faced from space weather with the objective of fortifying cooperation amongst the space and aviation stakeholder communities, alongside relevant legal and regulatory actors
- In preparation for UNISPACE+50, specifically thematic priority no. 4 on an International framework for space weather services, the open informal session of the 37th session of UN-Space in 2017[157] included discussion on space weather research and applications with the participation of members states and wider stakeholders to encourage dialogue on how the United Nations systems can best respond to the specific themes

3.2.3 International Organisations

While the UN is mainly involved in SWE matters from a policy perspective, its specialised agencies work on coordination aspects on the operational side. Their main activities are outlined below.

WMO

The World Meteorological Organisation (WMO) is a specialised agency of the UN with 191 member states. It acts as the expert group, and authoritative voice, on all meteorologically relevant issues concerning the Earth's atmospheric state and its interactions/behaviour with the oceans and lands, and importantly how this relates to resultant weather, climate, and water cycles.[158]

The WMOs interests in SWE can be characterised by two key components of concern for the WMO: (i) SWE's relevance to the observing function of meteorological satellites; (ii) and SWEs relevance to the delivery of meteorological services:[159]

- In regards to observation conducted via meteorological satellites, SWE has a two-fold relationship with WMO activity. First, SWE events can damage and disturb meteorological satellites and are the primary cause of in-orbit failure – such satellites being the main source of Earth-observation data that support weather forecasting and global climate monitoring; and sec-

[153] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:9
[154] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:9
[155] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:9
[156] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:10
[157] United Nations Office for Outer Space Affairs, 2018d
[158] World Meteorological Organisation, 2018a
[159] World Meteorological Organisation, 2008:6-8

ond, many of the meteorological observation systems in place can dually be utilised in SWE observation.

- Space weather and meteorological services are both key components of the environmental information necessary to ensure the safety and sustainability of several areas of socio-economic activity. In this sense, the integration of these two types of information provides a more reliable basis for environmental services, which can be of use in: aviation, spacecraft operations, electricity supply networks, GNNS, and human health.

As a global organisation with relevance and activity in many domains, the WMO has several motivations for engagement with SWE research and applications, to which its existing infrastructure and influence can be greatly beneficial. Accordingly, as SWE events can impact communities on a global and regional scale, and because SWE phenomena observation is best achieved through the coordination of multiple nations, "the global nature of WMO, as well as its intergovernmental status, its longstanding experience of operational coordination, its scientific basis, the potential synergy between meteorological and space weather related activities, the strong connection of WMO with the aeronautical sector through its Commission for Aeronautical Meteorology (CAeM),[160] and its engagement for the protection of life and property, are major assets enabling WMO to play a key role in this needed international coordination regarding space weather".[161] In this sense the existing WMO architecture provides a very strong basis for the integration of SWE activities, and is in a position to enable a global framework for operational SWE services through facilitating international coordination in SWE observations and forecasting in support of its objectives of protecting life, property, critical infrastructures and subsequently impacted economic activities. As such, the WMO has highlighted several "high-level goals" as part of its 'Four Year Plan for WMO Activities Related to Space Weather 2016-2019' in achieving the overarching objectives:[162]

- Promote the sustained availability, quality, and interoperability of the observations that are essential to support space weather warning and other services, while optimizing the overall cost of the observing system;
- Improve the collection, exchange and delivery of space weather data and information through open sharing, internationally agreed standards, and coordinated procedures taking advantage of the WMO Information System (WIS);
- Ensure that space weather analysis, modelling and forecasting methods allow the delivery of operational services on the best possible scientific basis; facilitate the transfer of technical and scientific advances from research to operations;
- Support the emergence and establishment of cost-effective and high-value services in identifying and addressing user requirements, focusing on the sectors where internationally coordinated responses are required, in coordination with aviation and other major application sectors
- Foster the production of high-quality end products and services by WMO Members, building on ISES centres and other examples of recognized services, in developing best practices, to improve the accuracy, reliability, interoperability, overall cost-efficiency of the provision of services;
- Improve the emergency warning procedures and global preparedness to space weather hazards in accordance with the WMO Strategy on Disaster Risk Reduction;
- Promote synergy between the space weather and the meteorological/climate communities and activities, and advance the understanding of space weather impacts on weather and climate processes;
- Support training and capacity-building, based on science and operational experience, to develop skills in the generation and interpretation of space weather products and services in order to allow WMO Members to utilize existing information in a meaningful way, build their own service capabilities, and facilitate user uptake of new products and services.

The potential WMO activity in the area of SWE has been characterised into three levels:

1. Strategic level – coordination, communication and advocacy
2. Products and services level – (i) Service requirements (user needs, feasibility, demonstration, prioritisation); (ii) Best practices for products and services in various domains of industry; (iii) Training and

[160] CAeM is one of the eight technical commissions of the WMO providing guidance and coordination for the WMO Aeronautical Meteorological Programme

[161] World Meteorological Organisation, 2016:5

[162] World Meteorological Organisation, 2016:5-6

capacity building (i.e. for new providers, users and products)

3. System level – (i) Observation (gap analysis, prioritisation, coordination, standardisation); (ii) Data management (data formats, meta data standards, data exchange); (iii) Science (analysis/forecasting, modelling, research-to-operations, interaction between weather and climate).[163]

In successfully managing a global framework for SWE observations and applications, the WMO will collaborate closely with many of the organisations detailed below, including: relevant agencies and offices of the UN (e.g. UNCOPUOS), ICAO and ICAO METP, CGMS, ISES, COSPAR, ICG, ITU etc.

ICAO

The International Civil Aviation Organisation (ICAO), a UN specialised agency, was set up in 1944 to manage the administration and governance of the Convention on International Civil Aviation. In doing so, ICAO works with the 192 Member States of the convention, as well as industry, to reach consensus on Standards and Recommended Practices (SARPs) a policy in international civil aviation[164]. The Meteorological Panel (METP) was established during the fifth meeting of the 197th Air Navigation Commission session (ANC 197-5), following reorganisation of the Secretariat and the ICAO panel structure[165]. It is METP's responsibility to "define and elaborate concepts and to develop ICAO provisions for aeronautical meteorological (MET) services consistent with operational improvements envisioned by the Global Air Navigation Plan (GANP) (Doc 9750) and in keeping with the Working Arrangements between the International Civil Aviation Organization and the World Meteorological Organization (Doc 7475)"; defining requirements, establishing standards, developing guidance and governance, and promoting international collaboration for globally harmonised meteorological services for international air navigation.[166] In April 2015, the Working Group for Meteorological Information and Services Development was established under the METP to assess user needs, specific shortfalls, develop operational concepts, and define the functional and performance requirements for new meteorological information.[167] This Working Group comprises five work streams to develop requirements, one of which is the Space Weather Work Stream – under which SARPs for the new space weather information service (proposed by the SWE Work Stream) were proposed during METP's second meeting in October 2016 and approved in March 2016 during the 204th session of the Air Navigation Commission (ANC).[168] In addition to SARPS, the ICAO (METP) participates in a process of designating space weather information service providers, supported by the WMO, and to be endorsed by the ANC.[169]

In recent times, ICAO has been progressing efforts to establish a space weather information service for international air aviation.[170] Following the work the Meteorology Panel (METP), established to "define and elaborate concepts and to develop ICAO provisions for aeronautical meteorological services"[171], space weather information standards are in development to be included within ICAO Annex 3[172] – *Meteorological Service for International Air Navigation*.[173] Accordingly, new space weather data and subsequent forecasting capabilities will be tailored for users with consistent information regarding the potential space weather impacts to aviation operations.[174] With aviation operators being required to develop mitigation plans for hazardous weather, the addition of space weather information will require further policy changes for both operators and regulators.[175] As such, the aforementioned Annex 3 of ICAO outlines four sets of requirements and guidance criteria for space weather information providers in the categories of: institutional criteria, operational criteria, technical criteria, and communication/dissemination criteria.[176] Advancing on this in 2018, the METP/3 meeting in April provided several recommendations for action on areas including the optimal number of space weather information providers and the strengths and weaknesses of the prospective providers.[177] In November 2018, the Council of ICAO eventually selected the United States, the Pan-European Consortium for Aviation Space weather User Services (PECASUS, see Box 6) and the consortium set up by Australia, Canada, France and Japan as the operators of the three global SWE service.[178] These selected providers shall now start production and

[163] World Meteorological Organisation, 2016:7
[164] International Civil Aviation Organization, 2018a
[165] International Civil Aviation Organization, 2018b
[166] International Civil Aviation Organization, 2018b
[167] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:11
[168] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:12
[169] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:12
[170] Romero, 2018
[171] International Civil Aviation Organization, 2016
[172] International Civil Aviation Organization, 2007
[173] International Civil Aviation Organization, 2016
[174] International Civil Aviation Organization, 2016
[175] International Civil Aviation Organization, 2016
[176] Romero, 2018
[177] Romero, 2018
[178] In addition, the ICAO Council agreed that two regional centres will be established by 2022. These will be led by

dissemination of SWE information for civil aviation. The success of this service will allow aviation operators to optimise flight paths and planning, as well as reducing radiation risks to crew and passengers. The provision scheme foresees free operation for 3 years, and then a fee levied on the passengers' tickets.

Box 6: PECASUS

The Pan-European Consortium for Aviation Space weather User Services (PECASUS) was established in response to the ICAO call for the establishment of global SWE centres. Nine European countries forms the consortium: Finland (lead), Austria, Belgium, Cyprus, Germany, Italy, Netherlands, Poland and the UK. PECASUS overarching objective is to provide "civil aviation with information on space weather that has the potential to affect communications, navigation and the health of passengers and crew" as specified by ICAO.[179] In order to fulfil this objective, PECASUS will:

- Deliver 27/7 manned observation of solar activities
- Issue appropriate ICAO advisories to the aviation sector
- Provide a guaranteed data flow and operational resilience
- Host ICAO training for SWE operators and aviation users
- Operate a robust and comprehensive IT infrastructure

In February 2018, PECASUS was successfully audited by SWE experts of the WMO and in July 2018 it organised a table top exercise with airlines and air traffic organizations "to test the existing procedures - such as loss of communication - against a realistic space weather scenario provided by the Met Office".[180] Thanks to its streamlined organisation and operational processes, which include specific feedback loops involving airlines and air-traffic organisations, PECASUS is already fully equipped to start operations as a global SWE centre as audited after the designation by ICAO.

IAEA

The International Atomic Energy Agency (IAEA) is the world's central intergovernmental forum for cooperation in the field of nuclear energy science and technology, playing a contributing role in achieving the UN's Development Goals.[181] In regards to SWE, cosmic radiation from solar and other celestial sources contributes around half of the total natural background radiation exposure that effects people and the environment on Earth, additionally posing significant threats to human health on manned space missions that go beyond the protection of the Earth's magnetosphere.[182] In line with this, the IAEA published 'Radiation Protection and Safety of Radiation Sources: International Basic Safety Standards —General Safety Requirements'[183] in 2014, outlining the measures and responsibilities necessary to be taken by governments in protecting human health and the environment in specific scenarios.[184] These standards have been jointly sponsored by a number of organisations in addition to the IAEA, including: the European Commission, the UN's Food and Agriculture Organisation, the UN's Environment Programme, the International Labour Organisation, the Nuclear Energy Agency of the Organization for Economic Cooperation and Development, the Pan American Health Organization, and WHO.

ITU

The International Telecommunications Union is the UN's specialised agency for information and communication technologies (ICTs) – allocating global radio spectrum and satellite orbits, developing technical standards to ensure interconnectivity, and improving access to ICTs worldwide[185]. Following the adoption in November 2015 of the World Radio Communication Conference's (WRC) resolution on spectrum needs and protection of space weather sensors, studies conducted by the Radiocommunication Sector of the ITU (ITU-R) will support the WRC's 2023 considerations of regulatory provisions necessary to provide protection to SWE sensors operating in the appropriately designated radio service. The WRC has also invited ITU-R to "document the technical and operational characteristics of space weather sensors… with the objective of determining what regulatory protection could be provided that would not place additional constraints on incumbent services".[186] ITU additionally has two Study groups (3[187] and 7[188]) working in SWE related areas.

South Africa and a China/Russia consortium respectively (FMI-SPACE, 2018)
[179] PECASUS, 2018a
[180] PECASUS, 2018b
[181] Internation Atomic Energy Agency, 2018
[182] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:12
[183] Internation Atomic Energy Agency, 2014
[184] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:12-12
[185] International Telecommunication Union, 2018a
[186] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:13
[187] International Telecommunication Union, 2018b
[188] International Telecommunication Union, 2018c

WHO

Set up in 1948, the overall objective of the World Health Organisation (WHO) is to improve human health across the globe. WHO has 194 member states globally, with offices in more than 150 countries, and works closely in partnerships with UN agencies, donors, foundations, academia, NGOs, and the private sector[189]. Working with UNOOSA as well as national and regional space agencies, WHO supports the use of space and technology in achieving health-related sustainable development goals and the targets for its member states. In this vein, in June 2015 WHO and UNOOSA held a meeting on the applications of space science and technology for public health, resulting in a reportthat identified the need for space science and technologies in achieving health goals.[190]

3.2.4 Coordination Group for Meteorological Satellites (CGMS)

The Coordination Group for Meteorological Satellites (CGMS) is an organisation that globally coordinates meteorological satellite systems through multi-lateral coordination and cooperation with all meteorological satellite operators and user entities[191]. This involves the operating agencies of meteorological, climate monitoring and environmental satellites functioning in accordance with requirements established by user communities such as the WMO. The CGMS effectively acts as a forum for its members aimed at planning, coordination, technical harmonisation and exchange of information towards safeguarding the long-term continuity of such satellite systems, their observations, and supporting operational applications.[192]

Regarding SWE, the CGMS's interests are two-fold – protecting satellite systems from SWE impacts, and guaranteeing the continuity and coordination of space-based meteorological and SWE satellite operations and their derived products and services.[193] To achieve these goals, the CGMS set up a Space Weather Task Team in 2015 to both identify CGMS priorities for SWE activity and to integrate SWE into CGMS activity, and in 2016 produced the CGMS High-Level Priority Plan 2016-2020 that included SWE objectives.[194] In 2018, SWTT was renamed as Space Weather Coordination Group, with a new Terms of Reference, better reflecting the permanent nature of the SWE activities within the scope of CGMS. Before this, in 2009 CGMS had established the International Radio Occultation Working Group (IROWG[195])– co-sponsored by CGMS and WMO to act as a forum for operational and research users of radio occultation data. IROWG hosts a Space Weather Subgroup on behalf of SWE users, facilitating the use of radio occultation missions for both atmospheric and ionospheric observations in support of research and operation for relevant users.

3.2.5 International Space Weather Initiative (ISWI)

The International Space Weather Initiative (ISWI) is a SWE-focused follow-up program to the IHY 2007, which focuses on international cooperation to progress SWE science[196]. ISWI's primary goals are to develop SWE science through instrument deployment, analysis and interpretation of derived data, in order to reconstruct, model and forecast. Its objectives also include educating, training, and conducting public outreach[197]. In accordance with these goals, ISWI has an open data policy that allows the free and open availability of all collected data and publications to the public.[198] In this regard, ISWI has developed appropriate data policy on:[199]

- *Data exchange and related products:* All data, associated documentations and tools created through ISWI are made freely and readily accessible to users worldwide. In line with this, there are no restrictions on data and knowledge exchanges between ISWI and its users so long as users explicitly acknowledge the appropriate sources in their products.
- *Data standards:* ISWI data and products are to be appropriately documented, presented and stored in standard formats. This minimises the amount of customisation necessary for different tools and interfaces for data access and exchange. Furthermore, the use of standardised metadata models is advantageous in that it allows ISWI data to be searched by other existing systems, e.g. NASA's heliophysics virtual observatories, furthering ISWI data dissemination capabilities

[189] World Health Organisation, 2016
[190] United Nations Committee on the Peaceful Uses of Outer Space, 2015b
[191] Coordination Group for Meteorological Satellites, 2018
[192] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:14
[193] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:14
[194] Coordination Group for Meteorological Satellites, 2016
[195] International Radio Occultation Working Group, 2018
[196] International Space Weather Initiative, 2018
[197] International Space Weather Initiative, 2018
[198] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:14
[199] International Space Weather Initiative, 2018

through compatibility with existing infrastructures.

- *Data archiving:* Two types of data, real-time (or near real-time) data, and retrospective data, fulfil ISWI's operational needs. The first, real-time data, is used for SWE forecasting or nowcasting, whilst retrospective data is used for modelling or research purposes. Real-time data, however, if archived properly – processed, documented, organised, stored, and maintained – can additionally be used for research and modelling purposes, and as such is important in ensuring the long-term value and utility of ISWI data.
- *Data distribution and accessibility:* To maintain ISWIS open and accessibly data policy, there are appropriate mechanisms for data exchange from various data distribution centres, placing a responsibility on such centres to provide adequate data services. The most efficient is direct access of data for users directly through data access portals on the internet, but real-time data-producing instruments also need effective broadcasting infrastructure, and methods of archiving to ensure data accessibility.

3.2.6 Research and Education: COSPAR and ILWS

Created in 1958 by the International Council for Science (ICSU), the Committee on Space Research (COSPAR) was established with the objective of promoting space science research internationally with a focus on the free exchange of information and results, as well as providing an open forum for scientists to discuss issues surrounding space research.[200] COSPAR established a Panel on Space Weather (PSW) in 1998 aimed at fostering cooperation in SWE research and bridging the gap between the research and application communities – additionally encouraging the development of predictive techniques for space environment change and acting as an expert advisory body to the COSPAR Scientific Commissions on SWE related matters.[201]

Together with the Steering Committee of International Living With a Star (ILWS) – an initiative established to "stimulate, strengthen and coordinate space research" with its contributing organisations – COSPAR led a strategic assessment for the advancement of space weather science with the intention of better meeting user needs.[202] The resultant publication, 'Understanding space weather to shield society: a global road map for 2015-2025 commissioned by COSPAR and ILWS', was an effective review of current SWE activities and served to identify priorities for SWE research and development, and provide recommendations on future developments, ground- and space-based data needs, scientific challenges, coordination and input on the transition to operational services – for the near-, mid-, and long-terms in order to improve overall SWE service provision to end users.[203]

In order to implement this road map, provide updates, and maintain flexibility to novel advancements and societal needs, the COSPAR Panel on Space Weather facilitates the establishment of International Space Weather Action Teams (I-SWAT).[204] I-SWAT is composed of teams and activities, multiple teams/activities are then further grouped into I-SWAT clusters (8 in total) – a grouping of teams/activities by certain criteria, areas of focus or objectives: e.g. by domain/physical phenomena, by impact, by timing of space weather information, by national/regional strategic plans etc.[205] I-SWAT has several overarching objectives to fulfil its supportive function to the ILWS-COSPAR roadmap:

- Providing a global hub for SWE community efforts
- Creating a working environment to encourage active participation and the formation of novel leads and ideas through inclusivity and information-sharing
- Enabling collaboration in SWE research, model and tool developing, testing, evaluation and utilization of SWE data
- Facilitating incorporation of the latest research in SWE into forecasting and analysis applications, rapidly addressing users' needs and improving upon services.
- Supporting a channel for the voices of the global community and a bottom-up approach to innovation and improvements.
- Incorporating the SWE community in strategic planning, i.e. roadmap updates,

[200] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:15
[201] National Aeronautics and Space Administration, 2018a; United Nations Committee on the Peaceful Uses of Outer Space, 2017b:15
[202] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:16
[203] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:16
[204] National Aeronautics and Space Administration, 2018b
[205] National Aeronautics and Space Administration, 2018c

based on user requirements and the latest scientific advances.[206]

An overview of I-SWAT's role and relations with other international entities operating in the field of SWE is provided in Figure 13.

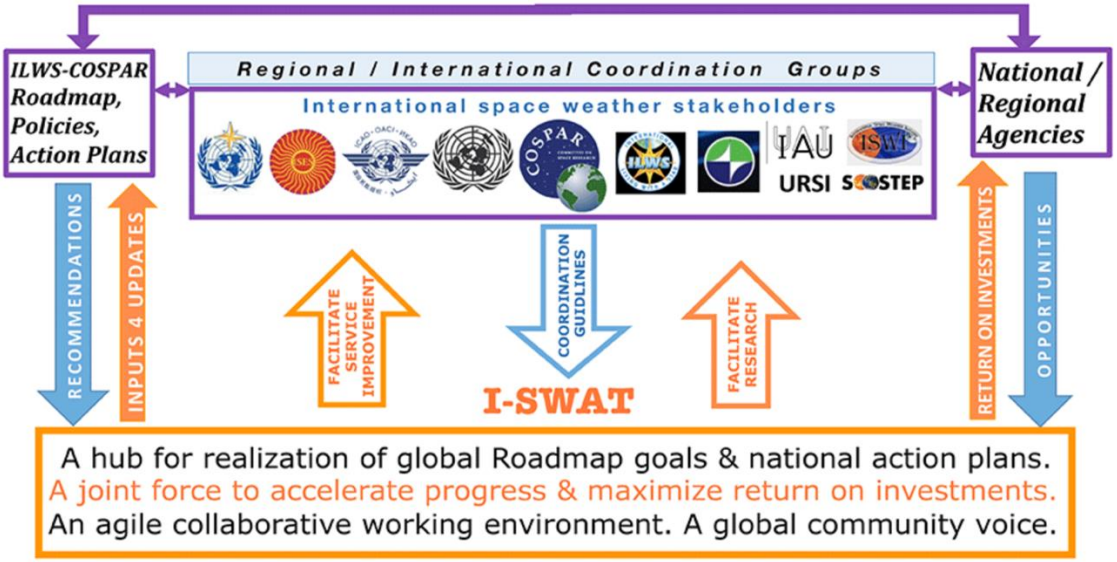


Figure 13: I-SWAT roles and relations with international entities (source: Kuznetsova, 2017)

3.2.7 Other International Service Providers

Finally, it is important to acknowledge the major national data & service providers contributing to international cooperation in the advancement of SWE services. A list is provided in Appendix A.5, but among them specific attention must be placed on U.S. policies and stakeholders, as the U.S. not only remains a benchmark for worldwide development but also plays an important role in the framework of international cooperation and exchange with Europe.

U.S. Policy and Strategy

In the U.S., the Department of State maintains a dual "coordination and clearance" role in international space cooperation; leading on government-government framework agreements in its coordination role, whilst developing agency-agency implementing agreements through its clearance role.[207] These active roles have been reflected in U.S. policy and strategy in recent years, with the US National Space Policy[208] published in 2010 expressing the objective of expanding international cooperation and the development of improved near-Earth SWE forecasting capabilities through space-based Earth and solar observations.[209] This has sparked multi-pronged efforts to increase coordination between U.S. agencies and internationally – e.g. FEMA,[210] the Federal Interagency Response Plan includes a long-term power-outage annex; and FERC[211] issued for the development of reliability standards for grids and geomagnetic disturbances.[212]

In terms of strategy, the National Space Weather Strategy[213], issued in 2015, specifies actions, and timescales for actions to be taken by U.S. Federal agencies to advance preparation and responsive capabilities to SWE storms. Six goals are outlined in this Strategy, including improving assessment, modelling and prediction of impacts on critical infrastructures, as well as cultivating international cooperation on SWE matters; e.g. collaboration on data sharing, research, products and services and extreme SWE preparedness.[214]

Furthermore, an Executive order was issued by the President of the **U.S.** on preparing the nation for SWE impacts, outlining the roles and

[206] National Aeronautics and Space Administration, 2018d
[207] Krausmann et al., 2016:6
[208] White House, 2010
[209] Krausmann et al., 2016:5
[210] FEMA: Federal Emergency Management Agency
[211] FERC: Federal Energy Regulatory Commission
[212] Krausmann et al., 2016:5
[213] National Science and Technology Council, 2015
[214] Krausmann et al., 2016:5-6

responsibilities of federal agencies in "preparedness, response and recovery and their authority to direct, suspend or control critical infrastructures before, during and after a space-weather event".[215]

NASA

The National Aeronautical and Space Administration (NASA) primarily conducts its SWE-related missions for scientific research purposes, although these missions provide substantial data and information that is additionally provided to, and used by, civilian and military customers. Whilst this provision of SWE data by NASA is invaluable, NASA does not provide SWE services itself.[216] Organisations such as NOAA's SWPC, the 557th Weather Wing (AFWA), as well as agencies outside of the U.S., disseminate the raw data foundations made available by NASA to generate SWE services.

NOAA

The National Oceanic and Atmospheric Administration (NOAA) is a scientific agency of the U.S. that conducts research on the climate, weather, oceans, and coasts to provide various products and services to relevant stakeholders[217]. The Space Weather Prediction Centre (SWPC) is NOAA's specialised SWE forecasting branch.[218] The SWPC is responsible for operational SWE products and services, acting as the official national source, providing watches, warnings, alerts and summaries for its user communities.[219] The SWPC provides over 39 types of operational products and services for worldwide users, utilising more than 1,400 types of SWE data from its own instruments as well as a network of partners – including NASA, the USAF and USGS's – ground- and space-based systems.

The services that SPWC provides, in cases of solar flares and geomagnetic storms, come in the form of texts and graphical products in accordance with their scaling categorisation systems (similar to how hurricanes or earthquakes are scaled). For major radiation storms, forecasts services can be provided within 24 hours of the event impact.[220] As mentioned, SPWC's services are provided on a subscription basis, created in 2005, which has seen a sharp customer increase as sectorial awareness of SWE associated risks has risen.[221] SPWC products are tailored to different user communities and infrastructure sectors.

Department of Defence – 557th Weather Wing

The 557th Weather Wing[222], formerly (pre-2015) known as the Air Force Weather Agency, is the U.S.'s leading military meteorology centre. It collects, analyses and provides accurate and timely situational awareness products and services to appropriate national military segments. To meet its user needs for the Department of Defence (DoD), the Air Force has access to a number of space-based observations systems, i.e. the Defence Meteorological program, as well as outsourcing additional data products from NOAA.

American Commercial Space Weather Association

Formed by commercial organizations as a result of the 2010 Space Weather Workshop, the American Commercial Space Weather Association (ASWA) represents private-sector commercial interests related to SWE[223]. As a formal association of commercial member companies, its objective is to promote SWE risk mitigation for critical infrastructure by:

- Providing SWE data and services necessary to mitigate risks to critical infrastructures;
- Taking a SWE advisory role for government agencies;
- Representing commercial SWE capabilities, both nationally and internationally;

In recent years ACSWA's membership has been rapidly growing, and as of Spring 2018 it maintained a roster of 19 companies. In terms of activity, ASCWA organises and participates in a number of meetings and sessions – e.g. the Annual NOAA-CSWIG/ACSWA Summit meetings, 'Growing the Space Weather Enterprise' sessions at SWW – in addition to successfully spurring action by making recommendations to NOAA's SWPC to expand its capabilities through NOAA's Small Business Innovative Research program, as well as sponsorship of various other initiatives [224].

[215] Krausmann et al., 2016:6
[216] National Research Council, 2008:36
[217] National Oceanic and Atmospheric Administration, 2018d
[218] National Oceanic and Atmospheric Administration, 2018e
[219] National Research Council, 2008:37
[220] Krausmann et al., 2016:10
[221] Krausmann et al., 2016:11
[222] 557th Weather Wing, 2018
[223] American Commercial Space Weather Association, 2018
[224] Ibid.

3.3 Summary: Status of Supply in the European and International Context

3.3.1 Europe

Over the past decade, European countries and institutions have been actively contributing to the advancement of SWE research and are now progressing on the research-to-operation (R2O) path. Most SWE-related activities and capabilities have been developed and carried out by individual national infrastructures/groups. These have remained somehow disconnected, with the exceptions of the initiatives and programmes led by the EU or ESA.

- Through its FP7 (2007-2013) and H2020 (2014-2020), the EU has funded over 30 research projects, which have led to key results in scientific research, models, and services development, and, more broadly, in supporting the R2O path.
- Through its SSA programme (SWE and LGR), ESA has been pulling existing assets and capabilities together into a federated virtual network for the delivery of SWE applications to end-users and has started the development of European SWE instrumentations/mission capabilities (space-based capability).

While the EU and ESA have generally operated separately in their SWE-related activities, their efforts have led to the development of important SWE assets in terms of ground- and space-based infrastructure, physics-based models, and databases, as well as data products providing applications to a variety of end-users. In pushing forward these developments, both the EU and ESA have adopted an integrated approach that makes simultaneous use of technology-push and user-driven models for service development. This approach has led to a clear understanding of the application domains, the expected users within these domains, and the different preliminary user requirements for the delivery of SWE services.

As dedicated service providers have not yet been identified within Europe, ESA maintains the responsibility for both development and current provision of pre-operational services. Whilst ESA will retain a development role once dedicated service providers are active, the identification of the operator and the procurement source has not yet been addressed. For these reasons, ESA currently has a primary role in all technological aspects of SWE service creation and provision, with the partial exception of some activities funded through H2020.

As of 2018, ESA's network has offered over 140 separate products providing scientific and pre-operational applications for 24 (out of the 39) services that are intended to be delivered to users. These services are now being managed in a "pre-operational" framework, with subscription services issuing alerts depending on the interest of the users. Critically, these notifications "provide information on what is happening in space but not how the space environment will affect, e.g., spacecraft operations".[225] Overall, European SWE services, at both national and pan-European level (ESA and EU), are still far from the level of maturity reached by their U.S counterparts. In meeting the requirements associated with the provision of fully-fledged services tailored to the needs of end users, there are still some technological, market, and organisational necessities that need to be fulfilled. These will be detailed in the next chapter.

3.3.2 International Context

International collaboration is a well-established tradition in the area of SWE research, but as stressed by the UN Expert Group on Space Weather, for an international framework for space weather services to become effective there is still a need "to strengthen the reliability of space systems and their ability to respond to the impact of adverse space weather; to develop a space weather road map for international coordination and information exchange on space weather events and their mitigation, through risk analysis and assessment of user needs; to recognize space weather as a global challenge and the need to address the vulnerability of society as a whole; to increase awareness through developed communication, capacity-building and outreach; and to identify governance and cooperation mechanisms to support this objective".[226] Beyond cooperative measures concerning the organisational, scientific and technical aspects of SWE science, observational data products, and services, sufficient progress also needs to be made in developing the market enablers, i.e. through increasing the awareness of potential user communities through capacity building and outreach.[227]

[225] Krausmann et al., 2016
[226] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:1
[227] Ibid:8

Currently, as outlined in 5.2.5, ISWI conducts activities to promote such collaboration on research, instrument and data interoperability, as well as education and outreach. Additionally, the UNCOPUOS has a history of promoting cooperation and collaboration on space activities, and since the agreement at the fourth-ninth session of the STSC of 2012, SWE has been introduced as a regular item on the agenda.[228] Building on these existing frameworks, and looking forward, UNSPACE+50 has been acknowledged as an ideal opportunity to strengthen action in accordance with global collaboration with relevant stakeholders for the period of 2018-2030.[229] In line with this, it has also been noted that there is a "pressing need to define a mechanism for a future coordinated approach for actions within and between States, United Nations entities, other international intergovernmental and non-governmental organizations and space weather stakeholders, including academia and industry".[230]

Granted that currently there is still fragmentation between the stakeholders working in the field of space weather, a mechanism for global coordination, monitoring and guidance is still necessary for the development of international operational SWE services. In achieving the objectives of Thematic Priority 4, it is clear that a mechanism is necessary to:[231]

- Stimulate and support scientific research with a view to achieving fast progress in the global ability to accurately predict space weather events;
- Stimulate cooperation among states for the free exchange of space weather data and forecasts;
- Increase communication between the scientific and space weather service communities, as well as industry and users;
- Promote the fast transition of new scientific research into improved and more accurate space weather services that meet user needs.

Such a mechanism should consider the promotion and facilitation of:[232]

- Oversight of coordination and communication between stakeholders in order to reduce duplicated efforts. This oversight must recognise that many SWE actors function under their states and national authorities, focusing on coordination and communication rather than governance or implementation.
- Establishing an international coordination group for SWE, which would report to COPUOS through the Scientific and Technical Subcommittee, effectively replacing the current position of the Expert group on Space Weather, expanding on its current role to additionally provide recommendations to the Scientific and Technical Subcommittee for the consideration of COPUOS. This group should include representatives from relevant international agencies and bodies of stakeholders to implement SWE services.
- The international coordination group should have a mandate to develop high-level coordination on SWE activity, to guide SWE policy, and promote the implementation of SWE guidelines and best practices.
- COSPAR should have a scientific support function to this new coordination group.
- An extension of the role of the Expert Group on Space Weather in actively organising outreaching meetings and workshops for the international SWE research and service communities.

[228] Ibid:10
[229] Ibid:10
[230] Ibid:11
[231] Ibid:13
[232] United Nations Committee on the Peaceful Uses of Outer Space, 2017b:11-13

4. Towards Operational SWE Services in Europe

Building on the status of European and international SWE activities in relation to the three main service enablers discussed in Chapter 2, (namely the technological, market, and organisational enablers) this chapter elaborates on the steps that need to be taken to ensure a smooth transition towards an operational SWE service delivery that is capable of meeting European stakeholders' needs. More specifically, this chapter will:

- Identify the technical conditions to be met for an operational service.
- Define the market conditions to be met for a commercial service.
- Reflect on the most appropriate organisational setting for the delivery of operational services
- Propose elements of a European roadmap in a user-driven approach.

4.1 Addressing the Technical Gaps

Whereas the achievements reached by European stakeholders in the span of the last decade (2008-2018) are certainly remarkable, in providing self-sufficient European SWE services, there are still several technological gaps that need to be addressed. In this section the analysis is focused mostly on ESA, which has been federating relevant capabilities of national member states. However, it is worth remembering that not all ESA MS have subscribed the SWE segment of ESA SSA programme and that at national level some countries do provide operational services to serve critical infrastructure. Therefore, national initiatives are mentioned were relevant. For an overview of SWE assets (data, models, instruments and services) in Europe please refer to the Space Weather Resource Catalogue.[233]

When performing a capability assessment by measuring the overall level of service maturity (here defined as the operational implementation of the service constituents) it becomes evident that the major issue is not simply associated with to the transition from the pre-operational to the operational stage; rather, it is first and foremost associated with the maturity of the constituent elements, namely the data, models, and data products underpinning the service (see Figure 14).

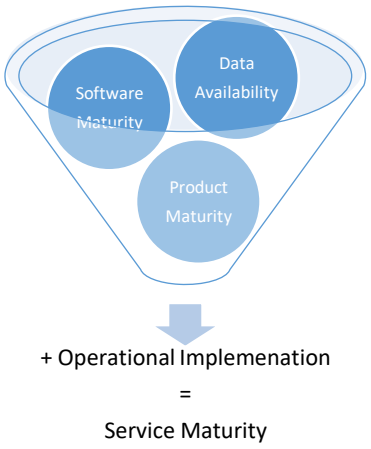


Figure 14: Advancing Service Maturity: Key Components

In order to provide fully operational SWE services to end-users, and to meet their requirements, there are several technological necessities that form the foundation of service provision. This primarily entails both ground- and space-based systems, used in SWE observation. Second, are the various systems utilised in regard to data acquisition, dissemination, processing and modelling, storage and archiving, and the interfaces (e.g. graphical) through which the end user interacts with such services. And third, are the processes in which this data is transformed into products and the products into services that the user receives.

For each of these building blocks (data, software and products) there is still an evident

[233] Committee on Space Research, 2018

mismatch in Europe in meeting the targeted objectives. In many instances, these mismatches fall within the scope of Period 3 (P3) of the ESA SSA Programme SWE segment development plan for 2017-2020 and H2020 activities, but often exceed them.

4.1.1 Filling Data Gaps

In terms of data availability, issues persist with respect to ground- and space-based observations. While Europe can count on a wealth of activity in regards to ground-based SWE observations, several Member States have voiced concerns about the ageing of the ground infrastructure and the need to replace and maintain it. In addition, there are some evident gaps in meeting the observational requirements, particularly in terms of space-based observations. The most important space-based observation /measurement requirements identified are presented in Table 15 below.

Observation /Measurement	Classification for service delivery	Instrument needed
Interplanetary Magnetic Field (IMF) properties and dynamics	High priority	Magnetometer
Solar wind velocity, bulk density and temperature	High priority	Plasma Analyser
Magnetic field mapping of the photosphere	High priority	Magnetograph
Intensity Mapping of the outer corona	High priority	Coronagraph
Intensity Mapping of the lower corona	High priority	EUV imager
Intensity Mapping of the Heliosphere	High priority	Heliospheric imager
X-ray flux monitoring	High priority	X-ray monitor
Energy distribution and flux dynamics with E> 10 MeV	High priority (from L1)	Radiation monitor
Detection of solar winds ions with E=30 KevV/nuc to 1 MeV/nuc	High priority (from L1)	Medium Energy particle spectrometer
Solar wind electron flux and energy distribution with E = 30 KeV to 8 MeV	High priority (from L1)	Medium Energy particle spectrometer
Detection of radio burst/flare signatures and associated outward expanding shocks	Enhancing	Radio burst spectrograph
Solar wind ion flux and energy distribution with E= 1 to 10 MeV/nuc	Enhancing	Medium Energy particle spectrometer
Solar wind ion flux and energy distribution with E > 10 MeV/nuc	Enhancing….	Radiation monitor

Table 15: Space-based observations and instruments requirements (source: Luntama et al., 2017)

Whilst many of these measurements and relative data are – and can be – made available by international partners, it is clear that if the objective of the ESA SWE programme is "to provide for its customers and end users a *non-dependent* source of space weather observed data and processed information based on relevant ground based and space-based sensors and appropriate data processing elements", then the deployment of new European instruments and systems becomes absolutely critical.[234]

This is not to downplay the relevant role of international cooperation – which will in any case remain key for Europe for improving timeliness, continuity, availability and reliability – but to highlight the importance of having guaranteed access to SWE data, also in case of changes in the current data policy of international partners, as well as an independent generation of SWE data not available to other foreign counterparts. In fact, it has been noted that the take-up of operational services (to serve critical infrastructure) at national and European level may be hampered if those are to be provided (on the basis of foreign service providers, that are not tailored to national specificities or requirements. In addition, it should not go unnoticed that many of the

[234] European Space Agency, 2011

measurements identified by ESA would provide novel data that will be used for service creation in both the European and international context.[235]

The above-listed requirements can be fulfilled through either hosting secondary payloads in larger missions or through dedicated missions, as summarised in Figure 15.

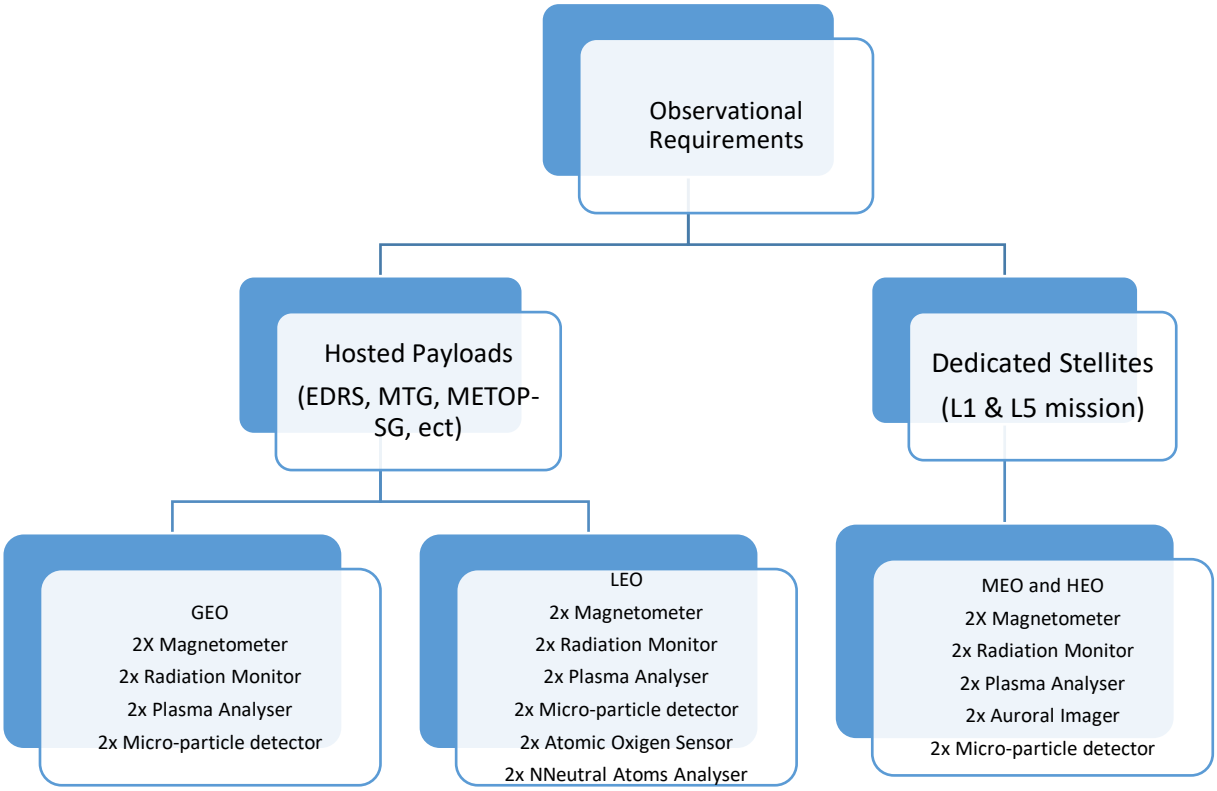


Figure 15: Instruments for Space-based Observational Requirements

More specifically, as mentioned in Chapter 3.3, ESA plans to fill these gaps with the envisaged L1 / L5 missions and the D3S.

Lagrange Missions L1 and L5

In the current phase, ESA's SSA Programme (P3, 2017-2020) activities are being coordinated with the U.S. (NOAA, NASA) to progress the development of the potential L1/15 SWE missions.[236] The availability of SWE observational data from these points, and fortifying collaboration with international partners to enable data continuity through these missions, is essential for future SWE service provision.

L1's position is situated in the solar wind upstream from the Earth, allowing for observational measurements of SWE coming towards the Earth.[237] L1's Sun-Earth Line (SEL) solar monitoring data (e.g. solar disc, solar corona, and SEPs associated with solar flares and CMES[238]) and in-situ data (e.g. solar wind speed, density, temperature, and pressure[239]) are considered to be "mandatory for SWE services"[240]. The continued availability of SEL data on L1 will potentially be consolidated through agreements with international partners such as the U.S.[241]

The other crucial SWE observation Mission on L5, located 60 degrees behind Earth, will provide a means of monitoring Earth-orientated CMEs from the side, as to allow for more precise estimates of the speed and direction of the CMEs[242]. L5 measurements will complement those of the L1 mission, additionally providing raw data on solar corona monitoring, heliospheric imaging, solar disc magnetic field, EUV imaging; as well as in-situ measurements of solar wind, magnetic field, charged particles and hot plasma.[243] Once implemented, the L5 mission will provide the necessary data to

[235] Luntama et al., 2017
[236] Bobrinksy, 2017
[237] European Space Agency, 2017b
[238] Ibid.
[239] Ibid.
[240] European Space Agency, 2016
[241] Luntama et al., 2015
[242] European Space Agency, 2017b
[243] Luntama et al., 2016; Bobrinsky, 2017

"substantially improve SWE forecasting capabilities".[244]

Distributed SWE Sensor System (D3S)

To gain a comprehensive view of the Earth-Sun environment, it will be also necessary to "capture the state of the magnetic field and the particle distribution in a sufficiently large number of sampling points around the Earth, such that it allows state-monitoring and modelling of the involved processes with sufficient accuracy and timeliness" (Kraft et al., 2018). In increasing the number of sampling points where SWE measurements are taken, ESA's SSA D3S system aims to produce a network of hosted payload, Small Sat, and potentially Cube Sat instruments, typically monitoring particles and fields within the magnetosphere and Auroral images[245]. These hosted payloads will take in-situ measurements in LEO, MEO or LEO, with candidate "host" satellites of spacecraft including METOP SG, MTG, Telecom, Galileo, in addition to dedicated Small Sat and Cub Sat missions[246]

Overall, the schedule plan of ESA's Lagrange missions and the D3S system is detailed in Figure 16.

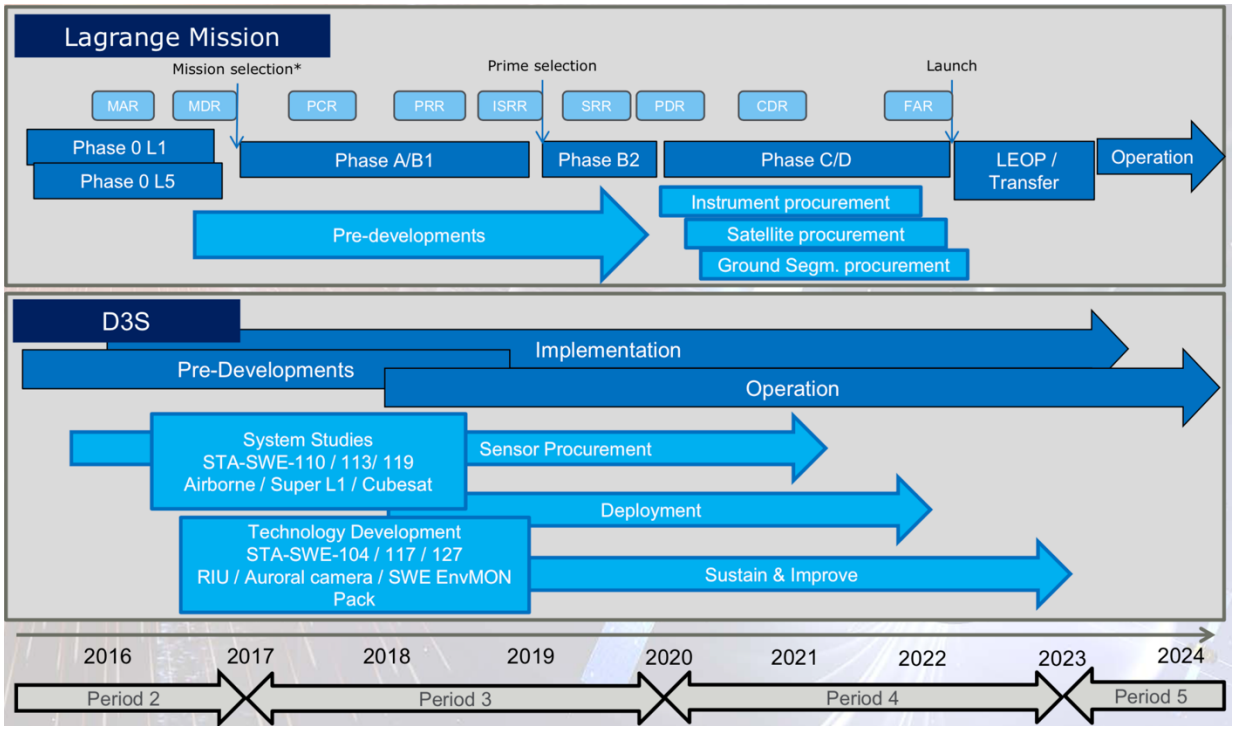


Figure 16: ESA SWE Space Segment Schedule Plan (credit: European Space Agency, 2016)

4.1.2. Improving Software Maturity

Whilst these observational data gaps are completely necessary for SWE service provision, from a service and end-user perspective, the raw data itself is essentially void of utility until processed and modelled into value-added products and services.

Modelling capabilities, at both national and European level, are still far from the level of maturity reached by the U.S. counterpart and needed to deliver reliable SWE services in Europe.

As discussed in Chapter 3, while there are many examples showing European countries' ability to provide operational services in certain sectors (e.g. the aviation services provided by the PECAUS consortium, ground infrastructure services by the FMI, space operations-related services by the French Air Forces, etc.), in most of these cases the reliability of the predictions remain limited. False alarms are very detrimental to users, such that many users prefer receiving a posteriori confirmation of an SWE event (to limit the effort in searching for the cause of a service downtime) rather than a false alarm. In this sense it can be stated that the current service

[244] European Space Agency, 2016
[245] European Space Agency, 2017b
[246] European Space Agency, 2016

provision in Europe is still in an observation/post-processing context rather than in a predictive one.

This is primarily due to the still small number of inputs used in the construction of models, as well as in the still stove-piped nature of current modelling activities.[247] Even for countries with strong modelling capacities, such as Finland (which is home to the Grand Unified Magnetosphere-Ionosphere Coupling Simulation),[248] providing SWE forecasting remain challenging.[249][250] Indeed, it can be even argued that no single model can yet fully predict SWE events in a reliable manner,[251] and that a breakthrough in the construction of physics-based models will be needed, as happened in the past for meteorological models.

Hence, in the coming years, all European stakeholders should ideally continue to conduct activities not only to further the development of empirical models, but also to make breakthroughs in physics-based models (e.g. heliospheric modelling, ionospheric scintillation, 3D modelling of the morphology of the ionosphere) and combine new physics-based and empirical models so as to improve still insufficient forecasting techniques. On the ESA side, efforts should also include the development of the required models & tools utilising L5 mission data. The use of L5 data in CME propagation models will indeed provide further insight into the benefits of the future L5 mission, whilst crucially demonstrating and refining heliospheric model development through assimilated data from multiple vantage points.[252]

More broadly, as also highlighted in a study led by the JRC, over the next few years: [253]

- Physical models should be improved or - where necessary - new models developed to allow a better prediction of CME arrival times, an earlier determination of the interplanetary magnetic field orientation, and an estimate of the probability and size of the likely impacts.
- Forecasting capabilities should be enhanced to provide regional or local forecasts on the severity and duration of extreme space weather to ensure the most appropriate operator response.

In addition to this, from the multitude of SWE observation systems and types of data there will be a continued need to put in place coordinated techniques for interoperability and federation, formatting and archiving, as well as methods for dissemination, presentation, and user interface platforms.[254]

In bridging gaps in these areas, the EU will open a H2020 call in 2019 (SU-SPACE-22-SEC-2019 - Space Weather) that will focus on addressing the "development of modelling capabilities and/or the delivery of prototype services able to interpret a broad range of observations of the Sun's corona and magnetic field, of the Sun-Earth interplanetary space, and of the Earth magnetosphere/ionosphere coupling reliance on existing observation capacities Proposals will address application domains that may include space as well as terrestrial infrastructure. Proposals will include architectural concepts of possible European space weather services in relation to the application domains addressed, and they will demonstrate complementarity to and, if relevant, utilize, precursor Space Weather services already available through the Space Situational Awareness programme of ESA, and take into account global space weather service developments by the WMO."[255] As with previous calls funding the development of modelling capabilities (see Chapter 3.1.3), the resources associated with this call have the potential to greatly improve forecasting capabilities by making a major leap forward in the use of physics-based and empirical models.

As for ESA, the SSA programme has planned further development of the federated data repository systems at the SWE Data Centre in Redu, the five ESCs across European countries, and the SWE SSCC located at the Space Pole in Belgium. Continuing through P3 and P4 of the SWE programme, efforts should made to consolidate SWE Data Centre enhancement and the utilisation of data products. This primarily entails improving data storage, browsing and retrieval capabilities, whilst fortifying links with federated data repositories[256]. Furthermore, development of level 1 processing chains is being forwarded for SWE hosted payload missions data, i.e. NGRM and SOSMAG missions, regarding data ingestion, processing, dissemination, and storage. While

[247] The different types of perturbations (X-ray flares, SEPs, CMEs, coronal holes) find their correspondence in rather separated modelling communities. Further splitting of modelling activity occurs for regions closer to Earth (magnetosphere, ionosphere / thermosphere, Earth atmosphere and surface) because of traditional scientific domains, specific customer needs, as well as the physical processes involved. Folini, 2018
[248] See Annex A.4
[249] Academy of Finland, 2018
[250] Palmroth et al., 2014
[251] Messerotti, 2017
[252] European Space Agency, 2017c:7
[253] Krausmann et al., 2016
[254] Interestingly this is one of the key recommendations put forward by the COSPAR Roadmap to 2025, according to which there is a need to "standardize (meta-) data and product metrics and harmonise access to data and model archives.
[255] European Commission, 2017a
[256] European Space Agency, 2016

these activities serve to strengthen the availability of data from European missions, allowing for data and product searching and retrieval, additional agreements for data from international missions (e.g. GOES, ACE, DSCOVR and GK2A) are being formulated.

As for the ESCs, extensions for the Solar Weather, Space Radiation, Heliospheric Weather, Geomagnetic Conditions and Ionospheric Weather ESCs are already scheduled to be implemented during 2019.[257] The efforts in developing empirical models and improving forecasting techniques through more accurate modelling tools, however, should ideally be continued in the next phase (P4) of ESA's SWE programme, together with the necessary validation and verification tasks. Indeed, "SWE services may be built upon diverse data sources, models, and processing tools, all relying on underpinning IT/network infrastructure for their availability. Understanding and monitoring the performance of all the different data sources, models, and processing tools, will also be an essential task to provide services which can be utilised with confidence by the intended user communities"[258]

Similarly, coordinated strategic investment into developing scientific and computational capability and know-how in the EU should be explored within the framework of the next MFF of the EU. A route that – in complementarity with the development of physics-based model – could be ideally explored to improve modelling capabilities may be offered by machine and deep learning, two increasingly advancing fields of computer science that use statistical techniques to give computer systems the ability to learn and improve the performance of data mining, prediction, and decision making. As also discussed during the 15th ESWW of 2017, "in the field of space weather, machine learning techniques may be applicable to some of the most intractable prediction problems such as solar eruption triggering, geomagnetic storm intensity, and solar energetic particle events".[259] While machine learning developments in the field of SWE are ongoing, efforts could ideally be made for moving forward into the realm of "deep learning," using very large neural networks with massive datasets such as the full SDO dataset and outputs from full-physics models of the Sun-Earth system.

4.1.3 Advancing Product and Service Maturity

Because the above-identified objectives in data availability and software maturity are yet to be reached, the overall maturity of SWE products – the derived data generated using SWE models/tools and core components contributing to the delivery of one or more SWE services – remains hindered.

This is not always the case when looking at national service provision (see Annex A.4 for an overview). However, when looking at the ESA network, which has thus far federated over 170 data products from 32 participating institutions of 14 European countries,[260] it emerges that several data products are still missing (e.g. products for some geographical regions such as the Artic and Mediterranean region) which, depending on the measurement, require further observational systems or increased computational capabilities from data already available.[261]

In addition, a number of currently available derived data products do not meet the targeted requirement (in terms e.g. of accuracy and timeliness) and still need to be combined and customised with software tools and technical reports to create the envisaged services. As a result, many of the pre-operational services already identified by ESA are yet to be introduced (see Figure 17 for a list of the services that are still pending and Chapter 3.2 for the ones currently available).

In this sense, further research and action to introduce, tailor, and combine new or existing data products into value-added services remains a key objective that should be pursued in the near future. In order to advance product maturity and ensure a match between available products and the targeted requirement, objectives to be reached during the next phase of ESA SWE (P4), should include not only the addition and combination of the products and tools not yet available, but also, and perhaps more importantly, the verification, validation and enhancement of existing products through coordination among different service providers that ensures consistency of e.g. forecasts (as done, for instance, by the MOSWOC, which coordinates with NOAA's SWPC) as well as through the pre-operational exploitation of the SWE System.

[257] European Space Agency, 2017c:4-6
[258] Glover et al., 2018
[259] Berger et al., 2018
[260] Note that additional data products are available both not only in the countries not participating in the ESA programme.
[261] For a list of current data products provided by the SSA SWE segment to respective service domains, as well as products, tools and services that are not yet provided, compare the product catalogue with the system requirement document: http://swe.ssa.esa.int/web/guest/user-domains

Towards this, the utilisation "of a process whereby SWE products and services are tested with real users in the loop" is – and will continue to be – of utmost importance, as the results of these test campaigns can lead to gap identification, and feed into longer term definition and development planning (see Figure 18) which, in turn will ultimately offer improved products and services.[262]

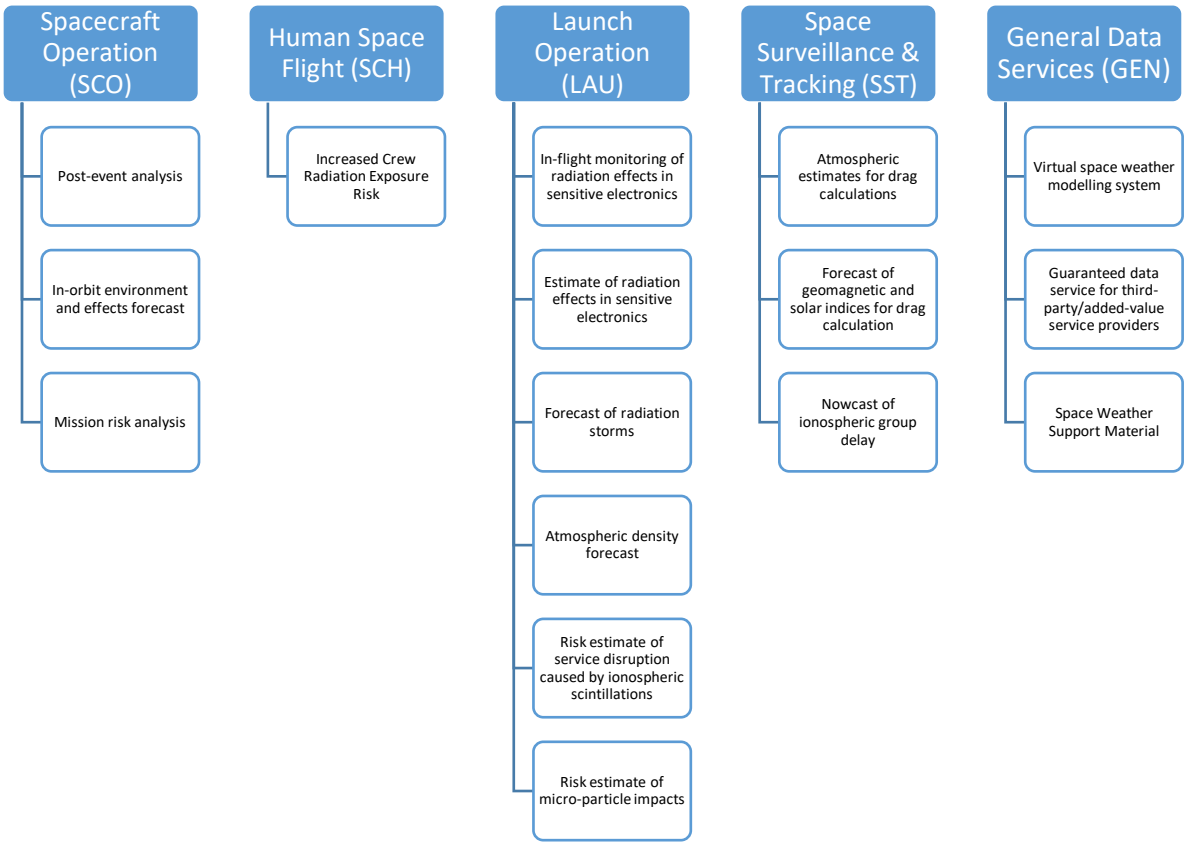


Figure 17: ESA SWE Services Under Development

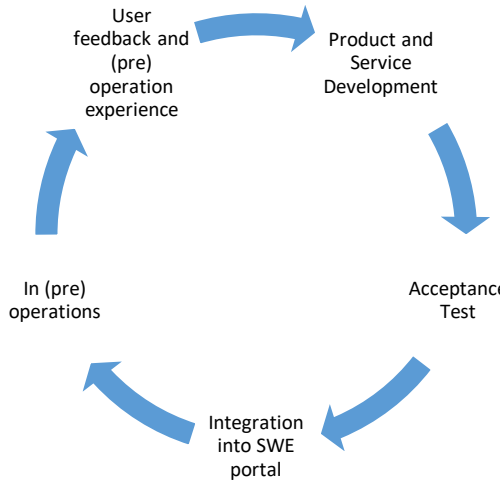


Figure 18: Product and Service Development Lifecycle (source: Glover & Luntama, 2017)

All ESA SWE centres will have a role to play in this process. The extension, and potential creation of additional, Expert Services Centres (ESCs) could play an increasingly significant role in increasing product and, consequently, service maturity, i.e. by conducting activities such as the tailoring of product presentation to best meet end user requirements.[263] Also the SWE Service Coordination Centre (SSCC) should reinforce its role in developing adequate user interfaces, working in conjunction with the ESCs to present data products in a communicable and utilisable form.[264]

The SSCC's role will be further detailed in the following section concerning market aspects, since it has a strong focus on dealing and developing interactions with user communities. What is important to highlight here is that once the above-identified gaps are addressed, the principal issues in reaching service maturity will be the operational implementation of the demonstrated services.

However, while moving from basic research and development to a demonstrated or pre-operational service is very much a matter of *technological readiness*, as highlighted on the

[262] Glover and Luntama, 2016; Glover and Luntama, 2017
[263] European Space Agency, 2017c: 4-5
[264] European Space Agency, 2017c: 4

vertical axis of Figure 19, transitioning from a demonstrated service to an operational one requires both *technological readiness* and more crucially its *sustainability*; a fully operational service is in essence both technologically feasible and sustainable in its wider organisational and funding prerequisites, as well as in terms of its market/user dimensions.[265]

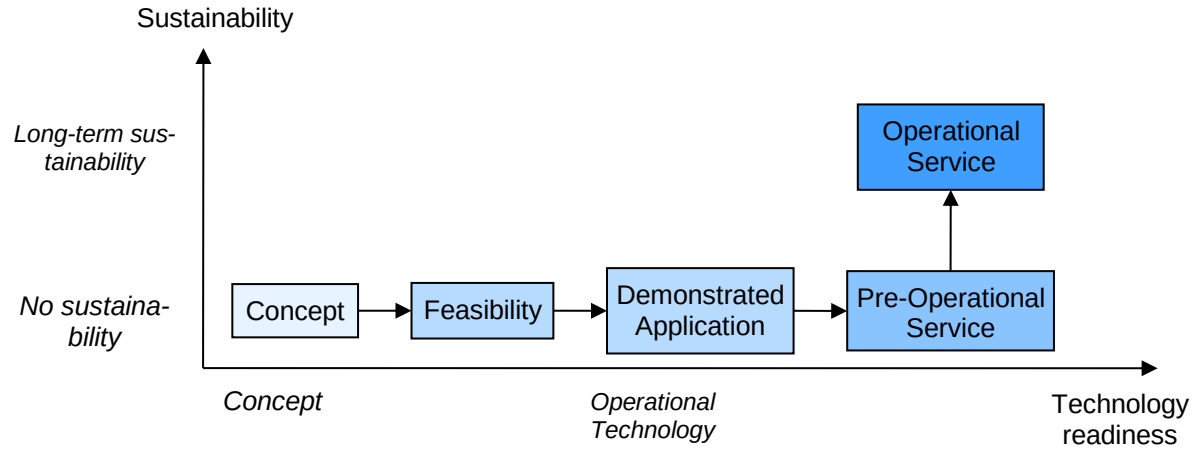


Figure 19: Operational service creation in terms of sustainability and technology readiness (source: Mathieu, 2009)

The issue of creating an operational service from a demonstrated application highlights the most current and prominent problems in the creation of fully operational SWE services, consequentially further attention will be paid to this particular transition of phases, addressing mostly the gaps beyond the technological aspects of operational service provision (i.e. the market and organisational gaps).

4.2 Addressing Demand/Market Requirements

Regarding the demand/market enablers, within the European context both the EU and ESA have adopted an integrated approach that makes simultaneous use of technology-push and user-driven models for service delivery. This approach has led to a clear understanding of the application domains, the expected users within these domains, and the different preliminary user requirements. This said, the continued research and understanding of SWE effects on specific domains, the awareness of SWE risks and impacts, and the benefits of SWE services, still constitute a gap that should be further addressed. Platforms for engagement with user communities, not only in promoting awareness of SWE impacts, but with feedback and extensive tailoring of services with and for user communities, are necessary.

Beyond the user-community development dimensions, advancing the market aspects of SWE services also raises questions concerning the structure of the value chain, ultimately "Who pays?", placing a distinction between the *user* and the paying *customer* of a service under alternative scenarios. These two core aspects are further addressed herein.

4.2.1 Fortifying Relations with End-Users

In regard to the user community, many actions that can be ideally taken by both the EU and ESA. On the EU's side, the awareness raising activities conducted by the JRC, which have greatly contributed to enhancing cross-sectoral discussion and communication, could be expanded in the coming years, for instance, through the organisation of table-top exercises with the civil protection community; through the development of framework for a more structured communication between the scientific community and end-users, and through to sustained collection of user feedback to feed into a European SWE roadmap.

As for ESA, in developing interaction, feedback, and help services to the user communities, a recent extension to the SWE Service Coordination Centre (SSCC) was presented within P3. Originally established in the Preparatory Programme, advanced more so during P2, the SSCC was set up to "provide the first line of user support with respect to the SWE precursor services, including helpdesk and

[265] Please refer to the general definition attached to services in Ch. 1

user management functions".[266] During P2 the SSCC had a continued mandate and role in strengthening its role with the user communities, maintaining an overview of the SWE service network, and providing a link between the ESCs. This role was reiterated in P3 as the SSCC was extended until the end of 2017, further collecting user feedback in order to refine end-user strategies and the forward-looking SWE service roadmap.[267]

Given that the technological and market aspects of SWE services are certain to grow and evolve in the coming years, it is apparent that a continued relationship with user communities is needed to provide continuous updates and tailoring of services as and when advancements are made. This places a necessity on the existence of a body with the role of the SSCC in facilitating the various dimensions of user interaction with developers and service operators. Whilst ESA will maintain its development role, the establishment of separate service provision entities may dictate a shift of this user engagement role to such an entity/entities. Otherwise, until external service providers are identified and whilst the organisational status quo persists, it can be recommended that the SSCC's role be continued through P4.

Irrespective of the eventual enactment of a different framework (which will be addressed in Section 4.3), what is important to highlight is the existence of a well-defined process for SWE capabilities assessment, improvement, and deployment that brings together scientists, application developers, mission specialists, and infrastructure engineers with end-users of SWE services.[268] Clearly, the more intertwined the various stakeholders, the more effective will be the offered services in meeting users' need.

Towards this goal, the Design Study Methodology (DSM) presented by Christel et al. (2017), provides an interesting framework for furthering development of SWE service design and provision through stakeholders' engagement. In fact, the process of service design and creation presented is extremely networked, both in the sense that it encourages cross disciplinary and user engagement, but also in that at any stage of development there is room to revert back to a prior stage as part of a continuous process of redesign and reassessment.

Although DSM is specifically applied to the design of climate services, the overall framework has implications for developers and governors to address the challenges in the creation of SWE services and broader space-based services in general. Some key features of recommendations presented by Christel et al., include: (1) identification of users through consultations about the accuracy and utility of services for users; (2) encouraging an interdisciplinary attitude throughout the service creation process, uniting scientists, engineers and users for mutual understanding of objectives; (3) forming adequate service provider-user interfaces (e.g. user engagement methods, surveys, consultations, interviews, design workshops) to increase mutual understanding of user requirements to reflect provider tasks and amendments to the service; (4) joining scientific soundness, functionality and aesthetics through visual representations; (5) an effective dissemination and engagement strategy, i.e. through visual design to help users "capture and understand the information provided by a climate service as simply and quickly as possible"; (6) inclusion of user feedback and co-design strategies, particularly in the development of user interface platforms and additional functionality necessities.[269] The three main phases identified by the DSM and various aspects of service design that can have feedback to a prior phase at any stage of development are summarised in Figure 20.

[266] European Space Agency, 2017c:4
[267] European Space Agency, 2017c:4
[268] Kuznetsova, 2017
[269] Christel et al., 2017:2

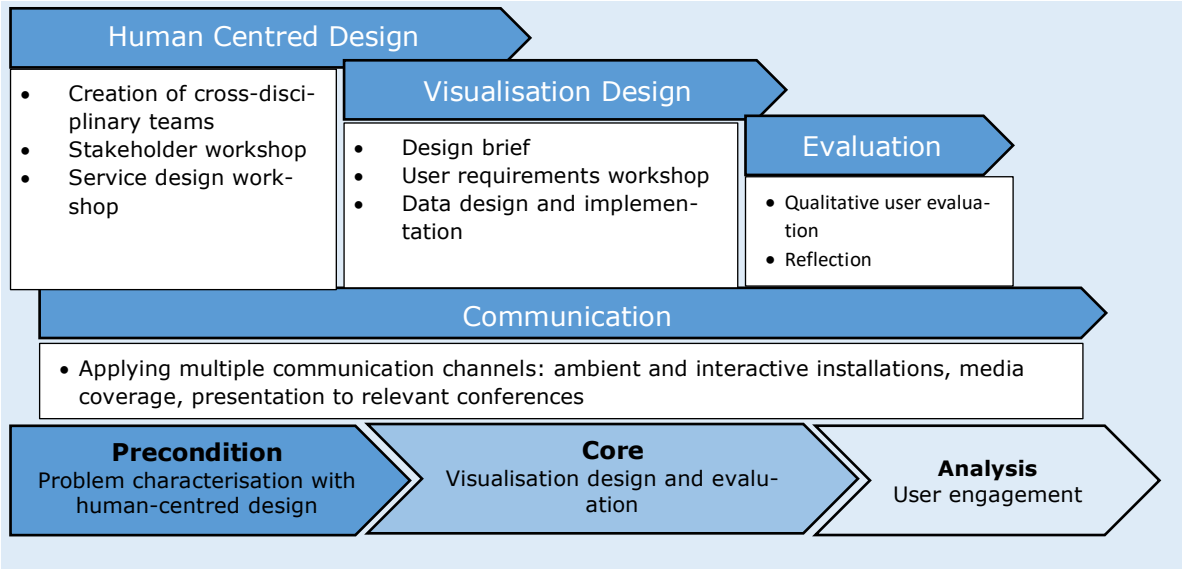


Figure 20: Author adaptation of the Design Study Methodology (DSM) (source: Christel et al., 2017).

* Note that feedback loops are permitted at any stage of this process – furthering flexibility to achieve increased clarity of objectives and effectiveness of results considering new advancements or inputs through e.g. user engagement.

4.2.2 Identifying Customers

Moving on from the user community towards the financial aspects of market development, as reflected in the ESA SSA Customer Requirement Document (CRD)[270], current efforts in demand analysis have focused purely on user identification and their requirements, however there is a lack of clarity as to who the customer a service will be, i.e. the entity procuring the SWE services. This outlines another gap to be addressed in SWE service provisions, the solution of which is heavily dependent on the model of funding that is applied; i.e. services provided on a public or a private basis.

From an operational point of view, the question of demand, or the market, is essentially a question of who pays for a service in order to ensure cash flow up the value-chain and ultimately its sustainability – in turn highlighting the public or private provision model of a service. The customer of a service might very well be a different entity to the end user of the service. This is most proinently the case in a public model for service provision; for example, the paying customer of a tailored service product could be a public funding programme or a public body, such as a National Meteorological Office, while the end-user of this service could be an operator (public or private) of a critical infrastructure (be it in space or on the ground). On the other hand, under a more commercially based model, a private entity would directly purchase access to a service from a service provider. In both cases, the monetary requirements for operations are met, even though with differences in the mechanisms for funding and data policy.

A model for commercial service provision requires two primary conditions. First, that there is an established model of SWE system development that provides sufficient data for services to be produced – the current trend being that public entities take this role (e.g. NOAA, ESA). Second, there has to be a market willing to pay for services – although public sector entities can still act as customer in this scenario. Crucially under this model, the service provider is a commercial entity, with cash-flow going up the value chain from a paying customer (public or private) – commercial actors will be incentivised to not make data publicly available and to charge for access (Nightingale et al., 2016).

Whether or not a public model for providing services is used depends on several issues that need to be answered. First, is the question of to what extent SWE services are deemed as a public good in order for public funding to sustain them? Second, is there a presence of a market that is willing, or is even able, to pay for service access? Third, do private service operators exist, with sufficient in-house expertise, to provide services in place of public funding and entities? Importantly under a public model, the service provider is a public entity, generally having an open data policy, free at the point of use for the end user.[271]

[270] European Space Agency, 2011

[271] Nightingale et al., 2016

A clear case in point is the U.S. SWE service setting, where there already exist both public (function under an open/free data policy for users, with some restrictions on military provisions) and private service operators (working on a subscription basis) – highlighting that both public and private frameworks for a value-chain can co-exist, with public entities generally holding a non-competitive role if a commercial actor is sustainable in its provision (Lautenbacher, 2014:3-4). Additionally, it is still the case in the U.S. that public organisations maintain the role of developers, providing the data and products necessary for private SWE services to function.

On the other hand, in the European context, the level of commercial uptake in the provision of SWE service is still very low. While ESA in its SWE policy has adopted a non-compete clause (NCC) with available commercial solutions, at this stage in the development of European-based SWE services it remains unclear whether it is possible to monetise such services from a commercial perspective. This is because, in general for all sectors, the development of a commercial market for SWE services is going to be considerably hampered by a number of factors. The most important include:

- The still low accuracy of nowcast and forecasts
- The understanding of SWE services as public goods with subsequent provision funded by governments for protection of their critical national infrastructure
- The availability of reliable free services on the Internet (e.g. NOAA SWPC)
- The still limited understanding of customer impact
- The lack of clear customer economic benefit related to SWE service provision, particularly with respect to centennial risks (benefits are perceived as too intangible or too distant in time)
- The lack of recent events creating significant impacts.[272]

In light of these factors, at the current state of development of SWE services within the European context, it is reasonable to suggest that, for the foreseeable future, SWE service provision will – and should – be conducted primarily under a public procurement model. It is also reasonable to assume that, as the SWE service status further develops, the emergence of commercial services will also develop increasing traction, in a similar fashion to what happened in the field of meteorology. While this may be a long-term scenario, to support the progressive uptake of commercial solutions and emergence of a customer base it will be necessary to tackle the issues identified above through a set of enabling actions, an overview of which is provided in Box 7.

Box 7: Enabling the Uptake of Commercial Market Solutions

The further development of the SWE market dimension in Europe will require:

- Tackling technological issues highlighted in Section 6.2.1 to increase the accuracy and reliability of forecasts
- Raising awareness of SWE impacts and associated costs through more accurate socio-economic impact assessments that *inter alia* identify hidden infrastructure vulnerabilities and interdependencies and address cascading effects and multiple stakeholders
- Continuing raising sectorial awareness of how SWE services might be beneficial to users through a reinforced dialogue with all stakeholders (authorities, operators, and also the public
- Increased research and understanding of SWE impacts for individual sectors
- Further collaboration with users/potential customers to identify specified requirements and service improvements, while educating them on the inherent limits of the services (so as to avoid them requesting e.g. a solar flare prediction with 99% confidence few days before the event)
- Seeking cooperation with industry for developing *ad-hoc* services tailored to specific domains (or even specific users) providing more value-adding information as compared to those already provided by government and openly accessible on the internet
- Identifying sectors (e.g. oil drilling, polar shipping; unmanned transport) and typologies of services where users may be incentivised to procure tailored services on a commercial basis because of clear economic benefits. In this respect it can be highlighted that businesses may be willing to invest in services assuring their business continuity, but only with respect to risks that may occur on a shorter timescale (i.e. some years) and have limited impact. Conversely, benefits may be perceived as too distant or too intangible and burdensome to

[272] PricewaterhouseCoopers, 2016

set up a market for large-scale, centennial SWE risks, which clearly calls for a sustained public investment.
- Identifying suitable commercial provision schemes in commercially viable sectors (e.g. fees levied on passengers' tickets as in the case of the ICAO aviation services, or a fee charged on energy operators and then cascaded onto the end-user energy bill)

4.3 Defining an Appropriate Organisational Setting

Since the inception of the broader European SSA programme, there has been a clear understanding that that the most challenging issue will be associated with the overall governance scheme.[273] Indeed, while the steps to address the technological and market gaps are in principle clear and detailed, many questions and known unknowns still hover above the broader institutional and financial framework to support decision-making across the multiple stakeholders, the rules for decision-making, and the mechanisms, to ensure conformance to the data policy rules and procedures.

As seen in Chapter 3, the current European architecture for SWE services is primarily centred on the individual activities led by most European countries and those of ESA. The Agency has done much to further develop and federate these capabilities, and its SWE Service Network is now well positioned to become the foundation of an operational European SWE infrastructure. However, it has been questioned whether ESA, as a research and development agency, can be the body responsible for the provision of operational services or should it, to the contrary, spin them out into an operational body (as it has done in building other operational systems such as, for instance, launchers, and meteorological satellite systems).

Indeed, the question of ESA's role in operational programmes has been broadly debated since the early 1970s, when questions about its possible role in the operations of the first meteorological satellites emerged. According to the ESA Convention and the 1977 Resolution on "The Agency and the Operational Systems", (the text of which is provided in Annex A.8 ESA and Operational Services), ESA may carry out operational activities, but only in the fields where organised users do not exist, or if requested by users to do so. This provision was mainly intended to avoid the situation where there is no operator for the systems it develops. Moreover, such involvement was intended to be temporary before a transfer of responsibility to external organisations. As for the source of funding for operational activities, this was meant to be different for R&D and operational activities, as clearly highlighted by Article V.2.

In short, "ESA's mandate does not prevent it from becoming involved in operational activities but its involvement in operational activities has been foreseen as temporary until another organisation can take over the responsibility for the operational systems".[274]

Consistently, already in the SSA programme proposal of 2008, the ESA Council stressed that "while ESA can be responsible for the development and validation of the European SSA system, its exploitation is expected to be assigned to a separate operational entity, which will operate it in line with the agreed governance, data policy and data security principles. Thus, once pre-operational qualification is achieved, the SSA system will be handed over to this operations entity".[275]

At the same time it was proposed that ESA would remain responsible for the future evolution of the SSA system through the adoption of a modular approach "for carrying out such system evolutions, qualifying them and then handing them over to the operations entity. These evolutions could include new instruments, enhancements of the ground and space infrastructure, new data products and services, as well as adaptation of the system to new technologies".[276] This approach is represented in Figure 21.

[273] European Space Agency, 2008
[274] Mathieu, 2009
[275] European Space Agency, 2008:11
[276] European Space Agency, 2008:12

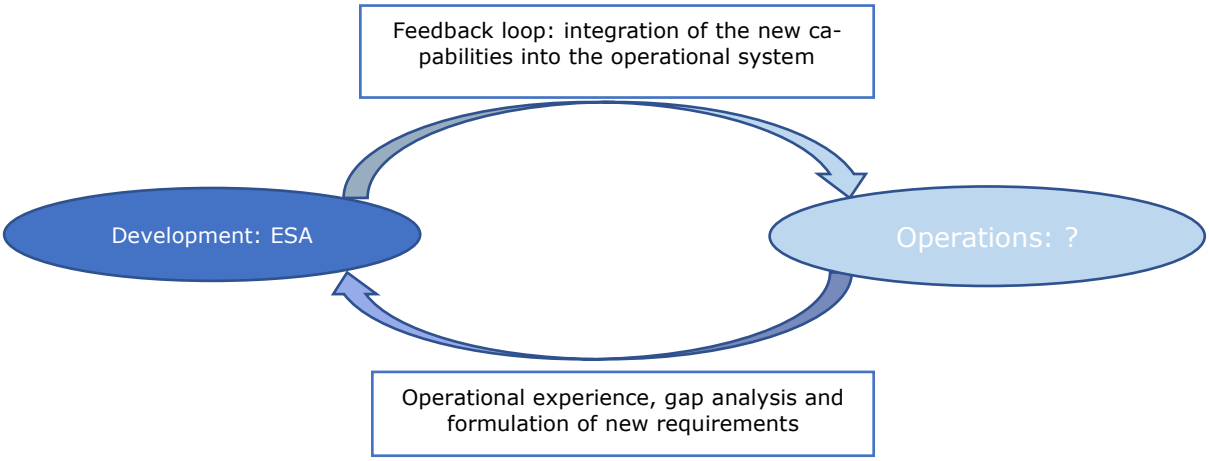


Figure 21: Feedback Mechanism for Operational Services (source: Fondation pour la Recherche Stratégique, 2008)

The organisation to which ESA would transfer the responsibility for SWE services and other SSA-related operations was discussed in the broader SSA governance study entrusted in 2008 by ESA to major European think thanks, consultancies, and law firms.[277] These first studies "have provided the necessary reference work for the selection of models to be confronted with in the space and non-space domains".[278]

However, as of mid-2018, no final decision had been reached and there is still a debate on whether the responsibility for operational services should eventually be transferred to another body, as well as whether the provision of such services should be detached from other SSA services (see Box 8).

Box 8: Rethinking Relations Among the SSA Components?

SWE research and services have been generally regarded as one of the contributing elements to the provision of the broader SSA services. In reality, however, there may be at least two sets of incentives to differentiate SWE services (and other space environment monitoring services such as NEOs) from SST services.

For one thing, as emerges from the assessment of SWE demand (see Annex A.3), the provision of SWE services encompasses a plethora of domains, some of which are not directly intended to support safe space operations (e.g. services to power grids operators, aviation, etc.). In this sense, SSA can be considered just one of the domains where SWE services can be provided. As SWE services gradually move from research to operations, they acquire a more autonomous dimension from SST-related activities. Therefore, the integration of the various SSA elements, although understandable and operationally useful from the perspective space-related users' (e.g. spacecraft operators, launch service providers), may not be the best way forward for the provision of SWE service to "terrestrial users".

In addition to the specialisation of SWE service-delivery in a variety of domains, another driver to differentiate between SST services and space environment monitoring services is that these two categories of services show clear differences in both governance and data policy. The former presents the most challenging political, security and liability-related issues, implying the identification of a number of specific issues to be addressed as a pre-requisite to any intra-European or international cooperation. By contrast, the second category of services may be considered more easily manageable due to the relatively low security and liability-related constraints. In this latter case, the demanding constraints applied to the first category of services may not apply and less constraining management and practical organisations can be found to better optimise SWE services' delivery also in terms of timeframe. It is in fact clear that keeping SST and SWE frameworks under the same roof may possibly slow down the objective of delivering operational SWE services as quickly as possible.

Overall, whereas these two considerations may call for rethinking the place of SWE services within the broader SSA programme, the rationales of, and incentives for, keeping an integrated approach

[277] See for instance Fondation pour la Recherche Stratégique, 2008

[278] del Monte, 2009

to services delivery in the field of SSA cannot be neglected, particularly as concerns the establishment of a common and comprehensive European framework for space safety and security. In light of these considerations, at this stage a viable pathway would be to envisage different services components with different governance structure within one overarching programme, as in the case of the Space Programme proposal of the EU. In any case, the eventual selection of a specific architecture over another will ultimately depend on the underlying objectives European stakeholders (i.e. Member States) will decide to pursue.

In addition, irrespective of who the operator will be, there are other key issues that will then need to be addressed from an organisational point of view. More specifically, there will be a necessity to**:**

- Guarantee the *availability* of the data and services delivered through an appropriate data policy tailored to address the requirements of the different stakeholders
- Guarantee *continuity* of such data and services particularly through international cooperation
- Guarantee the *sustainability* of the services by ensuring sustainable funding and allocation from users and ensuring programmatic overview

In this respect, different governance solutions, with different cascading implications for ensuring data/service availability, continuity, and sustainability, are discussed in the next section.

4.3.1 Scenarios for Operational SWE Services

When considering possible scenarios for the operationalisation of SWE services while taking into account the features identified above, the most pressing questions that European stakeholders will need to address include:

- Who will be in charge of system operation, maintenance and updates?
- Who will control the performance of the overall system (both technically and financially) and be responsible for a system error?
- Should an existing entity be picked or a new one created?
- Should the operator be public (civilian), military or commercial?
- Should SWE services be treated as an intergovernmental or supra-national function?
- Should the management of operational SWE services be detached from other SSA components (i.e. SST and NEOs) or should it be managed within the same operational body? (See Box-4)
- How can treated data be acquired, treated, disseminated, and under which conditions?
- How will the continuity of data be ensured?
- Where will the funding come from and who will ultimately pay for the services?

Depending on how SWE services are conceived (SWE services as private vs. public and SWE services as national vs. international), four broad scenarios for providing SWE services can be identified at European level. The scenarios are as follows:

- Self-provision by commercial entities (Approach A)
- Services provision by a public government agency at national or regional level (Approach B)
- Services provision by one or more public entities at European level (Approach C)
- Service provision through an international framework such as UNOOSA, or a dedicated international organisation (e.g. ICAO) (Approach D)

Each scenario presents different approaches to service management. Importantly, the four approaches here presented are not mutually exclusive. In fact, they are, and will continue to, develop simultaneously and even reciprocally. For example, the uptake of operational services at European level will not prevent or void service provision at national level; MSs will be keen to continue capitalising on their past investment to deliver services at national level to protect critical infrastructure and devise crisis management plans, for both operators and civil society. Beyond this, the solidifying roles of national/regional public entities regarding SWE service provision can provide a good basis, equipped with tools and expertise, for a complementary and collaborative relationship with international frameworks, as is the case of ICAO aviation services. At the same time, commercial entities may still able to act in their own interest in terms of service provision to ensure business continuity with

respect to risks that may occur on a timescale of some years and have limited impact.

However, what remains to be further discussed and assessed is precisely what architecture will take place at pan-European level.

From this standpoint, identification and development of a dedicated governance scheme for service delivery in Europe (Approach C) remain subject to different options (see Figure 22).

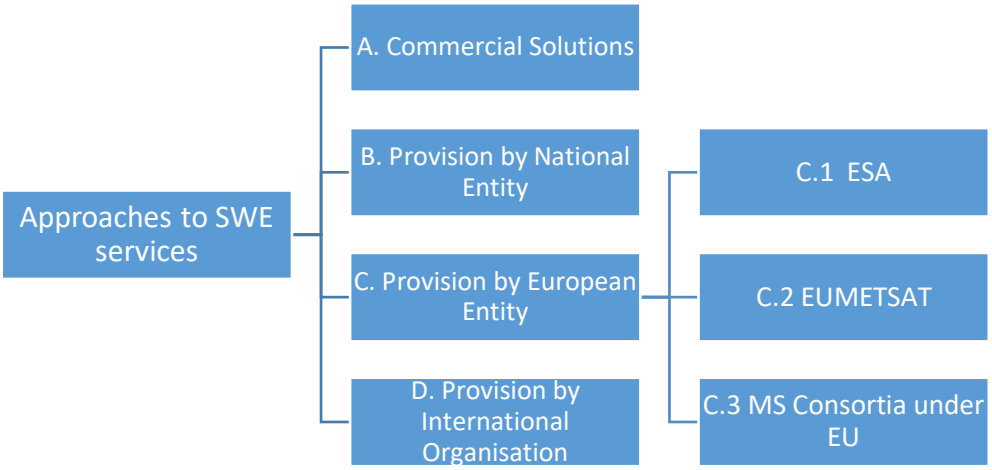


Figure 22: Approaches to Operational SWE Services

More specifically, under the present circumstances, three major institutional solutions come to the fore to act as the linchpin of an operational SWE system in Europe:

- ESA
- EUMETSAT
- the EU

It is essential to highlight that selecting one of the three configurations will neither automatically preclude complementary roles for the other two stakeholders, nor the potential involvement of member states, the private sector as well as international partners. Rather, such selection would only imply that that the provision of operational SWE services would be centred on the framework of one of these three pan-European institutions.

Scenario 1: SWE services provided through the ESA SWE network

A first option for the delivery of operational services would entail an expansion of ESA's current role in the development of pre-operational services into the delivery of fully operational services.

The Agency has already been entrusted with full responsibility for the design, development and exploitation of pre-operational SWE services, as the development of prototypes has been deemed to be the best way of advancing the associated technology and facilitating the transition to the operational phase. ESA is exercising this responsibility in consultation with potential users (mostly national SWE entities) and is identifying modalities to provide them with the technical and institutional assistance they may need to take over the management of these services and to organize their exploitation.

As stressed above, however, ESA's mandate does not prevent it from becoming involved in operational activities, particularly if organised users, i.e. an operator, do not exist and if requested to do so by member states. While such involvement may be considered as temporary before a transfer of responsibility to an external organisation, it is also possible that it could become permanent.

The main justification for enacting this route is the need to avoid a situation in which there is no operator for the SWE systems/services developed by ESA. Thus, under this option, the Agency would act as the linchpin of the SWE service governance framework by:

- Owning, managing and operating the space-based infrastructure, while coordinating the operations of the ground-based infrastructure owned by national member states;
- Providing data or data services in an operational manner to end users (either directly through its SWE Service Portal or via a service provider)
- Maintaining a programmatic overview of the development of new systems and their integration through relations with its stakeholders (e.g. industry and member states), and

- Ensuring the availability, continuity and sustainability of such services

In terms of service availability, ESA would be responsible for the implementation of suitable *data policies* and *distribution schemes* that include the definition of: categories of users and usage; data distribution methods; ownership and IPRs; and possible fees attributed to categories of users/usage. At this stage, preference is given to the adoption of an open data policy, where security and business confidentiality is, however, also ensured (e.g. spacecraft specifications). This is possible because an open access model for data usage, combined with low – or zero – costs, is deemed to hold the potential to drive the development of SWE services, particularly considering that commercial SWE markets are still fragmented. Still, variations in access and pricing policies may be considered on the basis of the type of service provided, i.e. raw data or processed & tailored products.

In terms of services continuity, ESA has already established bilateral *ad hoc* agreements for exchange of services and is in an optimal position to establish agreements with other SWE data/service providers, including NASA/NOAA, to secure the continuity of supply of space delivered data and services, and to avoid or negate system failure.

In order to ensure a sustainable service downstream to users, under this option ESA would be directly in charge not only of fulfilling all technical, operational and organisational requirements, but also of receiving sustainable funding for the operational service it delivers. This was, and remains, a key issue in the involvement of ESA in operational activities, as the source of funding for such activities needs to be different from R&D activities. Indeed, while funding can be sourced either publicly or privately and ensured by a cash flow from users to operators (usually via providers) across the service value chain, as clearly highlighted by Article V.2. of ESA Convention (see Annex. A.6), this funding must be allocated through an operational budget and not a research budget. From a monetary perspective, the other condition that must be fulfilled is that: "Member States that have contributed to the development of a space programme [are] to be equitably associated with the follow-up operational activities resulting from the programme in question, taking due account of any commercial constraints."

Therefore, under this option, it is more likely that member states (through their national agencies or designated entities – e.g. infrastructure protection authorities or National Meteorological Offices) would act as the providers and take the role of exclusive licencing agents for ESA's products to end users within their corresponding national regions. The necessary partnership between ESAs' Space Weather Coordination Centre (SWCC) as the operator and MSs as service providers can be formalised by their national entities having a role in the management of ESA, for instance by continuing to have representation in the ESCs and SWCC. In this way, the operator and service provider each have a real capability to shape and effectively meet user demand. The more intertwined the partners, the more sustainable is the value chain and therefore the service.

Scenario 2: Services Provided through EUMETSAT

A second option for managing operational SWE services would entail transferring the management of operational SWE services to EUMETSAT. As in the case of ESA, a scenario in which EUMETSAT takes the role of operator is an intergovernmental, institutional-oriented, and member states-driven model for SWE service governance.

The Darmstadt-based organisation has already voiced its interest in having a stronger and more defined role in the management of operational SWE activities and, thanks to its long and relevant experience in cost-effectively turning what is essentially the output of scientific research into operational services that are tailored to wide user communities, the rationale for EUMETSAT's involvement in the delivery of SWE services are in principle clear. Indeed, the current model for meteorology and climate services that EUMETSAT embodies is an attractive model for future SWE service operation because a number of parallels can be drawn regarding the types of activities to be conducted, the organisational structure, and the data policy. In addition, it should be noted that the EUMETSAT model developed from an ESA experimental satellite programme as an ad hoc operational organisation – with this approach proving its "validity over the years, to the satisfaction of its member states".[279] Importantly, unlike ESA's ESCs (which are organised according to the scientific domain) EUMETSAT's Satellite Application Facilities (SAFs) are tailored to individual domains and provide "users with operational data and software products, each one for a dedicated user community and application

[279] Fondation pour la Recherche Stratégique, 2008:13

area".[280] Additionally, the SAF model is fully rationalised and well-integrated with the user communities, as SAFs are situated within the National Meteorological Services (NMS), its member states, or other user-linked bodies[281].

Interestingly, a scenario in which EUMETSAT takes responsibility for SWE service provision finds an important parallel in the experience of the U.S., which today has a complete space weather monitoring and data service system operated by NOAA/SWPC in collaboration with NASA and USAF. Through NOAA/SPPC, many of the space weather data products and services are made available without any restriction to a wide variety of users. Also, protocols and procedures for international space weather data exchange and service coordination are established and implemented through NOAA's cooperation with ISES.

Reflecting this line of development in the case of EUMETSAT meteorological activities, in terms of the organisation of the service value chain, ESA would retain the role of the system developer, while EUMETSAT would act as the mediating operator, passing on its data products to its member states largely through National Meteorological Offices (NMOs) – or, alternatively, national civil protection authorities – who would in turn provide services to the end-users. End users would mainly be governmental entities in charge of national infrastructure protection but also commercial companies such as commercial spacecraft operators, airlines, etc.

It is important to note that in the delivery of SWE services, EUMETSAT would not act as the service provider, but as the system operator. Equally important, and in a similar fashion to EUMETSAT's involvement with the operation of Copernicus, EUMETSAT would have responsibility for the operation of the space segment only. Hence, the designation of a separate operator for the ground segment would be necessary.

In terms of relations between operators, providers and users, EUMETSAT would include representation of NMOs within its bodies, as in the case of meteorological services. The NMOs as service providers would subsequently be able to input EUMETSAT's decision processes, whilst also acting as a bridge between the operator and end-user.

From this operational role, EUMETSAT would be responsible for the dissemination of the resultant data products to national meteorological or civil authorities who would refine and provide services on an independent level, with potential support from EUMETSAT in doing so.

With regards to access, distribution, and IPR, as a meteorological system operator EUMETSAT would act as an appointed entity by ESA and its member states to set data and product prices, whilst NMOs would take the role of exclusive licencing agents for EUMETSAT's data within their corresponding national regions

While NMOs would be additionally responsible for marketing and commercial data distribution, it is foreseeable that, as in the case of meteorological and climate services, this distribution would function under an open data policy – the reason for which is that in this context the data products and services would be conceived as public goods, and ultimately be provided primarily for the benefit of EUMETSAT'S member states as well as wider users. This model depends on EUMETSAT's member states expressing their desire for EUMETSAT to take an operational role in SWE services, since EUMETSAT requires the mandate of its members and the certainty of an operational budget that would follow.

In obtaining sustainable funding for operations and development of new systems, EUMETSAT could receive contributions based on the GDP of its member states, as in the case of meteorological services. Furthermore, EUMETSAT would be awarded funding by national meteorological offices as service providers, additionally reinvesting any revenue made back intro infrastructure development.

EUMETSAT would own, manage and operate its own satellites. However, while EUMETSAT would conduct operations, its satellites would be procured by ESA. In this regard, there should be an agreement between ESA as the development agency and EUMETSAT as the operator for the development of novel satellite generations.

Overall, it is clear that if EUMETSAT were to take on the role as the operator for SWE services, it would need a clear mandate and resultant financial capabilities from its member states. To reach this stage, the dialogue in the European setting for SWE services needs to meet a critical mass in terms of convergence and consolidation across the board for stakeholders relevant to SWE services. This would primarily entail further platforms for discussion on not just the maturity of the scientific and technical aspects of SWE services, but crucially the organisational elements – namely

[280] European Organisation for the Exploitation of Meteorological Satellites, 2018c

[281] European Organisation for the Exploitation of Meteorological Satellites, 2018c

the designation of operators and service providers, the allocation of an operational budget, and the development of the user community.

Another critical aspect in the adoption of this option is that, notwithstanding the progressive expansion of EUMETSAT activities as provider of SWE-related data through the SWE instruments hosted on its satellites, the overall level of expertise remains minimal, with only a few in-house experts working on SWE matters. Therefore, adoption of this option would also entail a rather long timeframe for full operationalisation.

On a more positive side, it must be noted that EUMETSAT is in a good position for international cooperation on SWE matters, with solid links with international organisations such as WMO, CGMS and agencies like NOAA through years of cooperation in the fields of meteorology and climate. These international organisations are already interested and partaking in SWE activities, in some instances with far more maturity and progression towards full operation of SWE services. As such, EUMETSAT's pre-existing relationship with such entities would prove invaluable to the development of SWE services in the European context. One major factor would be ensuring the continuity and availability of data through mutual collaboration, sharing of data, and safeguarding satellite SWE observation capabilities should one's partner's satellites be defective or out of action (as seen with mutual support between NOAA and EUMETSAT in meteorology or in the case of GEOS and METEOSAT support between NOAA and ESA/EUTMETSAT during system failure events).

Scenario 3: Providing/Procuring Operational Services Under the EU

A third option for the operationalisation of pan-European SWE services is to organise the provision of operational SWE service under the umbrella of the European Commission.

In the Proposal for the space programme of the European Union[282] released in June 2018 and currently in co-decision by EU legislative authorities, the European Commission already expressed its resolve to cover the space weather element within the Situational Space Awareness component of its space programme with the aim of assessing and identifying SWE user needs, raising awareness of SWE risks and the delivery of operational and user-driven SWE services.

The rationale for this is the recognition that "all EU assets constituting the basis for EU space actions can be potentially impacted by space weather and therefore the development of a framework for space weather is necessary to protect them and ensure efficient and reliable services provided by other space activities".[283] While SWE services are being developed and coordinated by ESA and some member states, these are mostly focused on science and still need to be tailored to meet operational users needs. In other words, the EU recognises that the value stemming from its intervention would be higher than the value that would have otherwise been created by ESA or member states alone. Accordingly, the EU has proposed supporting the continuous provision of operational space weather services by building on, and in complementarity to, ESA and national activities.

With a nod to Article 59 of the June 2018 EU Space programme proposal, the European Commission would be primarily in charge of the procurement of operational SWE services to be delivered to the SWE users, according to identified user needs and relative technical requirements.

More specifically, building on the assessment and identification of the needs of users in pre-selected sectors[284] (the number of services and sectors can always be extended by means of implementing acts according to future needs of users, new technological capabilities, and future risk assessments) SWE services would be procured through calls for tenders open to a set of SWE consortia.

While further specifics of this *modus operandi* are not detailed in the proposal – the text of which may also likely change following the Council and Parliament amendments - some assumptions on the possible implementation of this scenario can be advanced.

For one thing, it is not precluded that the work of these SWE consortia could be organised around specific user domains (e.g. spacecraft, aviation, GNSS systems, electric power grids and communications, etc.) rather than scientific domains (solar weather, ionospheric weather, heliospheric weather, etc.). This would enable service provision to be more user driven and rationalised according to user needs. It is also possible that a specific stakeholder (e.g. a national research institute or an industry) may participate in multiple consortia, provided that its technical expertise and

[282] European Commission, 2018
[283] European Commission, 2018

[284] Article 59 identifies the following sectors: spacecraft, aviation, GNSSs, electric power grids and communications.

capabilities are relevant to the SWE service domain in question.

Overall, within this framework, the members of the consortia would have to make use of the necessary technical expertise and capabilities to ensure operational SWE services delivery. More specifically, they would be directly responsible for managing and operating their ground- and space-based systems and for processing the data, as well as for providing the necessary interfaces to centralise, store and make available SWE services to different types of users and ensuring the implementation of the data policy.

The role of the Commission would be to define the general guidelines for the governance of the SWE service framework, to coordinate the work of the consortia, and to ensure the funding for their activities. In line with Article 30.3 of the Proposal,[285] the Commission may entrust these and other tasks related to user uptakes of SWE services to the EU Agency for the Space Programme (the current GSA), although it is once again worth remembering that final version of the Regulation may differ from the Commission's proposal due to the Council and parliament amendments. Supposedly, the envisioned Agency would also be responsible for:

- Assuring, in cooperation with the SWE consortia, the *availability* and *continuity* of the SWE services through an appropriate data policy, distribution scheme, and international cooperative frameworks
- Assuring the *sustainability* of the delivered services by:
    - supporting, promoting and encouraging the use of the services across user communities;
    - gathering feedback to ensure the required alignment of services with user expectations;
    - ensuring a programmatic overview and providing direct reporting on the performance of the SWE services.

In order to guarantee the *availability* of the services, the national entities participating in the consortia would have to conclude implementing arrangements with the Agency to tackle the inevitable issues related to the ownership of data and the IPRs (because these data would remain national property, meaning that the decision to share them, and to what extent, is still up to states).

In terms of service *continuity*, the European Commission's proposal for an EU Space programme has already highlighted the importance of cooperation with international partners, in particular the U.S., international organisations, and other third parties to increase preparedness for the effects of extreme space weather events. Through the EEAS, the Union has in addition expressed its support for the creation of an International System for Operational Space Weather Service and a Coordination Committee within the UN to ensure the sharing of SWE data (and therefore continuity and quality of different SWE services) at international level.[286]

As for the sustainability of the service, a key issue for the eventual implementation of this option would be the source and extent of funding. In fact, the level of funding envisaged by the regulation to support SSA-related activities is arguably modest, especially considering that this budget would have to be split with GOVSATCOM and other SSA elements (SST and NEO). The EC has already acknowledged this issue and, accordingly, has called for prioritisation of the sectors to which the operational SWE services are to be provided taking into account the user needs, risks and technological readiness. In the long term, the needs of other sectors may be addressed. Whereas this incremental approach may entail an unclear implementation timeframe for the full delivery of SWE operational services, it is also likely that it will also enable the emergence of a self-sufficient and sustainable provision scheme, whereby in the future the consortia would obtain funding from service users through the operators, along the lines of what will be tested by the PECAUS consortium for the aviation sector.

4.3.2 Scenarios Assessment

Each of the above-discussed scenarios has very particular characteristics that carry different advantages but also raise possible drawbacks, as briefly outlined in Table 16.

	Pros	Cons
Scenario 1 (ESA)	• Demonstrated expertise in SWE	• Outside ESA's traditional mandate • Source of funding unclear

[285] The Commission may entrust other tasks to the Agency, including undertaking communication, promotion, and marketing of data and information activities, as well as other activities related to user uptakes with regard to the Programme's components other than Galileo and EGNOS

[286] EU Delegation, 2018

	• Quicker timeframe for implementation • Strong relations with SWE agencies worldwide	• Bias towards space operations-related services • Bias towards space-based capabilities
Scenario 2 (EUMETSAT)	• Long experience in managing operational activities • Suitable infrastructure for managing operational systems • Connection to the WMO • Strong relations with SWE agencies worldwide	• Minimal expertise in SWE • SWE detached from other SSA components • SWE space segment operations detached ground segment • Longer timeframe for implementation
Scenario 3 (EU)	• Political backing in the delivery of operational services • SWE embedded in a broader framework for space safety • Involvement of private sector in service delivery	• Relatively recent involvement in SWE activities • Unclear implementation timeframe • Still modest funding for implementation

Table 16: SWE Services Scenario Assessment

Clearly, selecting one scenario over another will eventually come down to the relative weight European stakeholders (most crucially, EU and ESA's member states) attribute to the different pros and cons vis-à-vis the political, financial and operational aspects of each scenario. In this sense, this report does not aspire to provide a conclusive answer regarding the best way forward, as any specific selection can only play out in the actual political decision-making process.

Irrespective of the possible enactment of one of the above-identified scenarios, what is, however, important to highlight is the need for public/private end-users of SWE services as well as national and European decision-makers to clearly recognise the stakes.

4.4 The Bottom Line: Enhancing Awareness and Preparedness

As highlighted in Chapter 2, there are strategic, economic, societal and environmental stakes associated with SWE service delivery. Recognising such stakes is an important, not to say the most important, prerequisite to ensuring the required political support to address the above-identified technical, market and organisational gaps, and eventually make the provision of operational services possible.

Thus far, however, this recognition has not been equally shared among all European countries and across all the different sectors potentially impacted by SWE. As also assessed by a study led by the JRC, awareness proves higher among sectors such as power grid operators and aviation, and, geographically, in countries located in northern and central Europe (Finland, Hungary, Netherlands, Sweden, UK and Norway), which have formally recognised the threat of extreme SWE by including it in their national strategic risk assessments.

However, as also highlighted by Mann et al (2018), "[I]n the 21st century, the infrastructure and economies of the world's nation states are increasingly and intimately connected, both regionally and globally. Therefore, even countries with a perceived low domestic space weather risk will benefit from a coordinated approach to mitigating space weather impacts.[287]

In order to achieve greater recognition that SWE can affect multiple infrastructures and overwhelm a single's nation response capacity, a pan-European vulnerability assessment of the interdependencies between critical infrastructures should be performed. Arguably, this would greatly help not only to identify criticalities and potential transboundary effects in case of SWE events, but also to raise greater awareness of the stakes associated with SWE service delivery. Eventually, the adoption of a comprehensive European policy statement in the domain of SWE seems a necessary step to ensuring a higher alignment among European stakeholders.

Of course, the delivery of SWE services will not per se be sufficient to mitigate against their

[287] Mann et al., 2018

effects or to derive socio-economic benefits. In fact, once SWE warnings or alerts are issued, it will also be essential to know what to do. Hence, "a set of best practices, operating procedures, and actions have to be identified by first assessing what risks and socio-economic impacts are present in each member state or region and then to recognize the correct and appropriate measures to be taken in order to mitigate the potential consequences."[288]

In the United States, the Obama administration "…. issued a National Space Weather Strategy that defines high-level strategic goals and actions for increasing preparedness levels". Vice-versa, in the European context, there is much uncertainty in matters related to SWE risk management. While there are protocols and preparedness plans at national level (which are also mostly based on single infrastructure disruptions), such documents are missing at the pan-European level. More broadly, to date there has been no European decision-making capability that could quickly respond to a SWE-caused disruption of critical infrastructures. Therefore, as also recommended by the JRC:[289]

- Protocols should be developed that define responsibilities and ensure good coordination between the stakeholders before, during and after an extreme event. This includes communication of the risks and potential impacts to the public.
- Emergency plans for extreme space weather should consider the full range of critical infrastructures possibly affected. Once drawn up, these plans need to be tested.
- The opportunity for organising a joint space-weather exercise at EU level should be explored to test existing response capabilities and identify critical gaps…. This or other space-weather exercises could be organised as a multi-national exercise under the Union Civil Protection Mechanism.

Ultimately, the existing European structures and mechanisms dealing with prevention, preparedness and response to possible hazards/disasters could cover SWE events to facilitate pan-European preparedness planning and coordination in case of a SWE event.[290] The European Response Coordination Centre (a 24/7 emergency operation centre that is progressively becoming the EC's hub for response to different types of crises) could offer a relevant framework in this respect. Once again, for this development to become feasible, political support from member states will need to be ensured – possibly through the Integrated Political Crisis Response (IPCR) arrangements[291] – due to the national sensitivities involved in critical infrastructure protection.

4.5 Elements for a European Roadmap

Over the next few years, the currently ongoing European efforts towards the delivery of operational services will face multiple commitments on multiple fronts. More specifically:

On the technical front there will be a need to:

- Secure access to SWE data through the development /deployment of European SWE missions and ad hoc agreements with international SWE service providers
- Improve physic-based model and computational tools that will enable to move from the current observation/post-processing capacity context into a predictive one
- Add and validate key products and services through coordination among different service providers well as through a process where SWE products and services are tested with real users in the loop.

On the market front, there will be a need to:

- Fortify interaction engagement with the end user communities, continuing to collect user feedback to feed into the SWE roadmaps
- Perform more accurate assessments of infrastructure vulnerabilities and interdependencies that address cascading effects and multiple stakeholders with a view to prepare strategic plans
- Progressively tailor services to the needs of specific user/customers (e.g. alerts not merely providing information on a SWE

[288] Mann et al., 2018 - Such studies need to recognize that geographic differences influence the severity of space weather effects on vulnerable infrastructure and technology systems. The COPUOS-approved space weather-related guidelines for LTS already include some additional elements relating to protection strategies. Their future implementation in member states would advance global space weather resiliency.

[289] Krausmann et al., 2016

[290] Ibid.

[291] The Integrated Political Crisis Response (IPCR) arrangements, sanctioned in June 2013 by the Council of the European Union, aim to "provide a flexible crisis mechanism for supporting the presidency of the Council of the European Union in dealing with major natural or man-made cross-sectorial disasters, as well as acts of terrorism". https://www.consilium.europa.eu/media/29699/web_ipcr.pdf

event but how the event will affect, e.g. spacecraft operations)

- Take steps to support the uptake of private actors in the delivery of commercial SWE services by mapping the potential customer base and European private capability and evaluating emerging business opportunities and challenges
- Identify the demand for new possible SWE services (e.g. in the field of STM)

On the organisational front, there will be a need to:

- Assess different governance solutions for the management of operational services with a view to establish possible synergies with other SSA services
- Entrust the identified operator with the responsibility for the operational system(s)
- Define an appropriate data policy and secure a sustainable funding and allocation scheme

An important step toward the fulfilment of these objectives will be the *2019 ESA Council at Ministerial Level*, which will be a key juncture point for the development of European operational SWE services for two reasons:

- On the one hand, the development, deployment and operation of ESA space-based SWE missions (L1/5 and D3S) will need to be approved,
- On the other, the activities for P4 will need to be defined, including the extension/reinforcement of the SSCC role in interacting with users and coordinating the work of the ESCs

Another important step toward the consolidation of a more robust European framework for SWE services is the current negotiations for the next *Multiannual Financial Framework 2021–2027* of the EU, approval of which will entail a number of important developments in this field in terms of new activities, means and ambitions.

In fact, if the European Commission's proposal for the EU Space Programme is agreed and adopted by the European Parliament and the Council, for the first time funding for SWE-related activities will not only be allocated for basic research through the next research and innovation framework programme (FP9 or Horizon Europe) but also for operational service delivery. Equally important, a new organisational setting may be introduced. While all this may well prove a crucial development toward the delivery of operational services, it remains unclear whether the currently foreseen funding will be sufficient to reach the level of resources required to establish an operational system for SWE service delivery, and whether a dedicated budget line (other than Horizon Europe), will thus be essential to reach the level of resources necessary to provide them.[292]

In parallel to these important schedules at European level, a third important element in the consolidation of a roadmap for SWE service delivery will be the future implementation of the proposals made in the UNISPACE+50 report under Thematic Priority 4 - Framework for International Space Weather Services. More specifically, European stakeholders will have to decide on the proposed creation of a *new International Coordination Group for Space Weather (ICSW)*, which would "lead international coordination between member states and across international stakeholders, monitor progress against implementation of guidelines and best practices, and promote coordinated global efforts in the space weather ecosystem spanning observations, research, modelling, and validation, with the goal of improved space weather services".[293]

In the envisaged schedule, an international space weather workshop should be held in the summer of 2019 to define the terms of reference, mandate, and formal structure of such a group. The ICSW will replace the UN COPUOS expert group for space weather in 2020 and become functional on approval by the COPUOS STSC and the plenary session in February and June 2020, respectively.

European support for the proposal may prove key to fulfilling the widely acknowledged need for further international collaboration to promote the implementation of space weather guidelines and best practices and minimize duplication of efforts, while filling key measurement or other service gaps.

[292] This is especially the case if the foreseen budget of €0.5 billion will have to be allocated also to the SST and GOVSATCOM initiatives.

[293] Mann et al. 2018

5. Findings and Recommendations

In the general progression towards a service-oriented European space policy and programme, an emerging issue-area is SWE services, here defined as the final output of the transformation of space weather data and products into practical applications for specific customers to mitigate the potentially harmful impacts of SWE.

Whereas SWE services are already used today in a number of countries and within a number of sectors (e.g. the commercial airlines, the satellite industry, drilling and surveying operations, power grid operators, users of satellite-based navigation systems, etc.), there is a general consensus that this demand will expand further in the future, requiring greater maturity of the services delivered to end users. Indeed, as dependency on space- and ground-based technological systems increases along with the sensitivity of our society and our economy to SWE, the stakes associated with the delivery of effective SWE services will become higher.

Such stakes are not only associated with the mitigation of the potentially catastrophic chain of impacts generated by rare major SWE events, but are more broadly connected to a series of strategic, commercial, societal and environmental objectives, including the development of end-to-end European capabilities, the advancement of basic and applied research in critical areas, and the overall positive impact on European economy. Whereas the achievements reached by European stakeholders in the span of the last decade (2008-2018) are certainly remarkable, there are still important issues in the delivery of operational SWE services.

Scientific and Technological Issues

These issues are, firstly, of a scientific and technological nature. Indeed, when performing a capability assessment by measuring the overall level of service maturity (here defined as the operational implementation of the service constituents), it becomes evident that the major hurdle is not simply associated with the transition from the pre-operational to the operational stage; rather, it is first and foremost associated with the maturity of the constituent elements, namely the data, models and data products underpinning the targeted service. Hence, in the coming years there will be a need to address the technological gaps, by filling data gaps (especially space-based observations), improving both empirical and physics-based modelling capabilities, and advancing product maturity and reliability.

Recommendations

- Funding for development and deployment of the L1/L5 mission and D3S needs to be approved at the next ESA Ministerial Council in order to ensure a non-dependent source of SWE data at European level.
- More synergies between the activities of ESA in P4 and those of the EU in the upcoming MFF need to be established for defining coordinated strategic investments into developing modelling capabilities and forecasting techniques.
- Additional efforts are needed to add and validate key products and pre-operational services through coordination among different service providers and testing with real users in the loop.

Demand/Market Issues

Other important issues persist in the demand/market dimensions of SWE services; more specifically in their source of operational funding and identification of customers. Current European efforts in demand analysis have focused purely on user identification and their relative requirements. However, there is still a lack of understanding as to who the customer of a service will be, i.e. the entity procuring the SWE services.

Within the European context, a number of factors still prevent the emergence of a business case for commercial procurement/delivery of SWE solutions, including: the still low accuracy

of forecasts, the availability of reliable free services by NOAA SWPC, the lack of clear economic benefits related to service provision, and the understanding of SWE as public goods, etc. In light of these issues, it is anticipated that, for the foreseeable future, SWE service provision will be conducted primarily under a public procurement model. However, it is also expected that, as the SWE demand further develops, businesses may become willing to invest in the procurement/ delivery of services that support their business continuity. This is especially the case for services related to risks that occur with high frequency and have limited impact. (e.g. radio black outs stemming from solar flares). Conversely, benefits may be perceived as too distant or intangible and burdensome to set up a market for highly disruptive but low-frequency SWE risks, such as large geomagnetic storms

Recommendations

- Sustained public investment will be needed to ensure operational service provision for the protection of national and European critical infrastructure, especially with respect to high-impact, low-frequency SWE events
- With respect to risks stemming from low-impact, high-frequency event, functional steps to support the progressive engagement of private actors in service provision and the emergence of a customer base for SWE services should be taken. These steps include:
  - Continuing raising sectorial awareness of how SWE services might be beneficial to users through a reinforced dialogue with all stakeholders (authorities, operators, and also the public) and through more accurate socio-economic impact assessments that sensitise operators of space- and ground-based infrastructure
  - Seeking cooperation with industry for developing *ad-hoc* services tailored to specific domains (or even specific users) providing more value-adding information as compared to those already provided by government and openly accessible on the internet
  - Identifying sectors (e.g. oil drilling, polar shipping; unmanned transport) where users may be incentivised to procure services on a commercial basis because of tangible economic benefits.
  - Identifying suitable commercial provision schemes in commercially viable sectors (e.g. fees levied on passengers' tickets as in the case of the ICAO aviation services, or fees charged on energy operators and then cascaded onto the end-user energy bill)

Organisational Issues

A third set of outstanding issues in delivery of operational SWE services in the European context concerns the definition of an appropriate organisational setting. As of 2018, there is still a lack of pan-European consensus on what institutional architecture should be entrusted with the responsibility for the operational services and the means to coordinate development efforts along the SWE service value-chain.

From this standpoint, the identification and development of a dedicated governance scheme is subject to different solutions, with different cascading implications for ensuring service *availability*, *continuity* and financial *sustainability*. More specifically, under the present circumstances, three major institutional solutions come to the fore to act as the linchpin of an operational SWE system in Europe, namely ESA, EUMETSAT and Member States Consortia under EU procurement. Each scenario has very particular characteristics that carry different advantages but also raise possible drawbacks. Therefore, selecting one scenario over another will eventually come down to the relative weight European stakeholders (most crucially, EU and ESA's member states) attribute to the political, financial and operational pros and cons of each scenario.

Recommendations

- The roadmap on the scientific and technological dimensions of SWE service delivery needs to be complemented by a strategic plan to define the roles of the key players in Europe. While in recent years several dedicated workshops, meetings and reports at expert level were held to consult stakeholders on specific needs on SWE, these were only of a technical nature. Thus, a joint policy workshop involving all stakeholders should be held to tackle outstanding issues related to the overall governance of SWE services (including data policy, procurement, funding schemes).
- Since in the Europe there is a multiplicity of approaches to SWE, different vulnerabilities and specific capacities which will inevitably drive significantly the future of the service provision in the European context, it will be essential to and capitalise on these existing capabilities and ensure effective coordination among

them in order to deliver consistent operational services
- Consensus among these stakeholders should be achieved on whether or not to integrate SWE as a key component of the European programme for space safety and security as well as an integral constituent of the European public policy against natural hazards.

International Cooperation

Whereas the uptake of European capacities in SWE service delivery is necessary to better address the specific vulnerabilities and requirements of the European context, there is also a clear understanding that "any European progress should notably be part of a global effort, very much in the sense of the recent ILWS/COSPAR Space Weather roadmap, which also has been adopted as the baseline for global space weather efforts as pursued and closely monitored by the UN-COPUOS Expert Group on Space Weather"[294].

Due to the nature and global impact of SWE events, improved coordination in the field of SWE "is especially relevant for filling key measurement gaps, securing the long-term continuity of critical measurements, advancing global forecasting and modelling capabilities, identifying potential risks, and developing practices and guidelines to mitigate the impact of space weather phenomena, including on long-term observation of climate change and risk events"[295].

Recommendations

- Global coverage from ground- and space-based observation systems is critical for ensuring the delivery of operational services and more mature forms of international coordination/cooperation in the exchange of SWE data/products should therefore be established.
- There is also an evident need for ensuring consistency in forecasting, and coordination of forecasts from different service providers is hence required.
- The prospected creation of an International Coordination Group for Space Weather (ICSW) should be supported by European stakeholders.
- In light of the envisaged engagement of the EU in the delivery of operational SWE services, there will be important opportunities to be leveraged in advancing the specific technical requirements expressed by the SWE scientific and research community through coherent European diplomatic initiatives (such as the one led by the EEAS in particular).

European Preparedness

The eventual delivery of operational SWE services will not *per se* suffice to mitigate against their effects. There is also a need to know what to do once forecasts or alerts are provided. In the European context, there is still much uncertainty in matters related to SWE risk management. While there are protocols and preparedness plans at national level (which are, however, mostly based on single infrastructure disruptions), such documents are missing at the pan-European level. More broadly, to date there has been no European decision-making capability that could quickly respond to a SWE-caused disruption of critical infrastructures.

Recommendations

- In order to enhance European preparedness for SWE events and support risk assessment and mitigation strategies by operators, impact models for different types of critical infrastructures, their interdependencies and ripple effects in society, should be developed in a coordinated manner.
- European-wide protocols defining responsibilities and actions of the various stakeholders before, during and after SWE events should be established.
- An exercise under the Union Civil Protection Mechanism should be held with a view to establishing preparedness plans and protocols that ensure coordinated and prompts actions before, during, and after SWE events.
- The European Response Coordination Centre should be entrusted with the responsibility to facilitate pan-European planning and coordination in case of SWE event.

[294] European Space Sciences Committee, 2017

[295] United Nations Committee on the Peaceful Uses of Outer Space, 2017b

Annexes

A.1 Explanation of Terms

Term	Definition
Alarm	A notification in near-real time of the occurrence of a SWE event.
Customer	A paying recipient of a service.
Data	Observational measurements of any space weather parameter, raw or processed
End User	An entity, i.e. a person, organisation or electronic system, that accesses/receives products or services.
Expert Group	An entity that provides expertise within a certain domain.
Forecast	A portrayal of the future space environment, based on historical and current data, proxies, and models.
Governance	The strategic, financial, regulatory or organisational aspects of a particular activity. Whilst governance is often associated with public institutions, these activities are also conducted by non-public entities.
Near Real-time	The data, product, or service is produced and delivered near, or as close as possible, to the same rate as which a space weather parameter is observed.
Nowcast	A representation of the current space environment based on data, proxies, and models.
Product	Data that is derived from a space weather model or tool, in some cases more than one. It has a defined format, is archived, reproducible, and can be utilised by one or more services.
Real-time	The data, product, or service is produced and delivered at the same rate as which a space weather parameter is observed.
Service	A collection of data products, software tools, technical reports and user support that fulfil the requirements of a specified user group.
Space Situational Awareness	Knowledge, understanding, and awareness of the status of space objects, the space environment, and posed threats/risks.
Warning	A notification in near real time of a potentially hazardous SWE event.
Data Policy	Rules and procedures for accessing, handling, storing and distributing both raw and processed data

The Near-Earth Orbit Environment

The Ionosphere

The Ionosphere is a region surrounding the Earth that is characterised by the ionised layer of the Earth's atmosphere through its interaction with solar and cosmic radiation[296]. The Ionosphere is primarily produced by solar radiation that energises particles in the Earth's atmosphere, causing them to lose an electron (i.e. ionisation), in turn forming a cloud of charge particles that is visible to the naked eye from space as a colourful emission known as airglow[297]. This region extends between 50 to 360 miles (80-600km)[298] above Earth's surface and is dynamic to variations in levels of solar and cosmic radiation[299]. This swelling of the ionosphere because of solar variation means that its exact shape changes throughout the day, with the day-side ionosphere always being proportionally larger than the night-side ionosphere[300].

The Magnetosphere

The magnetosphere refers to the spatial region surrounding the earth that is considered Earth's dominant magnetic field, as opposed to magnetic fields of interplanetary space, where the behaviour of charged particles is controlled by Earth's magnetic field.[301] The magnetosphere encompasses both the Earth's magnetic field - originating from contributions from the Earth's core, the lithosphere, and the coupling of electrical currents between the ionosphere and the magnetosphere[302] - and its interplay with solar wind[303]. This region takes the form a dipole, having north and south poles, with the pressure of solar wind causing a compression on the region of the Earth facing the sun (day-side) and a long tail extending from the Earth on the opposite side (night-side). The magnetosphere is extremely dynamic and responsive to solar variation[304].

The Thermosphere

The Thermosphere refers to the atmospheric layer around the Earth that extends between 53-375 miles (~750km/ 90-500km) in altitude[305]. Within this region, radiation of energised particles sourced from the Sun barrage oxygen and nitrogen molecules, which causes them to split into their constituent atoms to produce heat. Whilst temperature within the thermosphere increases with altitude because of decreasing abortion rates, it can also vary dependent on solar radiation levels[306].

[296] Stanford Solar Center, 2018
[297] National Aeronautics and Space Administration, 2016b
[298] Royal Academy of Engineering, 2013:61
[299] National Aeronautics and Space Administration, 2016b
[300] Garner, 2016
[301] Royal Academy of Engineering, 2013:61
[302] Royal Academy of Engineering, 2013:11
[303] National Oceanic and Atmospheric Administration, 2018f
[304] Zell, 2011
[305] Zell, 2013
[306] Zell, 2013

A.2 NOAA Space Weather Scales

NOAA Space Weather Scale for Radio Blackouts[307]

Radio Blackouts			GOES X-ray peak brightness by class and by flux*	Number of events when flux level was met; (number of storm days)
R 5	Extreme	HF Radio: Complete HF (high frequency**) radio blackout on the entire sunlit side of the Earth lasting for a number of hours. This results in no HF radio contact with mariners and en route aviators in this sector. Navigation: Low-frequency navigation signals used by maritime and general aviation systems experience outages on the sunlit side of the Earth for many hours, causing loss in positioning. Increased satellite navigation errors in positioning occur for several hours on the sunlit side of Earth, which may spread into the night side.	X20 ($2x10^{-3}$)	Fewer than 1 per cycle
R 4	Severe	HF Radio: HF radio communication blackout on most of the sunlit side of Earth for one to two hours. HF radio contact lost during this time. Navigation: Outages of low-frequency navigation signals cause increased error in positioning for one to two hours. Minor disruptions of satellite navigation possible on the sunlit side of Earth.	X10 (10^{-3})	8 per cycle (8 days per cycle)
R 3	Strong	HF Radio: Wide area blackout of HF radio communication, loss of radio contact for about an hour on the sunlit side of Earth. Navigation: Low-frequency navigation signals degraded for about an hour.	X1 (10^{-4})	175 per cycle (140 days per cycle)
R 2	Moderate	HF Radio: Limited blackout of HF radio communication on the sunlit side, loss of radio contact for tens of minutes. Navigation: Degradation of low-frequency navigation signals for tens of minutes.	M5 ($5x10^{-5}$)	350 per cycle (300 days per cycle)
R 1	Minor	HF Radio: Weak or minor degradation of HF radio communication on the sunlit side, occasional loss of radio contact. Navigation: Low-frequency navigation signals degraded for brief intervals.	M1 (10^{-5})	2000 per cycle (950 days per cycle)

* Flux, measured in the 0.1-0.8 nm range, in $W \cdot m^{-2}$. Based on this measure, but other physical measures are also considered.
** Other frequencies may also be affected by these conditions.

[307] National Oceanic and Atmospheric Administration, 2018

NOAA Space Weather Scale for Geomagnetic Storms[308]

Geomagnetic Storms			Kp values* determined every 3 hours	Nr. of storm events when Kp level was met; (nr. of storm days)
G 5	Extreme	Power systems: widespread voltage control problems and protective system problems can occur, some grid systems may experience complete collapse or blackouts. Transformers may experience damage. Spacecraft operations: may experience extensive surface charging, problems with orientation, uplink/downlink and tracking satellites. Other systems: pipeline currents can reach hundreds of amps, HF (high frequency) radio propagation may be impossible in many areas for one to two days, satellite navigation may be degraded for days, low-frequency radio navigation can be out for hours, and aurora has been seen as low as Florida and southern Texas (typically 40° geomagnetic lat.)**	Kp=9	4 per cycle (4 days per cycle)
G 4	Severe	Power systems: possible widespread voltage control problems and some protective systems will mistakenly trip out key assets from the grid. Spacecraft operations: may experience surface charging and tracking problems, corrections may be needed for orientation problems. Other systems: induced pipeline currents affect preventive measures, HF radio propagation sporadic, satellite navigation degraded for hours, low-frequency radio navigation disrupted, and aurora has been seen as low as Alabama and northern California (typically 45° geomagnetic lat.)**.	Kp=8, including a 9-	100 per cycle (60 days per cycle)
G 3	Strong	Power systems: voltage corrections may be required; false alarms triggered on some protection devices. Spacecraft operations: surface charging may occur on satellite components, drag may increase on low-Earth-orbit satellites, and corrections may be needed for orientation problems. Other systems: intermittent satellite navigation and low-frequency radio navigation problems may occur, HF radio may be intermittent, and aurora has been seen as low as Illinois and Oregon (typically 50° geomagnetic lat.)**.	Kp=7	200 per cycle (130 days per cycle)
G 2	Moderate	Power systems: high-latitude power systems may experience voltage alarms; long-duration storms may cause transformer damage. Spacecraft operations: corrective actions to orientation may be required by ground control; possible changes in drag affect orbit predictions. Other systems: HF radio propagation can fade at higher latitudes, and aurora has been seen as low as New York and Idaho (typically 55° geomagnetic lat.)**.	Kp=6	600 per cycle (360 days per cycle)
G 1	Minor	Power systems: weak power grid fluctuations can occur. Spacecraft operations: minor impact on satellite operations possible. Other systems: migratory animals are affected at this and higher levels; aurora is commonly visible at high latitudes (northern Michigan and Maine)**.	Kp=5	1700 per cycle (900 days per cycle)

* The K-index used to generate these messages is derived in real-time from the Boulder NOAA Magnetometer. The Boulder K-index, in most cases, approximates the Planetary Kp-index referenced in the NOAA Space Weather Scales. The Planetary Kp-

[308] National Oceanic and Atmospheric Administration, 2018

index is not yet available in real-time.** For specific locations around the globe, use geomagnetic latitude to determine likely sightings

NOAA Space Weather Scale for Solar Radiation Storms[309]

Solar Radiation Storms			Flux level of ≥ 10 MeV particles (ions)*	Number of events when flux level was met**
S 5	Extreme	Biological: unavoidable high radiation hazard to astronauts on EVA (extra-vehicular activity); passengers and crew in high-flying aircraft at high latitudes may be exposed to radiation risk. *** Satellite operations: satellites may be rendered useless, memory impacts can cause loss of control, may cause serious noise in image data, star-trackers may be unable to locate sources; permanent damage to solar panels possible. Other systems: complete blackout of HF (high frequency) communications possible through the polar regions, and position errors make navigation operations extremely difficult.	10^5	Fewer than 1 per cycle
S 4	Severe	Biological: unavoidable radiation hazard to astronauts on EVA; passengers and crew in high-flying aircraft at high latitudes may be exposed to radiation risk.*** Satellite operations: may experience memory device problems and noise on imaging systems; star-tracker problems may cause orientation problems, and solar panel efficiency can be degraded. Other systems: blackout of HF radio communications through the polar regions and increased navigation errors over several days are likely.	10^4	3 per cycle
S 3	Strong	Biological: radiation hazard avoidance recommended for astronauts on EVA; passengers and crew in high-flying aircraft at high latitudes may be exposed to radiation risk.*** Satellite operations: single-event upsets, noise in imaging systems, and slight reduction of efficiency in solar panel are likely. Other systems: degraded HF radio propagation through the polar regions and navigation position errors likely.	10^3	10 per cycle
S 2	Moderate	Biological: passengers and crew in high-flying aircraft at high latitudes may be exposed to elevated radiation risk.*** Satellite operations: infrequent single-event upsets possible. Other systems: effects on HF propagation through the polar regions, and navigation at polar cap locations possibly affected.	10^2	25 per cycle
S 1	Minor	Biological: none. Satellite operations: none. Other systems: minor impacts on HF radio in the polar regions.	10	50 per cycle

* Flux levels are 5-minute averages. Flux in particles·s^{-1}·$ster^{-1}$·cm^{-2}. Based on this measure, but other physical measures are also considered. ** These events can last more than one day. *** High energy particle measurements (>100 MeV) are a better indicator of radiation risk to passenger and crews. Pregnant women are particularly susceptible.

A.3 Space Weather Service Demand

This Annex details SWE impacts on various service domains, addressing the aspects of user identification and requirements for service provision, the potential benefits for each sector, and the technological systems needed to fulfil them.

[309] National Oceanic and Atmospheric Administration, 2018

Space Operations

Spacecraft Design

Spacecraft in orbit are exposed to a variety of SWE phenomena (e.g. UV irradiation, neutral particles, cold and hot plasma, particle radiation, and micro-meteoroids) that can cause significant damage or reduce the operational life of satellites through degradation of electronic components, thermal changes, contamination, excitation, spacecraft glow, charging, etc.[310] For example, it has been assessed that "one solar energetic particle (SEP) event may reduce the power from solar cells permanently by as much as 5%. If the number of SEP events is underestimated by more than two events, then some systems may have to be switched off in order to continue operating towards the end of the working life. This results in a reduced level of operations, reduced profitability, and the possibility of insurance claims".[311]

In order to prevent premature loss and provide sufficient protection against environment conditions, a number of measures can be undertaken, the most important of which are at design level. The traditional solution at this stage has been "to heavily over-engineer the operational satellites to radiation problems". However, because mounting pressures not to over-engineer solutions to keep costs down inevitably increase the susceptibility of spacecraft to damage from SWE effects, it is essential to design spacecraft in an optimal manner by considering both the short-term effects of single events and cumulative effects over the planned lifetime through worst case assessments of the space-environment. Therefore, a characterisation of the different SWE hazards and their possible impacts on satellite's systems is an essential task during the design of a space mission. The hazards and risks to nominal operation that must be considered during the design stage include erosion, leakage, charging, radiation damage, interference, SEU latch-up, and puncture. Most spacecraft modifications for these space environment effects include component selection and testing, subsystem design, shielding requirements, grounding, error detection and correction, and estimates of observation loss.

As reported by ESA, today, "there is an increasingly diverse and sophisticated body of software becoming available to study radiation effects on materials, and such effects tools may be used in conjunction with environment tools to construct a realistic representation of a piece of hardware intended for in-orbit operations. This could be a new technology detector or solar panel, for instance, and furthermore such an object can be encased within and surrounded by other objects and constructs to accurately represent either a single instrument or whole satellite. Such techniques are not restricted to radiation effects, and the effects on material due to interactions with space plasma and electric and magnetic field are readily studied. Having parameterised the orbit using the orbit generator, the hardware mock-up can then effectively 'be flown' for various epochs of the nominal mission, and the effects of the environment upon it quantified and analysed"[312].

These methods have been used extensively by several space agencies for their satellite systems, enabling them to avoid large cost overheads associated with the over-design of operational satellites as they are built. However, it is anticipated that the need for more tailored services for spacecraft design will increase, especially if one considers a series of trends making spacecraft more and more susceptible to SWE effects. Beyond the abovementioned pressure not to over-engineer solution to SWE problems, it must be noted that:

- Spacecraft are today designed for longer operational lifetimes of up to 20 years (or even more if in-orbit services become commercially available)
- There are commercial pressures to increase functionality and capacity, while reducing component mass, particularly of electronic components
- The continuous introduction of new technologies necessitates extensive testing to determine the survivability of the components and the level of tolerated error mitigation through worst-case assessment of the different SWE hazards and their possible impacts on satellite's systems.

Therefore, dedicated SWE services for spacecraft design will be increasingly needed in the future.

Users and Services Characterisation

End users procuring services primarily include spacecraft manufactures and space agencies, and more specifically the personnel involved in producing space environment specifications during the design of spacecraft. Their underlying needs can be summarised as follows:

- Minimise over-design and the associated costs

[310] European Space Agency, 201lk
[311] Horne, 2001
[312] European Space Agency, 2018l

- Define optimal design standards
- Ensure reliable operations
- Ensure maximum operational lifetime

Examples of services that can be provided on the basis of the needs of these users are summarised in Table 17, together with possible actions and benefits accruing from these services.

Service Examples	Possible Actions	Benefits
Leverage data archive to derive statistical information on the space environment (including long-term solar cycle prediction) and its effects on space systems (e.g. dose, single event upset, sensor background, cumulated charge, spacecraft anomalies, micro-particle impacts)	Tailor the design of space systems in relation to radiation protection, EMC and micro-particle impacts.	• Prevention of over design • Cost savings
Provide estimates of the environment's effects actually experienced by a spacecraft through in-orbit measurements of ionising radiation, plasma, micro-particles, atmosphere, UV and local magnetic field variations	Use information for the improvement of models and for in-flight validation of specifications of environments and effects.	• Validation of design standards • More reliable operations
Provide post-event analysis correlating the space environment at a given time and/or location with effects and anomaly events on specific spacecraft, equipment or components.	Identify failures related to SWE effects and determine the vulnerability of components, equipment or spacecraft based on the performed analyses could be used as input for future spacecraft models or versions.	• Improve future design models • Establishment of a new set of design standards • Cost savings

Table 17: Spacecraft design service examples, actions and benefits (source: European Space Agency, 2011)

Spacecraft Operations

The environment in which spacecraft operate can pose a wide range of potential risks to the safety and continuity of their operations, data transfer and service provision. As widely documented by ESA, for instance: "cosmic rays (GCRs) and solar energetic particles (SEPs) are known causes of single event upsets (SEUs) such as latch-ups in onboard electronics systems, often resulting in instruments and potentially platforms automatically going into safe mode. In the worst case, this can result in terminal damage. SEP events can disrupt telecommanding and data telemetry as a result of the interference in data systems, and the data itself is often worthless due to high levels of noise. Trapped radiation in the radiation belts leads to degradation of components as a result of prolonged dose, with processors, detectors and solar cells particularly vulnerable. A satellite passing through energetic charged plasma will experience a range of charging effects, both on the surfaces and internally within electrical systems, and these charge differentials can lead to sudden discharges and subsequent failure of electrical systems. Less energetic plasma also poses problems, with discharge and sputtering often leading to secondary electron emission and subsequent associated charging problems. Even the neutral atmosphere can be hazardous, with neutral atomic oxygen known to lead to surface erosion of the platform materials, potentially compromising the surface and leading to surface charging. A greater risk to surface integrity comes in the form of debris and micro-meteoroids which can compromise and puncture materials due to their high kinetic and potential energies".[313]

In addition, even when satellites remain active and functioning, ionospheric disturbances, such as scintillation, can influence or disrupt signal propagation, and more specifically the transionospheric radio link which bridges communication between ground and space systems. Ionospheric disturbances can impact the functionality of radio systems in different application areas, including satellite communications, earth observation and PNT systems.

According to a study by PwC, the spacecraft anomalies attributable to SWE amount to €642 million, with insurance claims for SWE anomalies totalling over €330 million (PricewaterhouseCoopers, 2016). It is important to highlight, however, that because of its broad array of service applications, the disruption or failure of satellite systems due to severe SWE events, have the most wide impact effect across a number of ground-based operations. This cascading effect subsequently impairs the functioning, in some cases completely, of ground-

[313] European Space Agency, 2018m

based sectors that are reliant on these networks for their operation, including:

- Telecom
- Broadcasting Satellite TV
- Weather forecast
- Road and maritime transportation
- Finance systems

Clearly the implications of interruptions of operations, data transfer and service provision are serious, in terms of both direct and indirect costs and effects. The provision of dedicated services offering forecasts and nowcasts of SWE events (e.g. solar flares, solar and magnetospheric energetic particle fluxes, geomagnetic storms, and total electron content) in combination with their effects on spacecraft operation (e.g. in terms of radio frequency interference, SEUs/latch-ups, radiation damage, charging, and telemetry signal propagation) is therefore quintessential to mitigate against all the potential risks involved in spacecraft operations.

Users and Services Characterisation

The primary entities making use of dedicated SWE services include flight control teams, operations support engineers, and science operations centre teams of national space agencies as well as public and private spacecraft operators. Their basic needs can be summarised as follows:

- Ensure the reliability of service provision
- Prevent the loss of data
- Be on alert ready to deal with problems
- Identify causes of satellite failures

Examples of services that can be provided on the basis of the needs of these users are summarised in Table 18, together with possible actions and benefits accruing from these services.

Services Examples	Possible Actions	Benefits
Provide near real-time estimate of the space environment (e.g. increased atmospheric drag) and its effects on spacecraft	Postpone or anticipate orbit manoeuvres	Better station keeping Save fuels and extend mission lifetime
Provide mission risk analysis based on forecast of the space environment conditions and mission susceptibility assessment	Switch off non-essential systems if the risk threshold is crossed	Protect satellite systems
Provide forecast of the occurrence risk of ionospheric disturbances (e.g. scintillations)	Re-route communications via other satellites and ground network	Prevent loss of data Ensure the availability and continuity for service provider

Table 18: Spacecraft operation service examples, actions and benefits (source: European Space Agency, 2011)

Human Spaceflight

Humans in orbit are constantly exposed to a number of hazards threatening their health and safety. Space radiation is a major health concern to astronauts. Even in LEO, where the effects of the Earth's magnetic shielding are still strong, astronauts are exposed to space radiation such as solar energetic particles (SEPs) and galactic cosmic rays (GCRs). These ionising radiations can generate both short-term physiological effects such as the 'flashes of light' experiences by the astronauts of the Apollo missions in the 1960s and 1970s, and long-term biological effects such as damage to the DNA and cell replication. ESA has estimated that "in just one week on the ISS, astronauts are exposed to the equivalent of one year's usual exposure at Earth's surface". Clearly, should major SEP events occur, the risk of higher exposure would swiftly increase. For instance, during the solar flare responsible for the SEPs and the large geomagnetic storm of 18 October 1989, "the astronauts and cosmonauts onboard the Mir space station received their full-year recommended dose within a few hours".[314]

With longer and more distant manned missions currently under the scrutiny of the international space community, astronauts will face even more risks, as they will completely lose the protection of Earth's magnetic field and will be fully exposed to SEPs and GCRs. To illustrate, "during the Apollo era, the risk of exposure was limited in time (less than 12 days), but for a Mars exploration it will be much

[314] European Space Agency, 2018n

longer (about 18 months) and solutions have to be found for the case when a life-threatening event will occur".[315]

Against these backdrops, dedicated services to circumvent these limiting factors for manned space missions and to ensure the health and safety of astronauts at all times are of paramount importance. There is now a requirement to develop models using real time data to predict the intensity and location of solar energetic particle events, and to study the longer-term effects of space radiation, so as to take appropriate preventive action.

Users and Services Characterisation

The primary entities making use of dedicated SWE services include the operation teams of human spaceflight during launch operations and activities inside and outside of the ISS, but also the flight operators for future space tourism and future human missions in outer space. Therefore, both public space agencies and private entities operating orbital or sub-orbital flights for space tourists (e.g. Virgin Galactic) can procure such services. Their requirements can be summarised as:

- Be aware and prepared to deal with potential SEPs and GCRs events
- Minimise radiation exposure threats to astronauts' health
- Comply with health and safety regulations
- Monitor astronauts' health

Examples of services that are or can be delivered for human spaceflight activities as per their needs are briefly presented in Table 19, together with possible actions and benefits accruing from these services.

Service Examples	Ensuing Actions	Benefits
Provide forecast of the risk of increased level of radiation along trajectory.	Delay human spaceflight launch Re-schedule EVA activities	Minimise risks for astronauts
Provide near real-time estimate of the radiation dose received by a person in space. Put staff and astronauts on alert in case of SEP events	End EVA activities or use protective shields	Reduce radiation exposure to astronauts Comply with health and safety regulations
Provide estimate of the past radiation dose accumulated by a person in space.	Use data to monitor the accumulated radiation dose	Monitor and assist astronauts' health

Table 19: Launch operations service examples, actions and benefits (source: European Space Agency, 2011)

Launch Operations

SWE events can exercise a considerable impact on launch operations. Just like terrestrial weather, adverse space environment conditions can force postponement of a launch, thereby causing important financial and logistical burdens. For instance, a launch delay can impair the ability of commercial satellite operators to ensure the continuity of services to their customers or compromise the one-off launch window available for an interplanetary mission. In addition to delaying a rocket launch, adverse space environment conditions can seriously endanger critical operations such as separation and orbit-insertion as well as "disrupt real-time communications between the ground and space segments".[316]

The major SWE hazard faced during the launch procedure is high energy radiation. "Energetic solar ions and protons pose a significant single event upset (SEU) threat to sensitive and complex electronics systems. The risk is highest during solar energetic particle (SEP) events, so for a given launch there must be a clearly-defined threshold beyond which the launch may not be considered". Dedicated tools and services must be in place to provide nowcast and forecast of SEP events, together with the likely energetic radiation profile for a given launch trajectory and an assessment of the potential effects on both launcher and payload must be made. Other potential tools for supporting launch operations against possible SWE hazards include risk assessments "from microparticle impacts and ionospheric effects on communications systems" as well as estimates "on atmospheric drag which is affected by current Space Weather conditions".[317]

Users and Services Characterisation

[315] European Space Agency, 2018n
[316] European Space Agency, 2018o
[317] European Space Agency, 2018o

End users making use of dedicated services in this domain primarily include the entities in charge of operating launch vehicles, and more specifically the personnel involved in launch operations. In the European context, this applies to the various stakeholders at the Guiana Space Centre: i.e. CNES, the launch site operator, ESA, the infrastructure operator, and Arianespace, the launch service operator, and also insurance companies. The requirements of launch operators can be summarised as:

- Schedule launch procedures in an optimal manner
- Determine the go/no-go threshold with higher precision
- Prevent hazards associated with SEP events
- Be aware of potential risks

Building on these needs, a non-exhaustive list of services that are or can be delivered to launch operators is presented in Table 20, together with the ensuing actions and benefits accruing from the provision of the service.

Service descriptions	Possible Actions	Benefits
Provide estimate of the risk of SEP events along the launcher trajectory.	Schedule launch operation	Higher confidence in SEE risk Go/no-go threshold
Risk estimate of service disruption caused by ionospheric scintillations between ground station and launch vehicle along the trajectory.	Anticipate possible disruptions in communication	
Forecast of the atmospheric drag for fairing ejection	Modify launch sequence	Optimisation of launch procedures
Provide estimate of the risk of impacts by micro-particles (objects with sizes below 1 mm)	Modify launch sequence	Awareness of potential impact risks
Provide near real-time estimate of the radiation effects on sensitive electronics along the launcher's trajectory.	In-flight monitoring of the functioning of the launcher's electronics.	Higher confidence of radiation effects
Provide estimate of past radiation effects on sensitive electronics along trajectory.	Retrieve information to analyse flight data	Assess resilience of the launcher's electronics Access to additional data

Table 20: Launch operations service examples, actions and benefits (source: European Space Agency, 2011)

Space Surveillance and Tracking

In addition to naturally occurring objects occupying Geospace, the Space Age has generated a vast number of man-made artefacts, travelling within Earth's orbit, that bring about potential hazards and risks. These risks include threats in space, i.e. collisions and resulting adverse effects on space systems, as well as ground threats, i.e. uncontrolled and high-risk re-entry. Whilst the object count in Earth's orbit is in its millions, only around 13,000 (>10 cm in size) are regularly tracked – tracked items are mainly launcher bodies and mission-related objects, with just 10% being active satellites[318]. Tracking objects involves the detection of objects and determining the orbit state and levels of uncertainty. SWE can impact on these measurements, for example an objects trajectory can be influenced by the density of the thermosphere, which itself changes dependent on altitude, latitude and longitude[319]. As such, predicting an object's trajectory through varying densities requires 3D modelling and forecasting of the thermosphere environment.[320]

Users and Services Characterisation

Producing such models and predictions necessitates reliable real-time data for nowcasts and forecasts, as well as data from archives. End-

[318] European Space Agency, 2018p
[319] European Space Agency, 2018p
[320] European Space Agency, 2018p

users utilising these data products and services include surveillance and tracking centre(s), stations and services, spacecraft operators, collision warning services, and re-entry risk assessment services. Their main requirements include:

- Adequate modelling of the geospace environment and factors that can influence object trajectory
- Improved precision of SST methods
- Prevention of damage to space systems
- Increased awareness and understanding of re-entry events.

Reflecting on these requirements, Table 21 presents a non-exhaustive list of services, possible actions that can be taken and benefits, that can be delivered to improve overall SST operation:

Service Example	Possible Actions	Benefits
Atmospheric estimates for drag calculations	The creation of models of object trajectory including drag effect	• More accurate object trajectory forecasting
Archive of geomagnetic and solar indices for drag calculation	Allows the user to input data on solar and geomagnetic indices into their own in-house models	• More accurate object trajectory forecasting
Forecast of geomagnetic and solar indices for drag calculation	Predict risk of losing track of objects and place staff on alert	• Management of and preparedness for loss of track events
Nowcast of ionospheric group delay	Estimation of ionospheric refraction of radio waves can be used to correct positions derived from radar tracking	• Increased tracking precision

Table 21: SST service examples, actions and benefits (source: European Space Agency, 2011)

In the future, it can be expected that SWE nowcasts and forecasts in this domain will also need to evolve to cope with the emerging requirements of an operational Space Traffic Management (STM) framework, rather than be limited to the provision of products for space surveillance.

Non-Space Operations

Aviation

The aviation industry is one of the many sectors that can be affected by disruptions in space-based GNSS services (discussed in 4.2 above). Beyond this, SWE alerts and warnings are being increasingly utilised and integrated into the operations of commercial airlines as the popularity of transpolar routes grows – these routes "offer the advantage of avoiding prevailing head winds as well as being on the great circle route for many destinations" thus reducing flight times, increasing fuel efficiency, and potentially reducing environmental impact.[321] Whilst there are clear economic and environmental benefits – i.e. reducing fuel usage and costs – from using transpolar routes, travelling through these areas takes aircraft to latitudes in which satellite communication cannot be used, thus becoming reliant instead on high-frequency radio communication for operations.[322] Dependency on radio communication when using transpolar routes can be disrupted by severe SWE events, i.e. during solar radiation storms, SEPs, generally protons accelerated by CMEs, travel down geomagnetic field lines to the polar ionosphere, increasing ionised gas density, which can interfere with radio wave propagation and cause radio blackouts.[323]

These events can sometimes last several days, in which case the diversion of aircraft routes to altitudes in which satellite communications can be used is necessary – as was the case in January 2005 when 26 aircraft of United Airlines had to be redirected to non-polar, or less viable polar routes to avoid radio blackout, causing increased flight time, extra landings and take-offs, increased fuel usage and costs, and delays to other flights.[324] In addition to this, solar particle and high-energy cosmic rays can produce a number of other high-energy particles through their interactions with the Earth's upper atmosphere. Due to proximity, the flux of ionising radiation for typical aircraft cruising altitudes being ~300 times that of sea level, exposure to such SWE radiation

[321] World Meteorological Organisation, 2008:6
[322] National Research Council, 2008:6
[323] National Research Council, 2008:7
[324] National Research Council, 2008:8

puts the health of aircraft passengers and crew safety at particular risk.[325] Increased integration of SWE services, such as warnings and alerts, in the commercial airline industry, similar to volcanic ash warnings, can potentially "improve efficiency of the end-to-end warning process through relying on established procedures and communication tools… and contribute to aeronautical safety".[326]

Users and Services Characterisation

SWE services for the aviation sector come in the form of forecasts, nowcasts and alerts for radiation events as well as disruptions to GNSS and communication systems, mostly in high latitude zones. These will be used primarily by airline operators for the scheduling, or rescheduling, of flight times and routes.

The primary entities of interest for SWE services in the aviation sector include regulatory bodies, operators, scientists and experts, and insurance companies. The service needs for the aviation sector can be characterised as follows:

- Optimisation of flight routes, flight scheduling and fuel efficiency
- Ensuring the health and safety of crew and passengers from radiation exposure
- Improved vehicle design
- Compliance with regulations and international standards

Several service examples for the aviation sector that build on these needs, including potential actions that can be taken, and benefits from service use are presented in Table 22.

Service Example	Possible Actions	Benefits
Cosmic ray dose forecasts & Radiation storm forecasts (5,12, 18 hours in advance) and warnings	Allows for appropriate mitigation procedures in the short-term, such as crew change and/or and flight path rerouting or rescheduling.	• Increased crew and passenger safety • Compliance with exposure limits • Computation of crew exposure • Optimisation of flight path and fuel efficiency
Post event information on radiation levels on a series of predefined routes used by commercial airlines (<1 week delay if significant activity).	Developing knowledge of flight paths for longer-term planning and route selection.	• Increases the ability of airline coordinators to plan flight-paths in the long term • Avoid last minute delays, rescheduling or rerouting and associated costs
A graphical forecast including intensity, onset, duration and boundary of degraded communications for polar routes (12-24 hours) in accordance with international standards	This information can be used by airlines to select optimal routes and develop necessary emergency response procedures.	• Improved route selection, management and emergency response • Compliance with international standards
Statistical information on the radiation environment at aircraft altitude for avionics.	Use input to improve avionics design for aircraft	Improved aircraft design
Radiation and ionospheric data for post-event analyses for aircraft operators	Support anomaly resolution and dose reconstruction in case of observed in-flight avionics errors.	Improved aircraft design and safety

Table 221: Aviation service examples, actions and benefits (source: European Space Agency, 2011)

Railways

Railway operations, as both long distance conductive networks and systems dependent on power, GNSS and radio communications, are susceptible to severe SWE events through direct impacts (e.g. GIC effects on rail tracks and transformers) and indirect impacts

[325] Royal Academy of Engineering, 2013:38

[326] World Meteorological Organisation, 2008:7

through their dependencies on other critical infrastructures used for operations (e.g. power, communication and navigation).[327] While cases of SWE impacts on rail operation have been well documented in Sweden and Russia, the railway sector lags behind other industries, such as aviation, in terms of awareness and preparedness to protect against these effects.[328] In line with this lack of certainty, a study conducted in the UK by Atkins, RAL and The University of York identified some of the key areas of impact that a severe SWE event could have on the rail sector[329]. In regard to power, as noted above, if GICs disrupt the power grids, normal rail operations would most definitely be affected. In terms of internal rail infrastructure, the failure of transformers – which distribute power up and down the lines – will cause similar effects; as well as potentially disrupting services at stations, e.g. lighting, lifts, ticket barriers and passenger information screens, posing additional threats to safety[330]. In terms of loss of GNSS services during GIC events, timing and synchronisation issues may occur for rail operations. These can impact the capabilities of positioning, rail condition monitoring and maintenance, telecommunications, power control and supplier spares tracking, essentially reducing or completely halting the operator's ability to command and control the rail network.[331] In addition to closing knowledge gaps on direct and indirect SWE impacts on rail operation, including re-analysis of historical data on potential SWE rail impacts, Atkins recommends "setting up of systems to notify track-side staff of space-weather related dangers to allow the implementation of appropriate safety measures".[332] The UK Association of Train Operators and Network Rail has also expressed an interest in improving the strategy of responding to SWE alerts and warnings, iterating the role of SWE services in protecting and mitigating SWE related system failure of the rail networks.[333]

Users and Services Characterisation

Railway operators will be the primary users of SWE services in this sector through forecasts and nowcasts of GICs that can affect the rail tracks. Second to this, train drivers and on board/station crew members will be able to develop procedures to react to warnings and alerts that may affect rail services.

The chief stakeholders in the rail sector to whom SWE will be of interest include railway operators, rail service staff, regulators, and scientists/experts. The main requirements for services within the railway domain include:

- Maintaining ability to command and control the network
- Safety for railway users and staff
- Post-event analysis of the network
- Identification of risks

Several examples of services that can be provided on the basis of the needs of these users in the railway sector are summarised in Table 23, together with possible actions and benefits accruing from these services.

Service Example	Possible Actions	Benefits
Providing forecasts or real-time warnings on geomagnetic induced current (GIC) events to railway operators	Notifications to railway staff members and operators during GIC events so that safety measures can be implemented	• Mitigating loss of signal • Reducing cost of delays • Improving rail operator's ability to command and control • Improving rail safety and quality of service for users
Post-event analysis of GIC effects on railways and transistors	Evaluation of risks to disruption to signalling and train control and devising engineering solutions	• Improve the resilience of railways to GIC events • Improved knowledge of GIC impacts on railways
GNNS disruption alerts		

Table 23: Railway service examples, potential actions and benefits (source: European Space Agency, 2011)

[327] Krausmann et al., 2016:8-9
[328] Krausmann et al., 2016:8-9
[329] McCormack, 2017
[330] McCormack, 2017
[331] Krausmann et al., 2015:6
[332] Krausmann et al., 2015:4
[333] Krausmann et al., 2015:4

Resource Exploitation

The geophysical resource exploration and extraction sector is heavily dependent on the precise positioning provided GPS/GNSS systems for the conduct of geophysical surveys and directional drilling operations. For geophysical surveying, it is commonplace to survey "geomagnetic field intensity, and to express the difference between the observed value and a notional value of the core-generated field as the "magnetic anomaly" field, which may aid in interpretation of subterranean structure and composition.[334] In the case of guided or directional drilling, precise GPS/GNSS measurements "are used to determine the orientation of the drill string and therefore to guide the direction of drilling".[335] SWE can impact both these dimensions of resource exploitation through the interference experienced during geomagnetic storm events that disturb the magnetic field, skewing survey data and reducing accuracy. Thus large businesses in this sector, e.g. BP and Shell, "seek information on near-time geomagnetic conditions so they can schedule surveys during quiet periods," typically avoiding surveys in such conditions as the results may be worthless.[336]

Users and Services Characterisation

In the resource exploitation industry, geophysical surveyors (primarily aeromagnetic surveys, but also ground-based), and directional drill operators, can make use of SWE services that provide forecasts for, and real-time data on, variations and disturbances in magnetic conditions. In the case of surveying (particularly aeromagnetic surveys[337]), there are notable cost implications for having to repeat geophysical surveys, creating a need for reliable geomagnetic forecast services for the use of planning survey operations.[338] In terms of drilling, there is economic balancing from weighing up the "cost of stopping drilling operations (costing many hundreds of thousands of dollars per day) against the costs that might arise from errors in the path of the drill string, particularly the risk of intersecting other well paths, which can lead to blow-outs"[339], whilst forecasting is still desirable in this case, there is more significance placed on the user need of reliable real-time monitoring services (now-casting).[340]

The primary parties interested in SWE services for resource exploitation include geophysical surveyors, drilling operators and extractors. Their central requirements include:

- Avoiding useless data and repeat surveys
- Enabling the continuity of drilling operations during SWE events

Based on the users and needs outlined here, Table 24 provides examples of services within the resource exploitation sector, potential actions to be taken, and the benefits of such services:

Service	Possible Action	Benefits
Nowcast and forecast (0-6hr, 24-48hr) of local geomagnetic activity for aerial survey operations	Suspending surveys during periods of strong geomagnetic activity, and rescheduling flights	• Prevents accumulating useless data • Reduced costs associated with repeating an aerial survey
Predictions of variations in the Earth's magnetic field and real-time magnetometer data for drilling operations	Enables drilling operators to apply magnetic corrections more accurately, and more frequently if necessary.	• Allows for the continuation of drilling operations with accuracy • Reduces costs associated with ceasing operations for any length of time

Table 24: Resource exploitation service examples, potential actions and benefits (source: European Space Agency, 2011)

[334] Clark, 2001:2
[335] Hapgood, 2010:19
[336] Hapgood, 2010:19
[337] European Space Agency, 2018q
[338] Clark, 2001:1
[339] Hapgood, 2010:19
[340] Clark, 2001:2

Power Grids

Rapid variations of the geomagnetic field, caused during geomagnetic storms and resultant ionospheric currents associated with the aurora, produce an electric field on the Earth's surface that in turn induces unwanted currents through electrical power grids and other grounded conductors.[341] Consequentially, these currents can affect the stability of power transmission networks and cause transformers to burn out. This happened during the geomagnetic storm of March 1989, which led to the Hydro-Quebec grid failure – the storm related geomagnetically induced currents (GICs) could not be mitigated by Hydro-Quebec's automatic voltage compensation equipment.[342] Electrical grid operators can mitigate or prevent damage from GICs by optimising design, introducing protective equipment, "or by redistributing and changing the power-generation resources so that fewer long-distance transfers are needed and more near-locally-generated power is available to counter frequency and voltage modulations." In contexts such as the UK, with comparatively compact grid systems, "bringing all available grid links into operation in order to maximize redundancy and to spread GICs over the whole system" can reduce the overall impact on individual elements".[343]

Users and Services Characterisation

For power grid operators to improve system protection, this industry sector has stressed the importance of and need for reliable forecasts for GICs, including their magnitude and duration, with at least half a day warning. Propagation of CMEs from the Sun to Earth typically take from 2-4 days (the fastest known events taking around 0.75 days), hence forecasts are necessary, and observationally possible, after the solar eruption has occurred, but before they reach the near-Earth environment.[344] The provision of forecasts and nowcasts for such GIC events can be compared to current meteorological support "provided to energy suppliers for the optimal exploitation and sustainability of their network, which encompasses multiple aspects such as medium-range and short-range temperature forecasting for anticipating heating user demand, nowcasting for lighting demand over large cities, and early warning of stormy and icing conditions for maintenance pre-alerts. It is anticipated that integration of Space Weather warnings could serve the overall efficiency of the end-to-end warning process and thus be beneficial to operators and end-users".[345] GIC events can ultimately cause damage to transformers, and increased generator capacity can be necessary to maintain power supply during such events, necessitating forecasts and real-time mapping of GIC impacts on the grid network. Additionally, improved knowledge of impacts, through analysis of historical and future events, can also improve this sector's resilience.

The major users of SWE services in the power grid sector include power grid operators, scientists and experts, insurance companies, and service providers. Services requirements within the domain of power grid operation include:

- Reduced damage to the network and transformers
- Identification of system status and risks
- Continuity of service

Table 25 outlines a few service examples for power grid operation, detailing possible actions and potential benefits from service provision:

Service Examples	Possible Actions	Benefits
A tailored service for generating Network maps showing geomagnetically induced currents throughout the power system including plotting local E-field and GIC by substation for registered users.	A networked map of GICs through the grid system can allow operators to carry out inspections of affected areas or promptly increase generator capacity if necessary	• Reduced downtime • Continuity of service • Reduce risk of transformer damage
A tailored service for specific users providing a table of modelled GIC values for the users	Recent, near real-time GIC data products can be used for anomaly identification and resolution	• Identification of system behaviour and risks • Continuity of service

[341] National Research Council, 2009:4; Royal Academy of Engineering, 2013:22
[342] National Research Council, 2009:4
[343] Schrijver et al., 2015:2755-2756
[344] Schrijver et al., 2015:2755-2756
[345] World Meteorological Organisation, 2008:7

network in the last minute and peak GIC in the last 60 minutes		
A global forecast of geomagnetic activity from 15 minutes up to 27 days ahead.	Advanced warning of GICs can allow grid operators to increase power generating capacity	• Maintain power supply during a geomagnetic storm

Table 25: Power grid service examples, potential actions and benefits (source: European Space Agency, 2011)

Pipeline Operations

Similar to electrical power grids and railways, wide ranging conductive networks such as long-distance oil and gas pipelines can be affected by GICs. GICs resulting from space weather, i.e. geomagnetic storms, interfere with the "cathodic protection systems" used by pipeline operators to reduce the corrosion rates on pipelines by applying "an electrical voltage opposite to that generated by the chemical processes that cause corrosion and thereby slow the corrosion rate," essentially hampering the protective effectiveness and reducing the longevity of a pipeline.[346] Negative impacts on pipelines running through high latitude areas, such as pipelines through Alaska and Finland[347] that are within the auroral zone, are commonly observed due to aurora associated electrical currents. Additional studies have also highlighted these effects occurring at mid- and low-latitudes[348]

Users and Services Characterisation

The main stakeholders with interests in SWE services within this domain include pipeline operators, pipeline developers, and scientists and experts. The key service requirements for pipeline operation entail:

- Identification of events and risks associated with pipeline corrosion
- Management of pipeline inspections and replacement
- Post-event analysis of the network for improved pipeline design and management

Examples of services that are or can be delivered for pipeline operations according to the user needs outlined are briefly presented in Table 26 together with possible actions and benefits accruing from these services.

Service Example	Potential Actions	Benefits
Global forecast of geomagnetic activity from 15 minutes up to 27 days ahead.	Allows pipeline companies to suspend routine maintenance and measurements of the cathodic protection during GIC events, and coordinate appropriate timings for inspections and replacements	• Improved management of corrosion risks • Planning of pipeline inspection and replacement
A tailored service for specific users providing pipe-to-soil potential difference (PSP) variations in the user's pipe network.	Monitoring and evaluation GIC impacts on the cathodic protection system on long-distance pipelines can allow for pipeline assessment and developing pipeline design	• Increased understanding of GIC impacts on the cathodic protection system • Improving pipeline design and resilience to reduce pipeline corrosion

Table 26: Pipeline operation service examples, potential actions and benefits (source: European Space Agency, 2011)

Auroral Tourism Sector

Auroras are a natural product of space weather events. Solar activity, such as coronal holes, flares and CMEs, causes geomagnetic disturbances in the near-Earth environment, which can cause the auroral lighting effects seen primarily at high latitudes.[349] There is an existing market within the tourism sector to see such auroral events, which provides po-

[346] Hapgood, 2010:18; European Space Agency, 2011:71
[347] Pulkkinen et al., 2001
[348] Marshall et al., 2010; Hapgood, 2010:19
[349] European Space Agency, 2018q

tential for reliable forecasts – the further in advance the better – to further develop the tourist market through provision to tourism companies and commercial airlines.[350]

Users and Services Characterisation

Users of SWE services for auroral tourism include tourist companies, airline companies and tourist themselves. The key service requirements for this domain involve:

- Prediction of conditions leading to optical aurora events,
- Early notification of aurora events

A service example that could be delivered for the auroral tourism sector is presented in Table 27, together with possible actions and benefits accruing from these services. Notably, there is a widely available mobile subscription service (Night Sky Alerts[351]) for auroral alerts.

Service Example	Actions	Benefits
Forecast of the probability of visible auroras (>12hours, >6hours).	Forecasts could be communicated between tourism companies, airlines and tourists to allow for appropriate advertisements and bookings to view aurora events.	• Targeted marketing • Develop the aurora tourism market

Table 27: Auroral tourism service example, potential actions and benefits (source: European Space Agency, 2011)

A.4 Selected National SWE Weather Activities in Europe

Austria

Within Austria, the international need for SWE services is recognised and implemented through the collaboration of a number of national institutions. The Institute of Physics at the University of Graz (ESA's Expert Service Centre for Heliospheric Weather) acts as a mediator and coordinator between different groups, addressing both SWE research and services. The institute of Physics itself provides real-time solar wind forecasting as well as CME forecasting tools, whilst its Kanzelhöhe Observatory (ESA's Expert Service Center for Solar Weather) provides real-time alerts and warnings for solar flare events.[352] The Conrad Observatory, in collaboration with the Seibersdorf Laboratories, deals with ground-induced currents and local magnetic field variations which consequence from solar activities, additionally part of ZAMG, the national weather services. The Seibersdorf Laboratories also serves as the ESA Expert Service Center for Space Radiation, providing real-time estimations of dose rates for aircrafts at different altitudes and dose rates at any desired location.[353] The Space Research Institute[354] in Graz is additionally active in dedicated SWE satellite missions.

Belgium

The Solar Dynamics Observatory (SDO) at The Royal Observatory of Belgium (ROB)[355] is host to a "rolling archive of the latest 6 months of data of the full AIA data, HMI magnetogram and HMI intensitygram, as well as a long-duration, low cadence data set, and a subset of the most frequently required events."[356] As part of this, the ROB facilitates access for external users to retrieve SDO data, accessible online via web-interfaces.[357] The Solar Influences Data Analysis Center (SIDC)[358] is part of ROB and a partner in the Solar Terrestrial Center of Excellence (STCE).[359] It is the mission of

[350] Horne, 2001:36-37
[351] Night Sky Alerts, 2018
[352] Temmer, 2018
[353] Ibid.
[354] Graz Institute for Space Research, 2018
[355] Royal Observatory of Belgium, 2018
[356] Committee on Space Research, 2018:57
[357] Committee on Space Research, 2018:57
[358] Solar Influences Data analysis Center, 2018
[359] Solar-Terrestrial Centre of Excellence, 2018

SIDC to "advance knowledge on the Sun and its influence on the solar system, through research and observations" additionally providing knowledge and expertise to the scientific community, government, industry and wider society, the provision of operational services and their dissemination at national and international levels.[360] The Novel EIT wave Machine Observing (NEMO – EIT wave detector) has also been developed and hosted at ROB, detecting EUV waves and coronal dimmings associated with CMES. Other institutions in Belgium that are active in SWE research and services include the Belgian Institute for Space Aeronomy (BIRA-IASB),[361] the Meteorological Institute (RMI)[362] and the Katholieke Universiteit Leuven.[363]

ESA's SSA Space Weather Coordination Centre (SSCC) is also located in Belgium at the Space Pole, marking the first European Space Weather Helpdesk that provides operators and experts to answer enquires regarding SWE conditions or the SWE precursor service network.[364]

Finland

At a national level Finland has several institutions and public agencies that focus on SWE. Notably here, the Finnish Meteorological Institute (FMI) has a multi-pronged approach to SWE, focusing on SWE research, operational services, and SWE customers.[365] In 2014, FMI created the national 24/7 SWE service in Finland, integrated into FMI's existing monitoring system for natural disasters.[366] FMI's Earth Observation unit, in collaboration with the Weather and Safety Centre and the Artic Research unit are responsible for the monitoring and forecasting of potentially dangerous SWE events, providing alerts and warnings on a 24/7 basis.[367] The FMI itself published 329 peer-reviewed papers in 2013 alone, making it first in the world in terms of publishing productivity. A large reason for this is historical, with the FMI having been established as early as 1938 when it began collecting ground magnetic records.[368]

Finland is also home to Europe's only space weather simulation. Beginning in 1993 on the ionosphere, with the magnetosphere added in 1996, the Grand Unified Magnetosphere-Ionosphere Coupling Simulation is coordinated through FMI and "models the dynamic effects of changing solar wind conditions in the near-Earth space. It is a global, three-dimensional magnetohydrodynamic simulation model of the Earth's magnetosphere, and it includes an electrostatic ionosphere model".[369] As mentioned, the FMI also hosts the Centre for Natural Disasters (LUOVA), which provides an early-warning system, conducts 24/7 monitoring, real-time risk assessments and forecasts, with regards to SWE events as well as terrestrial based causes of natural disasters.[370]

Funded by the Academy of Finland, the Research on Solar Long-term Variability and Effects (ReSoLVE) Center of Excellence was established in 2014 – a collaboration of five research teams from the University of Oulu and Aalto University, "focused on studying the long-term solar variability and is effects in near-Earth space, atmosphere and climate".[371] Other Finnish SWE projects include SOLE, focused on solar storms and their frequency), and SAFIR's Extreme weather and nuclear power plants (EXWE)[372] project, studying the impact of solar storms on nuclear safety.[373]

France

A large proportion of SWE activities in France are conducted by the French Space Agency (CNES) and the French Air Force (CDAOA/COSMOS). Under the coordination and management of CNES, France has also established a national group of over 30 experts from multiple institutions, i.e. universities and observatories, and public agencies e.g. CNES, Météo France, and the Ministry of Defence.[374] This expert group is tasked with providing an assessment of potential impacts on four key domains, including: defence, space, civil aviation, terrestrial technological infrastructure, and the promotion of sharing of SWE post-event analysis.[375] Other nationally led SWE activities include: a SWE system for the French Air force for military purposes; and ONERA, supported by CNES, which develops advanced SWE applications in the field of space radiation i.e. real-time nowcasting of the Earth's trapped particle environment using data assimilation techniques that allow the optimal

[360] Solar-Terrestrial Centre of Excellence, 2018
[361] Belgian Institute for Space Aeronomy, 2018
[362] Royal Meteorological Institute, 2018a
[363] Committee on Space Research, 2018
[364] Royal Meteorological Institute, 2018b
[365] Palmroth et al., 2014
[366] World Meteorological Organisation, 2018b
[367] European Commission, 2017b:65
[368] Palmroth et al., 2014
[369] Finnish Meteorological Institute, 2014
[370] Finnish Meteorological Institute, 2014
[371] Research on SOlar Long-term Variability and Effects, 2018
[372] The Finnish Research Programme on Nuclear Power Plant Safety, 2018
[373] European Commission, 2017b:66
[374] United Nations Committee on the Peaceful Uses of Outer Space, 2017c
[375] Ibid.

combination of in-situ measurements and physics-based models. Whilst there may be no truly operational SWE services in France, it does indeed possess many SWE assets, including two thematic poles – the Multi-Experiment Data and Operations Centre (MEDOC) and the Centre de Données de la Physique des Plasmas (CDDP) – as well as several serval data archives, e.g. the Neutron Monitor Data Base (NMDB), models, and individual developing services, e.g. Radiation belt models for the Earth's environment (CRATERRE).[376]

Germany

In Germany SWE research is conducted at several research institutes and universities, including the German Aerospace Center (DLR), the German Research Centre for Geosciences (GFZ), the Leibniz-Institut für Astrophysik Potsdam (AIP), the Max-Planck Institute for Solar System Research (MPS), the University of Goettingen, and the Christian-Albrechts-University (CAU) of Kiel.

As the national space agency, DLR has been supporting SWE-related missions (e.g. STEREO), and international initiatives such as the worldwide, near real-time network named GIFDS (Global Ionospheric Flare Detection System), and the educational project named SOFIE (SOlar Flares detected by Ionospheric Effects).[377] DLR also supports the activities of the German Space Situational Awareness Centre (GSSAC) and operates the Ionosphere Monitoring and Prediction Center (IMPC).

The IMPC offers a near real-time information and data service on the current state of the ionosphere as well as related forecasts and warnings of ionospheric disturbances. IMPC products include Total Electron Content (TEC) maps, scintillation indices and Rate of Change of TEC index, among others. IMPC products are disseminated via their website and users can subscribe to receive warning messages via e-mail. IMPC also conducts research activity in ionospheric science, including ionospheric perturbation detection, modelling and forecasting, empirical and physical modelling and 3D electron density reconstructions.[378]

Besides IMPC, the GSSAC, hosted by the German Air Force Operations Centre, provides SWE products and alerts for the support of national decision-making, the protection of German population, the support of German armed forces in theatre, and the protection of space infrastructure.[379]

Italy

SWE activities in Italy are conducted by several universities and research bodies such as INGV (National Institute for Geophysics and Volcanology) and INAF (National Institute of Astrophysics), as well as the Italian Space Agency (ASI) and Italian Air Force.[380] In 2014, an interest group (Space Weather Italian Community – SWICo) formed by scientists of universities, national research institutions and representatives of Italian Industries was established to better exploit and develop national expertise in observational, theoretical studies and modelling, as well as application in industrial sectors.[381]

In terms of service provision, INGV developed and manages a real time Ionospheric Weather Service able to provide now-casts and short-term forecasts on Ionospheric TEC maps over Italy and Europe; vertical sounding parameters and electron density profile over Rome and Gibilmanna (Sicily); ionospheric scintillation and TEC in mid and high European latitudes; geomagnetic indices (DST and Kp from Kyoto WDC and NOAA spwc); and solar activity (from NASA SDO).[382] INGV and INAF also provide real-time data and indices on the geomagnetic field and are developing forecasting services for GICs events. Additional service prototypes (SWERTO, FLARECAST, IPS) have been developed by Italian institutions within the EU's FP7 and H2020 framework.

Sweden

SWE activities in Sweden are mainly entrusted to the Swedish Space Weather Center (SRC), which is part of the Swedish Institute of Space Physics, and to the Swedish Civil Contingencies Agency (MSB), which is responsible for preventing and preparing for emergencies stemming from SWE events in collaboration with public and private stakeholders. MSB's work includes monitoring and defining contingency plans for the types of SWE events that have a low occurrence probability but could have major repercussions in space and on the ground.

In terms of SWE services, the SRC, which is also one of the 14 regional warning centres of ISES, provides nowcasts and postcasts on the effects of solar activity, including radiation hazards to astronauts, satellite problems, and navigation system problems stemming from proton storms as well as low frequency and HF communication problems stemming from radio

[376] Dudok de Wit, 2015
[377] Wenzel et al., 2017
[378] Ionosphere Monitoring and Prediction Center, 2018
[379] Braun, 2018
[380] Agenzia Spaziale Italiana, 2018
[381] Space Weather Italian Community, 2018
[382] Romano, 2017

blackouts.[383] Additionally, the SRC provides forecast services on the effects of solar wind and Coronal Mass Ejections (CMEs), including "geomagnetic storms (Kp >= 5 or Dst < -50 nT) and problems for electrical systems such as power grid systems and gas pipelines (forecasts of 3-hourly Kp-index and one-hourly Dst-index) and geomagnetically induced currents for one station in Southern Sweden: (30-minute forecasts of GIC)".[384]

UK

Within the UK, the Cabinet Office cooperates on the national SWE strategy with local first responders - the National Grid, rail networks, and the aviation sector. Beyond this, the UK Met Office has also set up the Met office Space Weather Operations Centre (MOSWOC), which is an "operational manned 24/7 forecaster" that, besides conducting observations and data analysis, provides general and tailored services, i.e. forecasts and alerts on solar activity, solar wind/geomagnetic activity, solar radiation etc., to user groups, i.e. UK critical infrastructure sectors and operators, the government and military so that they can prepare for impacts.[385] Similar to NOAA's SWPC, MOSWOC provides tailored services to different user groups, e.g. texts and graphics, which are customised information services addressing the needs of specific sectors.[386]

Other active public agency research programmes include the Natural Environmental Research Council's (NERC) British Geological Survey (BGS) which examines daily solar activity and provides forecasts and alters on potential geomagnetic storms with likely terrestrial impacts, available on the BGS website;[387] and the Science and Technology Facilities Council (STFC) and its Rutherford Appleton Laboratory (RAL) which funds solar-terrestrial physics research.[388] Besides these agencies, other UK based funders of SWE research and activity include a number of UK universities (e.g. UCL Mullard Space Science Laboratory).[389] Combining UK SWE activity hubs, in recent years three out of four teams – Airbus STFC RAL, and UCL – have been leading an ESA mission to reduce the impacts of SWE by placing a spacecraft at the 5th Lagrange (L5) point, significantly improving European SWE observation and forecasting capabilities.[390] The UK Space Agency (UKSA) is also taking a more prominent role in SWE activities. Despite its exact role in SWE during the time of its establishment being unclear,in 2016, €22 Million was committed by UKSA to ESA's SSA programme.[391]

A.5 Selected Worldwide Institutions Involved in SWE

Country	Institution
Argentina	• Comisión Nacional de Actividades Espaciales (CONAE) • Institute of Astronomy and Space Physics, University of Buenos Aires • National Council of Scientific and Technical Research
Australia	• Bureau of Meteorology (BoM) • SPACE Research Centre, RMIT
Austria	• Space Research Institute • Conrad Observatory • Institute of Physics, University of Graz • Kanzelhöhe Observatory
Belgium	• Royal Observatory of Belgium (SIDC) • Solar-Terrestrial Centre of Excellence in Belgium (STCE)
Brazil	• National Institute for Space Research (INPE)
Bulgaria	• Space Research and Technology Institute

[383] Swedish Institute of Space Physics, 2018
[384] Ibid.
[385] UK Met Office, 2018
[386] Krausmann et al., 2016:11
[387] British Geological Survey, 2018
[388] Rutherford Appleton Laboratory, 2018
[389] UK Parliamentary Office of Science and Technology, 2010:2
[390] UK Government, 2018
[391] UK Government, 2018

Canada	• Natural Resources Canada (NRCan)
China	• China Meteorological Organisation (CMA) • National Astronomical Obervatories, Chinese Academy of Sciences • National Space Science Centre, Chinese Academy of Sciences
Czech Republic	• Institute of Atmospheric Physics, Czech Academy of Sciences
Denmark	• DTU Space, National Space Institute
Egypt	• Space Weather Monitoring Centre
Ethiopia	• Washera Geospace and Radar Science Research Laboratory
Finland	• Sodankylä Geophysical Observatory • Finnish Meteorological Institute (FMI)
France	• Centre National d'Etudes Spatiales (CNES) • Centre Opérationnel de Surveillance Militaire des Objets Spatiaux (COSMOS)
Germany	• German Aerospace Centre (DLR) • German Air Force Operations Centre
Hungary	• Debrecen Heliophysical Observatory • Research Centre for Astronomy and Earth Sciences
India	• Space Physics Laboratory, Vikram Sarabhai Space Centre • National Remote Sensing Centre • Radio Astronomy Centre, National Centre for Radio Astrophysics, Tata Institute of Fundamental Research • Indian Institute of Geomagnetism
Indoneasia	• Space Science Centre, Indonesian National Institute of Aeronautics and Space (LAPAN)
Italy	• Istituto Nazionale di Geofisica e Vulcanologia (INGV) • Istituto Nazionale di Astrofisica (INAF) • Agenzia Spaziale Italiana (ASI)
Japan	• National Institute of Information and Communications Technology (NICT), Kyushu University • Japan Aerospace Exploration Agency (JAXA) • Planetary Plasma and Atmospheric Research Centre, Tohoku University
Kazakhstan	• Institute of Ionosphere
Kenya	• Technical University of Kenya
Malaysia	• Space Science Centre, Universiti Kebangsaan Malaysia • GNSS & Geodynamics Research Group, Universiti Teknologi Malaysia
Mexico	• Instituto de Geofísica, Universidad Nacional Autónoma de México • Mexican Space Agency
Nigeria	• National Space Research and Development Agency • Center for Satellite Technology Development • Center for Atmospheric Research
Netherlands	• Royal Netherlands Meteorological Institute
Norway	• Norwegian Centre for Space Weather • Norwegian Space Centre • Birkeland Centre for Space Science, University in Bergen
Peru	• Comisión Nacional de Investigación y Desarrollo Aeroespacial • Centro de Radio Astronomía e Astrofísica Mackenzie (CRAAM), Universidade Presbiteriana Mackenzi
Philippines	• Manila Observatory
Poland	• Space Research Centre, Polish Academy of Sciences • Astronomical Institute, University of Wrocław

Russia	• Federal Service for Hydrometeorology and Environmental Monitoring (Roshydromet) • Institute of Applied Geophysics
Slovakia	• Slovak Central Observatory • Astronomical Institute SAS
South Africa	• South Africa National Space Agency (SANSA)
South Korea	• Korea Meteorological Administration (KMA) • National Radio Research Agency (RRA)
Switzerland	• Institute for Astronomy, ETH Zurich
Spain	• Servicio Nacional de Meteorología Espacial, Spanish National Space Weather Service (SeNMEs, Universidad de Alcalá)
Tunisia	• Laboratoire de Spectroscopie Atomique, Moléculaire et Applications, University of Tunis El Manar
Ukraine	• Space Research Institute, Kyiv • Main Center of Special Monitoring, Gorodok
United Kingdom	• British Geological Survey (BGS) • STFC RAL Space • United Kingdom Meteorology Office (UKMO)
United States of America	• National Oceanic and Atmospheric Administration (NOAA) • National Aeronautics and Space Administration (NASA) • Air Force Weather Agency • American Geophysical Union • Los Alamos National Laboratory • 55th Space Weather Squadron

A.6 ESA's Expert Groups Overview

	Geomagnetic Conditions Product	Heliospheric Weather Products	Ionospheric Weather Products	Solar Weather Products	Space Radiation Products
Athens Neutron Monitor Station					2
BIRA-IASB Space Weather Services					3
Catania Astrophysical Observatory (INAF)				3	
Center for Space Radiations (CSR)					9
Centre de Données de la Physique des Plasmas (CDPP)		2			
Collecte Localisation Satellites (CLS)			12		
Department Radiation Biology (DLR-IAM)					4
Finnish Meteorological Institute (FMI)	5		2		
Helmholtz-centre Potsdam (GFZ)	7		4		
Hosted by the SWE Data Centre			1		3
Institute of 4D-Technologies (FHNW)				1	

Ionosphere Monitoring and Prediction Center (IMPC)			13		
Kanzelhöhe Observatory (KSO)				4	
Mullard Space Science Laboratory (UCL)					2
National Observatory of Athens (NOA)			7		
Norwegian Mapping Authority (NMA)			9		
Paul Buehler					5
RAL Space (STFC)		2			
Research Center for Astronomy and Applied Mathematics (RCAAM)				1	
Seibersdorf Laboratories					1
Solar Influences Data analysis Center (SIDC)	1			17	
Space Research Centre (SRC)			12		
Space Research Laboratory, Department of Physics and Astronomy, University of Turku					3
Swedish Institute of Space Physics (IRF)	1				
Technical University of Denmark (DTU)		1			
Tromsø Geophysical Observatory (TGO)	9				
UK Met Office (MET)		8			
Universidad de Alcalá (UAH)	8				
University of Graz (UNIGRAZ), Institute of Physics		2			

A.7 SWE Projects in EU Framework Programmes (FP7 and H2020)

Space Weather Projects Within FP7					
Year	**Project Research Theme**	**Project Title**	**Project EC Contribution**	**Project Total Cost**	**Project Objective**
2007	Space Science - SPA-2007-2.1-01	SOTERIA (SOlar-TERrestrial Investigations and Archives)	3.922.966,00	5.161.288,00	Aimed at making better use of current data and developed databases in terms of detail, time-resolution and methods of data access. A publicly available inventory of results is accessible via the SOTERIA website and European Space Weather Portal
2009	Exploitation of space science and exploration data - SPA.2010.2.1-03	SEPServer (Data Services and Analysis Tools for Solar Energetic Particle Events and Related Electromagnetic Emissions)	1.932.172,70	2.484.125,80	SEPServer is a data exploration project aimed at providing access to the latest observations and tools on SEP and EM emission events for the scientific community. SEPServer intends to facilitate the development of new predictive models for spacecrafts. Additionally, an online catalogue is provided the SEPServer website
2009	Exploitation of space science and exploration data - SPA.2010.2.1-03	HESPE (High Energy Solar Physics Data in Europe)	1.569.808,00	2.213.164,12	The objective of the HESPE project is to provide tools and products for the solar physics community through the formulation and implementation of computations methods regarding solar high-energy data
2009	Exploitation of space science and exploration data - SPA.2010.2.1-03	POPDAT (Problem-oriented Processing and Database Creation for Ionosphere Exploration)	1.374.209,00	1.712.528,20	Funded through the 2010 "Exploitation of space science and exploration data" call topic, POPDAT's objective is to add value to existing ionosphere mission data, providing the scientific community with new insight on wave processes in the ionosphere
2009	Exploitation of space science and exploration data - SPA.2010.2.1-03	ECLAT (European Cluster Assimilation Technology)	1.577.144,50	2.045.723,40	ECLAT intends to deliver novel tools and a database for space scientists through an upgrade to ESA's Cluster Active Archive (CAA) - a plasma physics data repository spanning 10 years of data from the ESA Cluster multi-spacecraft-mission
2009	Security of space assets from space weather events - SPA.2010.2.3-1	EURISGIC (European Risk from Geomagnetically Induced Currents)	1.056.184,00	1.561.175,40	EURISGIC aims to provide a risk assessment, worst-case scenarios, and a demonstration of GIC forecasting capabilities, regarding the impacts of Geomagnetically Induced Currents on European electrical power networks. Produced will be a European-wide forecast tool prototype, as well as a geographical map indicating the occurrence of rapid geomagnetic variation
2009	Security of space assets from space weather events - SPA.2010.2.3-1	ATMOP (Advanced Thermosphere Modelling for Orbit Prediction)	1.563.980,36	2.217.243,16	The objective of ATMOP is to develop a thermosphere model able to provide reliable thermospheric air drag calculations for satellite orbit tracking, track loss, collisions, and re-entry zone situations. The model will feature operational forecasting and nowcasting capabilities

Year	Project Research Theme	Project Title	Project EC Contribution	Project Total Cost	Project Objective
08/12/2009	Security of space assets from space weather events - SPA.2010.2.3-1	SPACECAST (Protecting space assets from high energy particles by developing European dynamic modelling and forecasting capabilities)	1.965.071,25	2.539.991,31	It is the intention of the SPACECAST project to develop European modelling and forecasting capabilities to protect space vehicles and manned spacecraft. SEP events and high-energy particles in the electron radiation belt are the focus of the project, poses the highest risk to manned and unmanned spacecraft within the region. Already developed under the project is a near-real-time (<3 hours) forecast of the high-energy electron radiation belt available online
2009	Security of space assets from space weather events - SPA.2010.2.3-1	SIDER (Radiation Shielding of Composite Space Enclosures)	1.067.329,00	1.440.726,20	SIDER is a project aimed at developing sufficient radiation shielding technology for satellite components. A nanomaterial is proposed as an alternative for shielding, and a high-density foil is also under consideration
2009	Security of space assets from space weather events - SPA.2010.2.3-1	PLASMON (A new, ground-based data-assimilative modelling of the Earth's plasmasphere - a critical contribution to Radiation Belt modelling for Space Weather purposes)	1.972.049,75	2.626.262,80	PLASMON aims to improve our understanding of plasmaspheric dynamics and its influence on radiation belts. Three ground-based measurements networks had been extended under this project with new stations, improving automatic detection and analysis algorithms. PLASMON involves European and wider international partners, contributing both expertise and geographical utilisation
2009	Security of space assets from space weather events - SPA.2010.2.3-1	COMESEP (COronal Mass Ejections and Solar Energetic Particles: forecasting the space weather impact)	1.798.718,00	2.518.021,40	The COMESEP project is designed to build an operational European Space Weather Alert system. This system will provide alerts and forecasters in an intelligible format in near-real-time, aimed at protecting against and mitigating the impacts of harmful space weather events
2009	Security of space assets from space weather events - SPA.2010.2.3-1	SWIFF (Space Weather Integrated Forecasting Framework)	1.559.005,56	1.991.474,08	Both the development of an integrated framework for mathematical-physical process, and the development of methods and software for linkage between different physics and processes occurring simultaneously or in cascade within specific regions, is a core issue to solve for the basis of SWE forecasting. SWIFF aims to address these issues, developing relevant models and algorithms, as well as providing an integrated software infrastructure for SWE forecasting

Year	Project Research Theme	Project Title	Project EC Contribution	Project Total Cost	Project Objective
2009	Security of space assets from space weather events - SPA.2010.2.3-1	AFFECTS (Advanced Forecast For Ensuring Communications Through Space)	1.999.893,00	2.550.245,00	In safeguarding terrestrial telecommunication and navigation system operations from solar , AFFECTS intends to develop an advanced prototype space weather warning system. An operational early-warning service is available on the SWACI AFFECTS website, designed with GNSS users and service operators in mind. Individual forecasting models and tools are also developed by AFFECTS, and integrated into the Forecast System Ionosphere (FSI)
2010	Exploitation of space science and exploration data - SPA.2011.2.1-01	eHEROES (Environment for Human Exploration and RObotic Experimentation in Space)	1.999.720,85	2.523.910,56	the eHeroes project utilises a synergistic approach with models, simulations and observational data in order to characterise the space environment and evaluate its impact on manned and unmanned space exploration. The project aims to combine different datasets from robotic explorations with current7past space- and ground-based observations, and other novel data, to achieve improved exploitation of scientific data
2010	Exploitation of space science and exploration data - SPA.2011.2.1-01	SHOCK (Solar and Heliospheric Collisionless Kinetics: Enabling Data Analysis of the Sun to Earth Plasma System with Kinetic Modelling)	1.998.104,00	2.602.739,60	The SHOCK project aims to progress scientific understanding of the fundamental processes occurring within the Sun-to-Earth plasma system - increasing usage of existing ESA mission data alongside modelling and kinetic simulation. The project has produced a prototype webtool visualisation of the Virtual Mission Laboratory
2010	Exploitation of space science and exploration data - SPA.2011.2.1-01	MAARBLE (Monitoring, Analyzing and Assessing Radiation Belt Loss and Energization)	1.995.042,90	2.845.504,37	MAARBLE utilises multi-spacecraft monitoring of the Geospace environment, alongside ground-based observations, to analyse and asses the physical processes leading to the radiation belt particle energisation and loss. Provided by the project is a database of properties of waves, made available to the scientific community. Using this database, a statistical model is under development
2011	Key technologies enabling observations in and from space - SPA.2012.2.1-01	STORM	1.998.200,00	2.655.900,00	The main objective of the FP7 project STORM was to make a systematic investigation of the in-situ space plasma data bases collected by ESA's missions launched in the solar system, as well as of data from other relevant satellite data bases

Year	Project Research Theme	Project Title	Project EC Contribution	Project Total Cost	Project Objective
2011	Key technologies enabling observations in and from space - SPA.2012.2.1-01	SOLID (First European Comprehensive SOLar Irradiance Data exploitation)	1.994.373,60	2.579.598,40	It is the aim of the SOLID project to provide a comprehensive analysis of solar irradiance variations - a crucial natural factor for effective climate modelling. The project will combine datasets to build a continuous and high level solar spectral irradiance record from 1979 onwards
2011	Key technologies for in-space activities - SPA.2012.2.2-02	SR2S (Space Radiation Superconductive Shield)	1.995.853,44	2.774.567,84	The mission of the SR2S project is to demonstrate the feasibility of an active magnetic shielding technology to protect astronauts from radiation in the space environment by using superconductive technology
2012	Exploitation of space science and exploration data - SPA.2013.2.1-01	HELCATS (Heliospheric Cataloguing, Analysis and Techniques Service)	2.499.833,15	3.168.901,60	The HELCATS strategy is to coordinate a range of observational and modelling studies of heliospheric phenomena to provide a foundation for enhancing the scientific discipline and the exploitation of European investment in the hardware involved. It is also a benchmark in the provision of facilities to understand the nature and development of solar transients in the heliosphere
2012	Exploitation of space science and exploration data - SPA.2013.2.1-01	F-CHROMA (Flare Chromospheres: Observations, Models and Archives)	2.204.174,50	2.811.687,60	The F-CHROMA project was a dedicated multi-mode, multi-wavelength study of solar flares as observed in the lower solar atmosphere, or chromosphere. F-CHROMA scientists led numerous successful campaigns to observed flares from ground-based solar telescopes, co-ordinated with space-based facilities. This campaign data, and other data, consisting of flare images and spectroscopy, were analysed in combination with state-of-the-art numerical simulations to deduce the structure and evolution of the flare chromosphere, and understand the mechanisms by which energy is transported through the solar atmosphere in a flare, and dissipated in the form of heat, ionisation, and radiation
2012	Space-weather events - SPA.2013.2.3-01	SPACESTORM (Modelling space weather events and mitigating their effects on satellites)	1.981.301,49	2.544.144,52	The goal of the SPACESTORM project was to model space weather events and mitigate their effects on satellites by developing better mitigation guidelines, forecasting, and by experimental testing of new materials and methodologies to reduce satellite vulnerability
2012	Space-weather events - SPA.2013.2.3-01	MISW (Mitigation of space weather threats to GNSS services)	1.968.231,00	2.882.063,80	MISW will research, develop and apply new solutions to compensate for ionospheric effects on GNSS. Measurements of actual extreme events will allow realistic estimates of the ionospheric delays and errors caused by scintillation
Total			**43.993.366,05**		Source: Research Executive Agency

Space Weather Projects within Horizon 2020					
Year	**Project Research Theme**	**Project**	**Proj. Maximum Grant Amount**	**Project Total Costs**	**Project Objective**
2014	Protection of European assets in and from space - Space Weather - PROTEC-1-2014	PROGRESS (Prediction of Geospace Radiation Environment and solar wind parameters)	2.358.230,50	2.359.235,00	PROGRESS aims to combine the individual strengths of current groups working in the field of space weather modelling in order to accurately predict the occurrence and severity of space weather events through developing a comprehensive set of forecasting tools
2015	Other Actions – GNSS Evolution, Mission and Services related R&D activities	IPS	669.000,00	n/a	The overall aim of the IPS project is to "design, develop and operate a service prototype platform to monitor and predict the ionospheric behaviour and the potential effects on the performances of GNSS based applications." It is in this sense that the IPS project is geared as a service towards stable performance for user bases of whom are vulnerable to GNSS disruptions from ionospheric disturbances
2014	Space Weather - COMPET-5-2017	FLARECAST (Flare Likelihood and Region Eruption Forecasting)	2.416.651,25	2.416.651,25	FLARECAST offers a flare prediction system with the objective of providing accurate and reliable space-weather monitoring and forecasting capabilities.
2014	Space Weather - COMPET-5-2017	HESPERIA (High Energy Solar Particle Events foRecastIng and Analysis)	1.101.456,25	1.208.956,25	HESPERIA aims to produce two novel forecasting tools based upon proven concepts, whilst additionally advancing the knowledge of the physical mechanisms which lead into high-energy solar particle events (SEPs) by exploiting novel datasets.
2017	Space Weather - COMPET-5-2017	TechTIDE (Warning and Mitigation Technologies for Travelling Ionospheric Disturbances Effects)	1.579.000,00	1.579.000,00	The project plans to design new viable Travelling Ionospheric Disturbances (TID) impact mitigation strategies for the technologies affected by TIDs and, to validate the added value of these techniques in close collaboration with operators of these technologies
2017	Space Weather - PROTEC-1-2014	SWAMI (Space Weather Atmosphere Model and Indices)	1.198.363,75	1.198.363,75	SWAMI aims at "developing improved neutral atmosphere and thermosphere models; make a major leap forward by combining these physics-based and empirical models; exploiting new geomagnetic; and improve the forecast of the activity indices"
2017	Space Weather - PROTEC-1-2014	ESC2RAD (Enabling Smart Computations to study space RADiation effects)	1.262.763,75	1.262.763,75	This project aims at establishing a fundamental and applied research program via the setup of a new "virtual modelling lab" which will open the path towards a change of paradigm in the modelling of Space Weather impact
2019	Space Weather - SU-SPACE-22-SEC-2019	n/a	9,000,000,00	n/a	n/a
Total			**19.585.465,50**		Source: Research Executive Agency

A.8 ESA and Operational Services

ESA Convention (approved on 30 May 1975)

Article V.2

In the area of space applications the Agency may, should the occasion arise, carry out operational activities under conditions to be defined by the Council by a majority of all Member States. When so doing the Agency shall:

a. place at the disposal of the operating agencies concerned such of its own facilities as may be useful to them;

b. ensure as required, on behalf of the operating agencies concerned, the launching, placing in orbit and control of operational application satellites;

c. carry out any other activity requested by users and approved by the Council.

Resolution on the Agency and its Operational Systems<< (adopted on 15 February 1977)

The Council, meeting at ministerial level,

CONSIDERING that, in addition to its task of developing space technology, the European Space Agency also has the mission, under the Convention for the establishment of a European Space Agency, of giving support for the development and management of European operational space systems,

RECOGNISING that the execution of operational activities will enable the Agency to exploit its capabilities and capital investments to the full and to achieve a better regulation of the workloads of the Agency and of industry, as well as to arrive at a better definition of its subsequent programmes in the light of the requirements of space-systems users,

CONSIDERING the importance, in the overall European economic context, of avoiding multiplication of space-related capabilities and capital investments,

JUDGING IT DESIRABLE, in consequence, to adopt a positive attitude in relation to the management of operational systems,

AGREES that the Agency's activities in the operational field must conform to the following principles:

1. "As regards the pre-operational systems which the Member States entrust to it for execution, the Agency will have full responsibility for design, development and exploitation. It will exercise this responsibility in consultation with potential users, particularly in cases where the development of prototypes is considered to be the best way of advancing the associated technology and facilitating the transition to the operational phase.

2. As regards operational systems:

   a: In the fields where organized users do not exist, the Agency will encourage the potential users of operational systems to take over the management of these systems and to organize their exploitation. In accordance with the Council's instructions, it will furnish them with all the technical and institutional assistance they may request to this end, including the making available of facilities.

   b. In the fields where organized users exist, the Agency will not undertake tasks unless so requested by them

3. Subject to any other activity requested by users and approved by the Council, the Agency will limit its operational activities to the launching, placing in orbit, and orbital control of satellites or space transport systems, and to the provision of technical assistance, in the design and exploitation of systems, either to the users themselves or to a body designated by them.

   The Agency will undertake operational activities only if it can do so without interfering with the effective discharge of the principal tasks for which it has been established.

4. The Agency will abstain from encroaching upon the acknowledged attributions of the user organisations. The principles set out in Article VII of the ESA Convention will be equally applicable to operational activities entrusted to the Agency

5. The interfaces between the Agency and the users will be defined precisely and will be the subject of appropriate arrangements.

6. The Agency's internal costs incurred through the execution of its operational activities will be limited as far as possible. To this end a charging policy relating to these costs will be defined.

7. The Agency's expenditure in connection with these activities will be charged to the users in accordance with terms to be determined in the arrangements referred to above. The Council will determine those cases in which the Agency may continue during a limited period to bear certain expenditure, notably in order to promote the constitution and starting up of user groups; in doing so the Agency will make every effort to reduce the amount of such expenditure.
8. No financial involvement shall arise for any Member State from operational activities without the specific approval of that Member State.
9. The Agency will set up a suitable internal management and accounting structure to permit clear identification and correct charging of activities in the operational sector.
10. The Agency will take care to remain within the framework of the privileges and immunities granted to it by the Member States in accordance with the provisions of Article 7.2 of Annex I of the ESA Convention.
11. The Agency will define and carry out a policy enabling the Member States that have contributed to the development of a space programme to be equitably associated with the follow-up operational activities resulting from the programme in question, taking due account of any commercial constraints.
12. The Agency will recommend to Member States all measures which allow harmonisation of the policies of user administrations or entities in their countries with the Agency's policy defined in the preceding paragraph.

The Council will take the necessary steps for the implementation of the above principles, which it will review from time to time in the light of experience gained.

A.9 Long-Term Sustainability Guidelines of Relevance to SWE

Guideline 16. Share operational space weather data and forecasts

16.1 States and international intergovernmental organizations should support and promote the collection, archiving, sharing, intercalibration, long-term continuity and dissemination of critical space weather data and space weather model outputs and forecasts, where appropriate in real time, as a means of enhancing the long-term sustainability of outer space activities.

16.2 States should be encouraged to monitor space weather continuously and to share data and information with the aim of establishing an international space weather database network.

16.3 States and international intergovernmental organizations should support the identification of data sets critical for space weather services and research and should consider adopting policies for the free and unrestricted sharing of critical space weather data from their space- and ground-based assets. All governmental, civilian and commercial space weather data owners are urged to allow free and unrestricted access to and archival of such data for mutual benefit.

16.4 States and international intergovernmental organizations should also consider sharing real-time and near-real-time critical space weather data and data products in a common format, promote and adopt common access protocols for their critical space weather data and data products, and promote the interoperability of space weather data portals, thus promoting ease of data access for users and researchers. The real-time sharing of these data could provide a valuable experience for sharing in real time other kinds of data relevant to the long-term sustainability of outer space activities.

16.5 States and international intergovernmental organizations should further undertake a coordinated approach to maintaining the long-term continuity of space weather observations and identifying and filling key measurement gaps, so as to meet critical needs for space weather information and/or data.

16.6 States and international intergovernmental organizations should identify high-priority needs for space weather models, space weather model outputs and space weather forecasts and adopt policies for free and unrestricted sharing of space weather model outputs and forecasts. All governmental, civilian and commercial space weather model developers and forecast providers are urged to allow free and unrestricted access to and archival of space weather model outputs and forecasts for mutual benefit, which will promote research and development in this domain.

16.7 States and international intergovernmental organizations should also encourage their space weather service providers to:

(a) Undertake comparisons of space weather model and forecast outputs with the goal of improved model performance and forecast accuracy;

(b) Openly share and disseminate historical and future critical space weather model outputs and forecast products in a common format;

(c) Adopt common access protocols for their space weather model outputs and forecast products to the extent possible, to promote their ease of use by users and researchers, including through interoperability of space weather portals;

(d) Undertake coordinated dissemination of space weather forecasts among space weather service providers and to operational end users.

Guideline 17. Develop space weather models and tools and collect established practices on the mitigation of space weather effects

17.1 States and international intergovernmental organizations should undertake a coordinated approach to identifying and filling gaps in research and operational models and forecasting tools required to meet the needs of the scientific community and of the providers and users of space weather information services. Where necessary, this should include coordinated efforts to support and promote research and development to further advance space weather models and forecasting tools, incorporating the effects of the changing solar environment and evolving terrestrial magnetic field as appropriate, including within the context of the Committee on the Peaceful Uses of Outer Space and its subcommittees, as well as in collaboration with other entities such as the World Meteorological Organization and the International Space Environment Service.

17.2 States and international intergovernmental organizations should support and promote cooperation and coordination on ground- and space-based space weather observations, forecast modelling, satellite anomalies and reporting of space weather effects in order to safeguard space activities. Practical measures in this regard could include:

(a) Incorporating current and forecast space weather thresholds into space launch criteria;

(b) Encouraging satellite operators to cooperate with space weather service providers to identify the information that would be most useful to mitigate anomalies and to derive recommended specific guidelines for on-orbit operations. For example, if the radiation environment is hazardous, this might include actions to delay the uploading of software, implementation of manoeuvres etc.;

(c) Encouraging the collection, collation and sharing of information relating to ground- and space-based space weather-related impacts and system anomalies, including spacecraft anomalies;

(d) Encouraging the use of a common format for reporting space weather information. In relation to the reporting of spacecraft anomalies, satellite operators are encouraged to take note of the template proposed by the Coordination Group for Meteorological Satellites;

(e) Encouraging policies promoting the sharing of satellite anomaly data related to space weather-induced effects;

(f) Encouraging training on and knowledge transfer relating to the use of space weather data, taking into account the participation of countries with emerging space capabilities.

17.3 It is acknowledged that some data may be subject to legal restrictions and/or measures for the protection of proprietary or confidential information, in accordance with national legislation, multilateral commitments, non-proliferation norms and international law.

17.4 States and international intergovernmental organizations should work towards the development of international standards and the collection of established practices applicable for the mitigation of space weather effects in satellite design. This could include sharing of information on design practices, guidelines and lessons learned relating to mitigation of the effects of space weather on operational space systems, as well as documentation and reports relating to space weather user needs, measurement requirements, gap analyses, cost-benefit analyses and related space weather assessments.

17.5 States should encourage entities under their jurisdiction and/or control to:

(a) Incorporate in satellite designs the capability to recover from a debilitating space weather effect, such as by including a safe mode;

(b) Incorporate space weather effects into satellite designs and mission planning for end-of-life disposal in order to ensure that the spacecraft either reach their intended graveyard orbit or de-orbit appropriately, in accordance with the Space Debris Mitigation Guidelines of the Committee on the Peaceful Uses of Outer Space. This should include appropriate margin analysis.

17.6 International intergovernmental organizations should also promote such measures among their member States.

17.7 States should undertake an assessment of the risk and socioeconomic impacts of adverse space weather effects on the technological systems in their respective countries. The results from such studies should be published and made available to all States and used to inform decision-making relating to the long-term sustainability of outer space activities, particularly with regard to mitigating the adverse impacts of space weather on operational space systems.

A.10 List of External Contributors to the Research

Name	Position	Organisation
Alexi Glover	Space Situational Programme Office	European Space Agency (ESA)
Andrej Rozkov	Project Officer	European Commission /Research Executive Agency
Andrew Monham	Spacecraft Operations Manager	European Organisation fort the Exploitation of Meteorological Stellites (EUMETSAT)
Brigit Blasch	Deputy Head of Unit	European Commission/Research Executive Agency (REA)
Elisabeth Krausmann	Scientific Officer	European Commission / Joint Research Centre (JRC)
Eric Guyader	Administrator – Galileo Programme	European Commission
Ewa Oney	Policy Officer	European Commission
Harm Greidanus	Scientist	European Commission/ Joint Research Centre (JRC)
Juha-Pekka Luntama	Space Weather Manager	European Space agency (ESA)
Masha Kutseova	Panel on Space Weather Chair	Committee on Space Research (COSPAR)
Mike Hapgood	Head of the Space Environment Group	Rutherford Appleton Laboratory (RAL)
Mike Williams	Head of Flight Operations	European Organisation fort the Exploitation of Meteorological Stellites (EUMETSAT)
Paul Counet	Head of Strategy and International Relations	European Organisation fort the Exploitation of Meteorological Stellites (EUMETSAT)

List of Acronyms

Acronym	Explanation
ACE	Advance Composition Explorer
AFWA	Air Force Weather Agency (now the 557th Weather Wing)
ANC	Air Navigation Commission
ASWA	American Space Weather Association
BIRA-IASB	Royal Belgian Institute for Space Aeronomy
CAeM	Commission for Aeronautical Meteorology (WMO)
CGMS	Coordination Group for Meteorological Satellites
CMA	China Meteorological Administration
CME	Coronal Mass Ejection
CNES	French Space Agency
COMPET	Competitiveness of the European Space Sector
COSPAR	Committee on Space Research
COST	European Cooperation in Science and Technology
COST	Cooperation on Science and Technology
D3S	Distributed SWE Sensor System (SSA)
DLR	German Aerospace Center
DoD	Department of Defence (U.S.)
DRAO	Dominion Radio Astrophysical Observatory
DRM	Disaster Risk Management
DSCVR	Deep Space Climate Observatory
DSM	Design Study Methodology
EC	European Commission
ECI	European Critical Infrastructure
EISCAT	European Incoherent SCATer Scientific Association
EO	Earth Observation
ESA	European Space Agency
ESC	Expert Service Centre
ESPI	European Space Policy Institute
ESTEC	European Space Research and Technology Centre
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
FEMA	Federal Emergency Management Agency
FERC	Federal Energy Regulation Commission

Acronym	Explanation
FLARECAST	Flare Likelihood And Region Eruption foreCASTing
FP7	7th Framework Programme on Research and Innovation
GCR	Galactic Cosmic Rays
GDP	Gross Domestic Product
GEO	Group on Earth Observation
GEOS	Global Earth Observation Systems
GIC	Geomagnetically Induced Current
GMDN	Global Muon Detector Network
GNSS	Global Navigation Satellite System
GONG	Global Oscillation Network Group
GPS	Global Positioning System
H2020	Horizon 2020
HESPERIA	High Energy Solar Particle Events forecasting and Analysis
I-SWAT	International Space Weather Action Teams
IAEA	International Atomic Energy Agency
ICAO	International Civil Aviation Organisation
ICG	International Committee on Global Navigation Satellite Systems
ICT	Information and Communication Technologies
IGY	International Geophysical Year
IHY	International Heliophysical Year
IMPC	Ionosphere Monitoring and Prediction Centre (DLR)
INTERMAGNET	International Real-time Magnetic Observatory Network
IPR	Intellectual Property Rights
ISES	International Space Environment Service
ISS	International Space Station
ISWI	International Space Weather Initiative
IT	Information Technology
ITU	International Telecommunications Union
JRC	Joint Research Centre (EC)
L1 & L5	Langarian Points of the Earth-Sun System
LEO	Low Earth Orbit
LTSSA	Long-Term Sustainability of Outer Space Activities
METP (ICAO)	Meteorological Panel
MOSWOC	UK Met Office Space Weather Operations Centre
NASA	National Aeronautical and Space Administration
NEO	Near Earth Object
NGO	Non-Governmental Organisation
NMDB	Real-Time Neutron Monitor Database
NOAA	National Oceanic Atmospheric Administration

Acronym	Explanation
NOAA-CSWIG	Commercial Space Weather Interest Group
NRC	National Research Council
PNT	Position, Navigation and Timing
PSP	Pipe-to-Soil Potential difference
PwC	PricewaterhouseCoopers
RAE	Royal Academy of Engineering
RAL	Rutherford Appleton Laboratory
REA	Research Executive Agency (EC)
RWC	Regional Warning Centres
SCD	Spacecraft Design
SCO	Spacecraft Operation
SDC	Space Weather Data Centre
SDG(s)	Sustainable Development Goal(s)
SEP	Solar Energetic Particles
SEU	Single Event Upset
SOHO	Solar and Hemispheric Observatory
SSA	Space Situational Awareness
SSA SSCC	SSA Space Weather Coordination Centre
SST	Space Surveillance Tracking
STEREO	Solar Terrestrial Relations Observatory
STFC	Science and Technology Facilities Council
SuperDARN	Super Dual Auroral Radar Network
SWAMI	Space Weather Atmosphere Model and Indices
SWAD	Space Weather Awareness Dialogue
SWE	Space Weather
SWPC	Space Weather Prediction Centre (NOAA)
SWWT	Space Weather Working Team
TEC	Total Electron Content
TGO	Tromsø Geophysical Observatory
UIP	User Interface Platform
UN	United Nations
UN PSA	United Nations Programme on Space Activities
UN STSC	United Nations Scientific and Technical Subcommittee
UNCOPUOS	United Nations Committee on the Peaceful Uses of Outer Space
UNISPACE+50	50th anniversary of the first United Nations Conference on the Exploration and Peaceful Uses of Outer Space
UNOOSA	United Nations Office for Outer Space Affairs
URSI	International Union of Radio Science
US SEC	United States Space Environment Centre
USAF	United States Air Force

Acronym	Explanation
USGS	United States Geological Survey
UV	Ultraviolet
WG	Working Group
WHO	World Health Organisation
WIGOS	WMO Integrated Global Observing System
WIS	World Meteorological Organisation Information System
WMO	World Meteorological Organisation

References

557th Weather Wing, 2018. "About Us." 557th Weather Wing. 2018. http://www.557weather-wing.af.mil/About-Us/Space-Weather/.

Academy of Finland, 2018. 'Space Weather Forecasting Still Difficult Due to Insufficient Observations'. AKA. 2018. http://www.aka.fi/en/about-us/media/press-releases/2013/Space-weather-forecasting-still-difficult-due-to-insufficient-observations/.

Agenzia Spaziale Italiana, 2017. 'Space Weather, the Italian Strategy'. A.S.I. - Agenzia Spaziale Italiana. 2017. https://www.asi.it/en/news/space-weather-italian-strategy.

Aliberti, M., Ferretti S., Hulsroj P., and Lahcen, A., 2016. "Europe in the Future and the Contributions of Space." ESPI Report 55. Vienna, Austria: ESPI.

American Commercial Space Weather Association, 2018. "About ACSWA." American Commercial Space Weather Association. 2018. http://acswa.us/about.html.

Armet, Dave, 2017. "NSO/GONG: About GONG." National Solar Observatory, Global Oscillation Network Group (GONG). March 31, 2017. https://gong.nso.edu/info/fact_sheet.html.

Baker, D. N, 1998. "What Is Space Weather?" *Advances in Space Research*, Solar-Terrestrial Relations: Predicting the Effects on the Near-Earth Environment, 22 (1): 7–16. https://doi.org/10.1016/S0273-1177(97)01095-8.

Belehaki, A., Cander, Lj., Zolesi, B., Bremer, J., Juren, C., Stanislawska, I., Dialetis, D., and Hatzopoulos, M., 2005. "DIAS Project: The Establishment of a European Digital Upper Atmosphere Server." *Journal of Atmospheric and Solar-Terrestrial Physics*, Space weather and RF communications: monitoring and modelling, 67 (12): 1092–99.

Belehaki, Anna, 2010. "Developing Space Weather Products and Services in Europe ES0803." presented at the Space Weather Working Team (SWWT) Plenary Meetings, Brussels, Belgium. http://swe.ssa.esa.int/DOCS/SWWT/m27/COST%20ES0803%20APC%20Presentation_2010.pdf.

Berger, T., Mcgranahan, R., and Camporeale, E., 2018. "TOPICAL DISCUSSION MEETING - MACHINE LEARNING SYSTEMS FOR SPACE WEATHER PREDICTION." presented at the 15th European Space Weather Week, Leuven, Belgium. http://www.stce.be/esww15/program/tdm.php?nr=30.

Bobrinsky, Nicolas, 2017. "Space Situational Awareness." presented at the Swiss Space Industry Days Lausanne, 15-16 June 2017.

Braun, Gerald, 2018. 'German Space Situational Awareness Centre (GSSAC): Status & Perspective'. http://www.jsforum.or.jp/stableuse/pdf/11%20SSA-Tokio-10-03-2018-englisch-final-GERALD.pdf.

British Geological Survey, 2018. 'Space Weather Alerts'. 2018. http://www.geomag.bgs.ac.uk/data_service/space_weather/alerts.html.

Castells, Manuel, 2011. *The Rise of the Network Society*. Vol. 12. John Wiley & Sons.

Chiarini, Paola, 2013. "Space Weather in the EU's FP7 Space Theme - Preface to the Special Issue on 'EU-FP7 Funded Space Weather Projects.'" *Journal of Space Weather and Space Climate* 3: E01. https://doi.org/10.1051/swsc/2013054.

Chisham, G., Lester, M., Milan, S. E., Freeman, M. P., Bristow, W. A., Grocott, A., McWilliams, K. A., et al., 2007. "A Decade of the Super Dual Auroral Radar Network (SuperDARN): Scientific Achievements, New Techniques and Future Directions." *Surveys in Geophysics* 28 (1): 33–109.

Christel, I., Hemment, D., Bojovic, D., Cucchietti, F., Calvo, L., Stefaner, M., and Buontempo, C., 2018. "Introducing Design in the Development of Effective Climate Services." *Climate Services*, Climate services in practice: what we learnt from EUPORIAS, 9 (January): 111–21.

Christian, E. R., and Davis, A. J., 2017. "Advanced Composition Explorer (ACE) Mission Overview." Space Radiation Lab at California Institute of Technology. 2017. http://www.srl.caltech.edu/ACE/ace_mission.html.

Clark, Toby, 2001. "A Review of the Effects of Space Weather on Ground Based Technology." British Geological Survey. https://ssa-be-vm-fe-08p.ssa.esa.int/spweather/workshops/SPW_W3/PROCEEDINGS_W3/ESTEC_gb_review.pdf.

Committee on Space Research, 2018. 'COSPAR Roadmap for Space Weather Activities: Asset Catalogue'. http://www.spaceweathercatalogue.org/COSPARCatalogue.pdf.

Community Research and Development Information Service, 2015. "PROTEC-1-2014 - Space Weather." CORDIS | European Commission. 2015. https://cordis.europa.eu/programme/rcn/664644_en.html.

——— 2017a. "COMPET-5-2017 - Space Weather." CORDIS | European Commission. https://cordis.europa.eu/programme/rcn/701815_en.html.

——— 2017b. "Prediction of Geospace Radiation Environment and Solar Wind Parameters | Projects | H2020." CORDIS | European Commission. 2018 https://cordis.europa.eu/project/rcn/193480_en.html.

——— 2017c. "High Energy Solar Particle Events FoRecastIng and Analysis | Projects | H2020." CORDIS | European Commission. 2018. https://cordis.europa.eu/project/rcn/193483_en.html.

——— 2017d. "Flare Likelihood and Region Eruption Forecasting | Projects | H2020." CORDIS | European Commission. 2018.

https://cordis.europa.eu/project/rcn/193702_en.html.

——— 2017e. "Warning and Mitigation Technologies for Travelling Ionospheric Disturbances Effects | Projects | H2020." CORDIS | European Commission. 2018. https://cordis.europa.eu/project/rcn/212425_en.html.

——— 2018. "Space Weather Atmosphere Model and Indices | Projects | H2020." CORDIS | European Commission. 2018. https://cordis.europa.eu/project/rcn/212434_en.html.

Coordination Group for Meteorological Satellites, 2007. "CGMS Global Contingency Plan." CGMS. http://www.wmo.int/pages/prog/sat/documents/CGMS_Contingency-Plan-2007.pdf.

——— 2016. "CGMS HIGH LEVEL PRIORITY PLAN (HLPP) 2016 - 2020." Coordination Group for Meterological Satellites. https://www.cgms-info.org/documents/CGMS_High_Level_Priority_Plan_2016-2020.pdf.

——— 2018. "Coordination Group for Meteorological Satellites." Coordination Group for Meteorological Satellites. 2018. https://www.cgms-info.org/index_.php/cgms/page?cat=ABOUT&page=INDEX.

Crosby, Norma, 2010. "Space Weather Working Team Plenary Meeting 27 Report." Brussels, Belgium: Space Weather Working Team (SWWT) Plenary Meeting. http://swe.ssa.esa.int/DOCS/SWWT/m27/SWWT-PlenaryMeeting27-18Jun2010.pdf.

Dudok de Wit, Thierry, 2015. 'Space Weather: Facilities and Services'. presented at the Meudon.

European Commission. 2017b. 'COMMISSION STAFF WORKING DOCUMENT: Overview of Natural and Man-Made Disaster Risks the European Union May Face'. SWD(2017) 176 final. Brussels, Belgium. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=COM%3A2018%3A447%3AFIN.

European Commission, 2011. "Space-Weather Awareness Event Explores the Road from Understanding to Action." EU Science Hub. 2011. https://ec.europa.eu/jrc/en/news/space-weather-awareness-event-explores-road-understanding-action-7134.

——— 2015. "Service Contract: Ionosphere Prediction Service - Growth - European Commission." 2015. /growth/content/service-contract-ionosphere-prediction-service_en.

——— 2016a. "Horizon 2020 Work Programme 2016 - 2017." Horizon 2020. 2016. https://www.h2020.md/en/compet-5-2017-space-weather.

——— 2016b. "Communication From the Commission to the European Paliament, the Council, the European Economic and Social Committee and the Committee of the Regions - Space Strategy for Europe." COM (2016) 705 final. Brussels, Belgium: European Commission. https://ec.europa.eu/transparency/regdoc/rep/1/2016/EN/COM-2016-705-F1-EN-MAIN.PDF.

——— 2017a. "Horizon 2020 - Work Programme 2018-2020 Leadership in Enabling and Industrial Technologies - Space." WP 2018-2020 5.iii. http://ec.europa.eu/research/participants/data/ref/h2020/wp/2018-2020/main/h2020-wp1820-leit-space_en.pdf.

——— 2017b. "COMMISSION STAFF WORKING DOCUMENT: Overview of Natural and Man-made Disaster Risks the European Union may face." SWD(2017) 176 final. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=COM%3A2018%3A447%3AFIN

——— 2018. "REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL Establishing the Space Policy Programme of the European Union, Relating to the European Union Agency for Space and Repealing Regulations (EU) No 1285/." Brussels, Belgium: EC. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=COM:2018:447:FIN.

European Cooperation in Science and Technology, 2002. "COST Action 724 Draft Memorandum of Understanding." COST 248/02. COST European Cooperation in Science and Technology.

——— 2008. "COST Action ES0803 | Developing Space Weather Products and Services in Europe." COST European Cooperation in Science and Technology. 2018. http://www.cost.eu/COST_Actions/essem/ES0803.

European Global Navigation Satellite Systems Agency, 2018. "New EC Service Monitors Ionosphere for GNSS Users | European GNSS Service Centre." 2018. https://www.gsc-europa.eu/news/new-ec-service-monitors-ionosphere-for-gnss-users.

European Incoherent SCATer Scientific Association, 2018. "EISCAT – Scientific Association." EISCAT. 2018. https://www.eiscat.se/.

European Organisation for the Exploitation of Meteorological Satellites, 2016. "EUMETSAT Strategy Challenge 2025." *EUMETSAT*. https://www.eumetsat.int/website/wcm/idc/idcplg?IdcService=GET_FILE&dDocName=PDF_BR_COR02_EN&RevisionSelectionMethod=LatestReleased&Rendition=Web.

——— 2017. "EUMETSAT Data Policy." EUMETSAT. https://www.eumetsat.int/website/wcm/idc/idcplg?IdcService=GET_FILE&dDocName=PDF_LEG_DATA_POLICY&RevisionSelectionMethod=LatestReleased&Rendition=Web.

——— 2018a. "Metop — EUMETSAT." 2018. https://www.eumetsat.int/website/home/Satellites/CurrentSatellites/Metop/index.html.

——— 2018b. "SEM — EUMETSAT." 2018. https://www.eumetsat.int/website/home/Satellites/CurrentSatellites/Metop/MetopDesign/SEM/index.html.

——— 2018c "SAFs — EUMETSAT." 2018. https://www.eumetsat.int/website/home/Satellites/GroundSegment/Safs/index.html.

European Space Agency, 2005. "Cluster Mission Summary." ESA, Science and Technology. http://sci.esa.int/cluster/31258-summary/.

——— 2008. "Space Situational Awareness Preparatory Programme Proposal." ESA/C(2008)142. European Space Agency Council. https://www.ffg.at/sites/default/files/SSA_Programme_Proposal.pdf.

——— 2011. "Space Situational Awareness - Space Weather Customer Requirements Document Issue 4 Revision 5 Author ESA SSA Team Date Approved by Date." SSA-SWE-RS-CRD-1001. European Space Agency.

——— 2013. "Space Situational Awareness - Space Weather System Requirements Document." SSA-SWE-RS-RD-0001. European Space Agency.

——— 2016. "The SSA Programme and SWE Segment Status + View towards Period 3." presented at the SWWT Steering Board meeting, 15th November 2016. http://swe.ssa.esa.int/DOCS/SWWT/m

37/SWWT_PM-37%2020161116-Minutes_Annex%202_slides%20JPL.pdf.

——— 2017a. "Space Weather and Its Hazards." European Space Agency. 2018. https://www.esa.int/Our_Activities/Operations/Space_Situational_Awareness/Space_weather_and_its_hazards.

——— 2017b. "Monitoring Space Weather." European Space Agency. 2017. https://m.esa.int/Our_Activities/Operations/Space_Situational_Awareness/Monitoring_space_weather.

——— 2017c. "European Space Agency Industrial Policy Committee Space Situational Awareness Programme Period 3 - SSA Procurement Plan for 2017 (First Half of Year)." ESA/IPC(2017)47. Paris, France: European Space Agency Industrial Policy Committee.

——— 2018a. "What Are Solar Flares?" European Space Agency. 2018. https://m.esa.int/Our_Activities/Space_Science/What_are_solar_flares.

——— 2018b. "Space Weather Effects." European Space Agency. 2018. https://www.esa.int/spaceinimages/Images/2018/01/Space_weather_effects.

——— 2018c. "ESA Expands Space Weather Services." European Space Agency. 2018. https://www.esa.int/Our_Activities/Operations/Space_Situational_Awareness/ESA_expands_space_weather_services.

——— 2018d. "ESA - Space Situational Awareness Space Weather - SWE." European Space Agency, Space Situational Awareness, Space Weather. 2018. http://swe.ssa.esa.int/.

——— 2018e, "About SSA." European Space Agency Space Situational Awareness. 2018.

http://m.esa.int/Our_Activities/Operations/Space_Situational_Awareness/About_SSA.

——— 2018f. "SSA Programme Overview." European Space Agency Space Situational Awareness. 2018. http://m.esa.int/Our_Activities/Operations/Space_Situational_Awareness/SSA_Programme_overview.

——— 2018g. "Space Weather Segment." European Space Agency Space Situational Awareness. 2018. http://m.esa.int/Our_Activities/Operations/Space_Situational_Awareness/Space_Weather_Segment

——— 2018h. "SSA Space Weather Activities." European Space Agency Space Situational Awareness. 2018. http://swe.ssa.esa.int/ssa-space-weather-activities.

——— 2018i. "Expert Service Centres." European Space Agency Space Situational Awareness. 2018. http://swe.ssa.esa.int/web/guest/expert-centres.

——— 2018j. "SSA Space Weather Network Service Product Catalogue Summary." SSA-SWE-SSCC-TN-0011. Brussels, Belgium: ESA. http://swe.ssa.esa.int/DOCS/SSA-DC/SSA-SWE-SSCC-TN-0011_Product_Catalogue_summary.pdf .

——— 2018k. "SSA Space Weather Service User Domains." 2018. http://swe.ssa.esa.int/web/guest/user-domains.

——— 2018l. "Spacecraft Design." European Space Agency, Space Situational Awareness, Space Weather. 2018. http://swe.ssa.esa.int/web/guest/spacecraft-design-info.

——— 2018m. "Spacecraft Operation." European Space Agency, Space Situational Awareness, Space Weather. 2018. http://swe.ssa.esa.int/spacecraft-operation.

——— 2018n. "Human Space Flight." European Space Agency, Space Situational Awareness, Space Weather. 2018. http://swe.ssa.esa.int/human-space-flight.

——— 2018o. "Launch Operation." European Space Agency, Space Situational Awareness, Space Weather. 2018. http://swe.ssa.esa.int/launch-operation.

——— 2018p. "Space Surveillance and Tracking." European Space Agency, Space Situational Awareness, Space Weather. 2018p. http://swe.ssa.esa.int/web/guest/space-surveillance-and-tracking-info.

——— 2018q. "Non-Space System Operation - Service to Resource Exploitation System Operators." European Space Agency, Space Situational Awareness, Space Weather. 2018. http://swe.ssa.esa.int/nso_res.

——— 2018r. "ESA - Space Situational Awareness Space Weather - Auroral Tourism Sector." Non-Space System Operation - Service to Auroral Tourism Sector. 2018. http://swe.ssa.esa.int/auroral-tourism-sector.

European Space Science Committee, 2018. 'European Space Science Committee'. European Space Science Committee, European Science Foundation Expert Board. 2018. http://www.essc.esf.org/.

Fondation pour la Recherche Stratégique, 2008. "Study on Suitable Governance and Data Policy Models for a European Space Situational Awareness (SSA) System." FRS.

Finnish Meteorological Institute, 2014. 'GUMICS - a Global Magnetosphere–Ionosphere Coupling Simulation'. 2014. http://gumics.fmi.fi/en/index.html.

FMI-SPACE, 2018. 'PECASUS Has Been Selected to Become One of Three Global Space Weather Service Providers for ICAO'. 2018. http://space.fmi.fi/2018/11/20/pecasus-has-been-selected-to-become-one-the-three-global-space-weather-service-providers-for-icao/.

Folini, Doris. 2018. 'Climate, Weather, Space Weather: Model Development in an Operational Context'. *Journal of Space Weather and Space Climate* 8: A32. https://doi.org/10.1051/swsc/2018021.

Garner, Rob. 2016. "Earth's Pulsating Ionosphere." National Aeronautical and Space Administration (NASA). 2016. http://www.nasa.gov/image-feature/goddard/2016/earths-pulsating-ionosphere.

——— 2017. "SOHO - Solar and Heliospheric Observatory." National Aeronautical and Space Administration (NASA). 2018. http://www.nasa.gov/mission_pages/soho/index.html.

Glaber, Max, 2014. "The Difference Between Flares and CMEs." National Aeronautical and Space Administration (NASA). September 2014. http://www.nasa.gov/content/goddard/the-difference-between-flares-and-cmes.

Glover, A., Bingham, S., Dierckxsens, M., and Angling, M., 2018. "TOPICAL DISCUSSION MEETING - METRICS, VERIFICATION AND VALIDATION: FROM INITIAL CONCEPT TO OPERATIONAL SPACE WEATHER SERVICES." presented at the 15th European Space Weather Week, Leuven, Belgium. http://www.stce.be/esww15/program/tdm.php?nr=28.

Glover, A., and Luntama, J, 2016. "SSA Space Weather Radiation Expert Service Centre in Period 3 (2017-2019)." SSA Programme Office, ESOC, Darmstadt, Germany. https://indico.esa.int/event/188/contributions/1620/attachments/1551/1785/SWE_Space_Radiation_Expert_Service_Centre_P3.pdf.

——— 2017. "Metrics for Addressing ESA SSA End User Needs." ESA SSA Programme Office, ESOC, Germany.

Government of Canada National Research Council, 2017. "Dominion Radio Astrophysical Observatory - National Research Council Canada." National Research Council Canada. 2018. https://www.nrc-cnrc.gc.ca/eng/solutions/facilities/drao.html.

Graz Institute for Space Research, 2018. 'The Graz Institute for Space Research (IMF)'. 2018. http://www.iwf.oeaw.ac.at/home/.

Guhathakurta, Madulika, 2011. "Space Weather Super-Storm Not IF but WHEN and Extreme Solar Minimum." Presentation to UNCOPUOS Meeting, Vienna, Austria, February 10.

Hapgood, Mike, 2001. "Space & Ground Segments for a European Space Weather Programme." "Space Weather Workshop: Looking Towards a Future European Space Weather Programme", 17-19 December 2001 ESTEC, Noordwijk, The Netherlands. http://swe.ssa.esa.int/TECEES/spweather/workshops/SPW_W3/PROCEEDINGS_W3/esws_space_seg.pdf.

——— 2010. "Space Weather: Its Impact on Earth and Implications for Business (Lloyd's and RAL)." Lloyd's 360 Risk Insight. https://www.lloyds.com/news-and-risk-insight/risk-reports/library/natural-environment/space-weather.

Horne, Richard B, 2001. "Benefits of a Space Weather Programme." WP1100. ESA Space Weather Programme Study. Alcatel Consortium: British Antartic Survey, National Environment Research Council, Cambridge, U.K.

INTERMAGNET, 2017. "List of IMOs and Responsible GINs." INTERMAGNET. July

5, 2017. http://www.intermagnet.org/imos/imotblobs-eng.php.

——— 2018. "International Real-Time Magnetic Observatory Network (INTERMAGNET)." INTERMEGNET. May 10, 2018. http://www.intermagnet.org/index-eng.php.

International Atomic Energy Agency, 2014. "Radiation Protection and Safety of Radiation Sources: International Basic Safety Standards." http://www-pub.iaea.org/books/IAEABooks/8930/Radiation-Protection-and-Safety-of-Radiation-Sources-International-Basic-Safety-Standards.

——— 2018. "Overview." Internation Atomic Energy Agency. 2018. https://www.iaea.org/about/overview.

International Civil Aviation Organization, 2007. "Meteorological Service for International Air Navigation." Annex 3 to the Convention on International Civil Aviation. https://www.wmo.int/pages/prog/www/ISS/Meetings/CT-MTDCF-ET-DRC_Geneva2008/Annex3_16ed.pdf.

——— 2010. "Space Weather Effects in Regard to International Air Navigation." IAVWOPSG. International Civil Aviation Organization. https://www.icao.int/safety/meteorology/iavwopsg/Space%20Weather/Space%20Weather%20Effects%20in%20regard%20to%20International%20Air%20Navigation.doc.

——— 2016. "PROVISION OF SPACE WEATHER INFORMATION IN SUPPORT OF INTERNATIONAL AIR NAVIGATION." A39-WP/121. ASSEMBLY — 39TH SESSION TECHNICAL COMMISSION. International Civil Aviation Organization. https://www.icao.int/Meetings/a39/Documents/WP/wp_121_en.pdf.

——— 2018a. "About ICAO." International Civil Aviation Organization. 2018. https://www.icao.int/about-icao/Pages/default.aspx.

——— 2018b. "The Meteorology Panel (METP)." International Civil Aviation Organization. 2018. https://www.icao.int/airnavigation/METP/Pages/default.aspx.

International Radio Occultation Working Group, 2018. "International Radio Occultation Working Group (IROWG)." International Radio Occultation Working Group. 2018. http://irowg.org/.

International Space Weather Initiative, 2018. "International Space Weather Initiative." International Space Weather Initiative Secretariat. 2018. http://newserver.stil.bas.bg/ISWI/.

International Telecommunication Union, 2018a. "About International Telecommunication Union (ITU)." International Telecommunication Union. 2018. https://www.itu.int/en/about/Pages/default.aspx.

——— 2018b. "Study Group 3 (SG 3)." International Telecommunication Union. 2018. https://www.itu.int/en/ITU-R/study-groups/rsg3/Pages/default.aspx.

——— 2018c. "Study Group 7 (SG 7)." International Telecommunication Union. 2018. https://www.itu.int/en/ITU-R/study-groups/rsg7/Pages/default.aspx.

Ionosphere Monitoring and Prediction Centre, 2018. 'IMPC: IMPC Concept'. 2018. https://impc.dlr.de/index.php?id=12.

Koch, Christina, 2017. "Horizon 2020 Space." La Paz, September 12.

Kraft, S., Adriano, L., and Luntama, J., 2018. "ESA'S Distributed Space Weather Sensor System (D3s) Utilizing Hosted Payloads for Operational Space Weather Monitoring." *Acta Astronautica*, February 2018.

Krausmann, Elisabeth, 2011. "The Space-Weather Awareness Dialogue: Findings and Outlook," no. JRC 67605.

Krausmann, E., Andersson E., Russell T., and Murtagh, W., 2015. *Space Weather and Rail: Findings and Outlook*. Luxembourg: Publications Office of the European Union. https://ec.europa.eu/jrc/en/publication/eur-scientific-and-technical-research-reports/space-weather-and-rail-findings-and-outlook.

Krausmann, E., Andersson E., Gibbs M., and Murtagh W., 2016. *Space Weather & Critical Infrastructures: Findings and Outlook*. Publications Office of the European Union.

Kuznetsova, M. 2017, "Addressing the Need for Coordinated Assessment, Development and Deployment of Space Weather Prediction Capabilities." presented at the 14th European Space Weather Week, Ostend, Belgium.

Lautenbacher, Conrad C., 2014. "Growing the Space Weather Enterprise - Roles and

Contributions." presented at the Space Weather Workshop, Boulder, CO, April 2018. http://www.acswa.us/presentations/Lautenbacher%202014%20SWW%20Presentation.pdf.

Lilensten, Jean, 2008. *COST 724 Final Report: Developing the Scientific Basis for Monitoring, Modelling and Predicting Space Weather*. Office for Official Publications of the European Communities.

Liu, S., and Gong, J., 2015. "Operational Space Weather Services in National Space Science Center of Chinese Academy of Sciences." *Space Weather* 13 (10): 599–605.

Luntama, J., Bobrinsky, N., Glover, A., Keil, R., Kraft, S., and Lupi A., 2015. "SSA SWE Segment Status and Prospects." presented at the 12th European Space Weather Week 23-27 November, 2015, Ostend, Belgium. http://www.stce.be/esww12/contributions/public/K23-K1/K23-K1-01-LuntamaJuha-Pekka/Luntama-SSASWESystemStatusandProspects.pptx.

Luntama, J., Glover, A and Kraft, S., 2017. "ESA SSA Space Weather Network." ESA SSA Programme Office.

Mann, I. R., Di Pippo S., Opgenoorth H. J., Kuznetsova, M., and Kendall, D.J., 2018. "International Collaboration Within the United Nations Committee on the Peaceful Uses of Outer Space: Framework for International Space Weather Services (2018–2030)." *Space Weather* 16 (5): 428–33.

Marshall, R. A., Waters, C. L., and Sciffer, M. D., 2010. "Spectral Analysis of Pipe-to-Soil Potentials with Variations of the Earth's Magnetic Field in the Australian Region." *Space Weather* 8 (5).

Mathieu, Charlotte, 2009. "Space-Based Services in Europe: Addressing the Transition Between Demonstration and Operation." ESPI Report 17. Vienna, Austria: ESPI.

Mavromichalaki, H., Papaioannou, A., Plainaki, C., Sarlanis, C., Souvatzoglou, G., Gerontidou, M., Papailiou M., et al., 2011. "Applications and Usage of the Real-Time Neutron Monitor Database." *Advances in Space Research*, Recent Advances in Space Weather Monitoring, Modelling, and Forecasting - 2, 47 (12): 2210–22.

McCormack, Les, 2017. "Rail Network Disruption from Outer Space." Atkins Global. April 27, 2017. http://www.atkinsglobal.com/en-GB/angles/all-angles/rail-network-disruption-from-outer-space.

Messerotti, M., 2017. 'An Operational Scheme for Establishing a National Space Weather Infrastructure'. presented at the 14th European Space Weather Week, Ostend, Belgium.

Monte, Luca del, 2009. "Possible Governance Models for a Future European SSA System." presented at the International Interdisciplinary Congress on Space Debris 7-9 May 2009, McGill University, Montreal, Canada.

Moranta, S., Pavesi, G., Perrichon, L., Plattard, L., and Sarret, M., 2018. "Security in Outer Space: Rising Stakes for Europe." Report 64. Vienna, Austria: European Space Policy Institute (ESPI).

National Aeronautics and Space Administration, 2012. "Solar Energetic Particles." NASA's Cosmicopia, Goddard Space Flight Centre. 2018. https://helios.gsfc.nasa.gov/sep.html.

——— 2016a. "ACE | Science Mission Directorate." NASA Science. 2018. https://science.nasa.gov/missions/ace.

——— 2016b. "NASA Viz: Earth's Ionosphere." NASA Visualization Explorer. 2016. https://svs.gsfc.nasa.gov/12220.

——— 2017. "STEREO Overview." Text. NASA. 2017. http://www.nasa.gov/mission_pages/stereo/mission/index.html.

——— 2018a. "COSPAR Panel on Space Weather." Community Coordinating Modeling Centre. 2018. https://ccmc.gsfc.nasa.gov/psw/.

——— 2018b. "I-SWAT (International Space Weather Action Teams)." Community Coordinating Modeling Centre. 2018. https://ccmc.gsfc.nasa.gov/iswat/.

——— 2018c. "I-SWAT Clusters." Community Coordinating Modeling Centre. 2018. https://ccmc.gsfc.nasa.gov/iswat/clusters.php.

——— 2018d. "I-SWAT Goals." Community Coordinating Modeling Centre. 2018. https://ccmc.gsfc.nasa.gov/iswat/vision-goals.php.

National Oceanic and Atmospheric Administration, 2014. "U.S Space Weather Products and Services, Lesson 2, Summary." National Oceanic and Atmospheric Administration, National

Weather Service, National Weather Service Training Centre. 2018. https://training.weather.gov/nwstc/spacewx/wmo/lesson2/review2.html.

——— 2015. "U.S Space Weather Products and Services Course, Lesson 2, Types of Space Weather Events." Nation Oceanic and Atmospheric Administration, National Weather Service National Weather Service Training Centre. 2018. https://training.weather.gov/nwstc/spacewx/wmo/lesson2/events.html.

National Oceanic and Atmospheric Administration, 2018a "Solar Flares (Radio Blackouts) | NOAA / NWS Space Weather Prediction Center." National Oceanic and Atmospheric Administration, National Weather Service, Space Weather Prediction Centre. 2018. https://www.swpc.noaa.gov/phenomena/solar-flares-radio-blackouts.

——— 2018b. "Coronal Mass Ejections | NOAA / NWS Space Weather Prediction Center." National Oceanic and Atmospheric Administration, National Weather Service, Space Weather Prediction Centre. 2018. https://www.swpc.noaa.gov/phenomena/coronal-mass-ejections.

——— 2018c. "Space Weather Phenomena | NOAA / NWS Space Weather Prediction Center." National Oceanic and Atmospheric Administration, National Weather Service, Space Weather Prediction Centre. 2018. https://www.swpc.noaa.gov/phenomena.

——— 2018d. "About Our Agency." National Oceanic and Atmospheric Administration. 2018. http://www.noaa.gov/about-our-agency.

——— 2018e. "National Space Weather Portal." Nation Oceanic and Atmospheric Administration, National Weather Service National Weather Service Training Centre Space Weather Predication Centre. 2018. https://www.swpc.noaa.gov/portal/.

——— 2018f. "Earth's Magnetosphere | NOAA / NWS Space Weather Prediction Center." National Oceanic and Atmospheric Administration, National Weather Service, Space Weather Prediction Centre. 2018. https://www.swpc.noaa.gov/phenomena/earths-magnetosphere.

National Research Council, 2008. *Severe Space Weather Events: Understanding Societal and Economic Impacts: A Workshop Report*. https://doi.org/10.17226/12507.

National Science and Technology Council. 2015. "National Space Weather Strategy." NSTC. https://obamawhitehouse.archives.gov/sites/default/files/microsites/ostp/final_nationalspaceweatherstrategy_20151028.pdf.

Night Sky Alerts, 2018. "Night Sky Alerts." 2018. https://nightskyalerts.com/.

Nightingale, E., Lal, B., Weeden, B., Picard, A., and Eisenstadt, A., 2016. "Evaluating Options for Civil Space Situational Awareness (SSA)." IDA Paper NS P-8038. Institute for Defense (IDA) Science and Technology Policy Institute. https://www.ida.org/idamedia/Corporate/Files/Publications/STPIPubs/2016/P-8038.pdf.

Palmroth, M., I Honkonen, P. Janhunen, A. Viljanen, S. von Alfthan, and G. Facsko, 2014. 'Finnish Meteorological Institute 24/7 Space Weather Service'. presented at the 7th CCMC Community Workshop Presentations, Annapolis, Maryland, United States. https://ccmc.gsfc.nasa.gov/RoR_WWW/workshops/2014_7thCCMCworkshop/presentations/honkonen_FMIspaceweather.pdf.

PECASUS, 2018a. 'PECASUS: Providing Civil Aviation with Information on Space Weather That Has the Potential to Affect Communications, Navigation and the Health of Passengers and Crew'. 2018a. http://pecasus.eu/dissemination/PECASUS%20info%202018.pdf.

——— 2018b. 'PECASUS Fact Sheet'. http://pecasus.eu/dissemination/PECASUS%20info%202018.pdf.

Peitso, P., and Tanskanen, E., 2013. "Space Weather Instruments and Measurement Platforms." Aalto University School of Electrical Engineering.

Pinheiro, F., Barata, M., and Fernandes, J., 2016. "Comparison of Space Weather Services: Information Systems, Activity and Forecasts." *J. Comp. Int. Sci* 7 (2): 3–24.

PricewaterhouseCoopers, 2016. "Space Weather Study Results." PwC. https://esamultimedia.esa.int/docs/business_with_esa/Space_Weather_Cost_Benefit_Analysis_ESA_2016.pdf.

Pulkkinen, A., Pirjola, R., Boteler, D., Viljanen, A., and Yegorov, I., 2001. "Modelling of Space Weather Effects on Pipelines." *Journal of Applied Geophysics* 48 (4): 233–56.

Research on SOlar Long-term Variability and Effects, 2018. 'ReSoLVE - Center of Excellence'. 2018. https://spaceclimate.fi/resolve/index.html.

Romano, Vincenzo, 2017. 'Italian Contribution to Space Weather'. presented at the UN COPUOS 54rd Session STSC, Vienna, Austria. http://www.unoosa.org/pdf/icg/2017/iswi-sc/14-7._17.pdf.

Romero, Raul, 2018. "Establishment of Space Weather Information Service For Interrnational Air Navigation." presented at the INTER-PROGRAMME TEAM ON SPACE WEATHER INFORMATION, SYSTEM AND SERVICES, SECOND SESSION, Tokyo, Japan. http://www.wmo.int/pages/prog/sat/meetings/IPT-SWeISS-2/documents/3.1_SW-ICAO-Tokyo%20%202018.pdf.

Rothwell, Roy, 1994. "Towards the Fifth-generation Innovation Process." *International Marketing Review* 11 (1): 7–31.

Royal Academy of Engineering, 2013. "Extreme Space Weather: Impacts on Engineered Systems and Infrastructure." London, United Kingdom: Royal Academy of Engineering. https://www.raeng.org.uk/publications/reports/space-weather-full-report.

Royal Belgian Institute for Space Aeronomy, 2018a. 'Belgian Institute for Space Aeronomy (BIRA-IASB)'. 2018a. http://www.aeronomie.be/en/index.htm.

——— 2018b. 'SSA Space Weather Coordination Centre (SSCC)'. 2018b. http://www.aeronomie.be/en/services/sscc.htm.

Royal Meteorological Institute, 2018. 'Weather in Belgium: Forecasts - RMI'. Meteo. 2018. https://www.meteo.be/meteo/view/en/65239-Home.html.

Royal Observatory of Belgium, 2018. 'Time Lab: The ROB's Time-Frequency Laboratory'. 2018. https://betime.be/en/.

Russu, A., Gómez-Herrero R., Prieto, M., Monstein, C., Ivanov, H., Rodríguez-Pacheco, J., Blanco, J., 2015. "A Year of Operation of Melibea E-Callisto Solar Radio Telescope." *Journal of Physics: Conference Series* 632 (1): 012078.

Rutherford Appleton Laboratory, 2018. 'RAL Space Space Weather'. 2018. https://www.ral-space.stfc.ac.uk/Pages/Space-weather.aspx.

Schrijver, C., Kauristie, K., Aylward, A., Denardini, C., Gibson, S., Glover, A., Gopalswamy, N., et al., 2015. "Understanding Space Weather to Shield Society: A Global Road Map for 2015–2025 Commissioned by COSPAR and ILWS." *Advances in Space Research* 55 (12): 2745–2807..

Schwenn, Rainer, 2006. "Space Weather: The Solar Perspective." *Living Reviews in Solar Physics* 3 (1): 2.

Shaw, Andy. 2001. "Market Analysis Final Report Highlights." presented at the ESA Space Weather Study, Newton Conference Centre, ESTEC, Noordwijk, December 6.

Solar Influences Data analysis Center, 2018. 'SIDC - Solar Influences Data Analysis Center'. 2018. http://sidc.be/index.php.

Solar-Terrestrial Centre of Excellence, 2018. 'STCE'. 2018. http://www.stce.be/.

Space Weather Italian Community, 2018. 'Assemblea Nazionale SWICO 2018'. 2018. http://www.swico.it/.

Space Weather Working Team, 2016. "SWWT Plenary Meeting 37." Ostend, Belgium. http://swe.ssa.esa.int/DOCS/SWWT/m37/SWWT-PM-37-20161116-minutes-v.2.pdf.

Stanford Solar Center, 2018a. "Tracking Solar Flares." Stanford Solar Center. 2018. http://solar-center.stanford.edu/SID/activities/index.html.

——— 2018b. "Tracking Solar Flares." Stanford Solar Center. 2018. http://solar-center.stanford.edu/SID/activities/ionosphere.html.

Steinegger, M., Hanslmeier, A., Otruba, W., Freislich, H., Denker, C., Goode, P., Marquette, W., et al, 2000. "An Overview of the New Global High-Resolution H-Alpha Network." *Hvar Observatory Bulletin* 24: 179.

Swan, J., and Scarbrough, H., 2005. "The Politics of Networked Innovation." *Human Relations* 58 (7): 913–43.

Swedish Institute of Space Physics, 2018. 'Regional Warning Center Sweden of International Space Environment Service'. 2018. http://www.lund.irf.se/rwc/.

Telespazio, 2018. "IPS Project - Ionosphere Prediction Service." 2018. https://ips.telespazio.com/web/guest/ips-project.

Temmer, Manuela, 2018. 'Space Weather Activities in Austria'. presented at the Weather Expert group Meeting and Workshop, Vienna, Austria, January 31. Jan 31, 2018.

The International Space Environment Service, 2018. "The International Space Environment Service (ISES)." 2018. http://www.spaceweather.org/index.jsp.

The Finnish Research Programme on Nuclear Power Plant Safety, 2018. 'SAFIR2014 - The Finnish Research Programme on Nuclear Power Plant Safety 2011-2014'. 2018. http://safir2014.vtt.fi/.

UK Government. 2018, 'UK to Play a Major Role in Space Weather Mission Concept'. GOV.UK. 2 February 2018. https://www.gov.uk/government/news/uk-to-play-a-major-role-in-space-weather-mission-concept.

UK Met Office, 2018. 'Space Weather Current Status'. 2018. https://www.metoffice.gov.uk/public/weather/space-weather/.

UK Parliamentary Office of Science and Technology, 2010. 'Space Weather' POST-PN-361 (July). https://researchbriefings.parliament.uk/ResearchBriefing/Summary/POST-PN-361.

United Nations Committee on the Peaceful Uses of Outer Space. 2015a. "Report of the Scientific and Technical Subcommittee on Its Fifty-Second Session." A/AC.105/1088. Vienna, Austria: United Nations Committee on the Peaceful Uses of Outer Space. http://www.unoosa.org/oosa/oosadoc/data/documents/2015/aac.105/aac.1051088_0.html.

———. 2015b. "Report on the Meeting on the Applications of Space Science and Technology for Public Health Organized by the WHO and UNOOSA." A/AC.105/1099. United Nations Committee on the Peaceful Uses of Outer Space. http://www.unoosa.org/oosa/oosadoc/data/documents/2016/aac.105/aac.1051099_0.html.

———. 2016. "Report of the Scientific and Technical Subcommittee on Its Fifty-Third Session." A/AC.105/1109. United Nations Committee on the Peaceful Uses of Outer Space. http://www.unoosa.org/oosa/oosadoc/data/documents/2016/aac.105/aac.1051109_0.html.

———. 2017a. "Thematic Priority 4. International Framework for Space Weather Services." A/AC.105/1171. A/AC.105/. United Nations Committee on the Peaceful Uses of Outer Space. http://www.unoosa.org/oosa/oosadoc/data/documents/2018/aac.105/aac.1051171_0.html.

——— 2017b. "Space Weather: Special Report of the Inter-Agency Meeting on Outer Space Activities on Developments within the United Nations System Related to Space Weather." A/AC.105/1146. United Nations Committee on the Peaceful Uses of Outer Space. http://www.unoosa.org/oosa/oosadoc/data/documents/2017/aac.105/aac.1051146_0.html.

——— 2017c. 'Space Weather: Views of France'. A/AC.105/C.1/2017/CRP.24. Scientific and Technical Subcommittee Fifty-Fourth Session. Vienna, Austria: United Nations Committee on the Peaceful Uses of Outer Space. http://www.unoosa.org/res/oosadoc/data/documents/2017/aac_105c_12017crp/aac_105c_12017crp_24_0_html/AC105_C1_2017_CRP24E.pdf.

United Nations Office for Outer Space Affairs, 2018a. "UNISPACE+50 Thematic Priorities Booklet." United Nations Office for Outer Space Affairs. 2018. http://www.unoosa.org/documents/pdf/unispace/plus50/thematic_priorities_booklet.pdf.

——— 2018b. "Roles and Responsibilities." United Nations Office for Outer Space Affairs. 2018. http://www.unoosa.org/oosa/en/aboutus/roles-responsibilities.html

——— 2018c. "UN Programme on Space Applications." United Nations Office for Outer Space Affairs. 2018.

http://www.unoosa.org/oosa/en/ourwork/psa/index.html.

——— 2018d. "UN-Space: Open Informal Session: 13th, The Transformative Potential of Space Technology for Development: Approaches and Opportunities in the UN System." United Nations Office for Outer Space Affairs. 2018. http://www.unoosa.org/oosa/en/ourwork/un-space/ois/13th.html.

Veronig, A., and Pötzi, W., 2016. "Ground-Based Observations of the Solar Sources of Space Weather." *ArXiv:1602.02721 [Astro-Ph]*, February. http://arxiv.org/abs/1602.02721.

Walter, Nicolas, 2017. 'The European Space Sciences Committee'. presented at the SSB Meeting, Irvine, CA, November 1. http://sites.nationalacademies.org/cs/groups/ssbsite/documents/webpage/ssb_182864.pdf.

Wenzel, D., Berdermann, J., Jakowski N., 2017. 'Space Weather Activities in Germany'. In. Boston. https://elib.dlr.de/118302/.

White House, 2010. "Presidential Policy Directive 4: National Space Policy of the United States of America." *Presidential Policy Directive 4 (PPD 4)*, June. https://www.nasa.gov/sites/default/files/national_space_policy_6-28-10.pdf.

World Health Organisation, 2016, "The Global Guardian of Public Health." World Health Organisation. 2018. http://www.who.int/about/what-we-do/global-guardian-public-health/en/.

World Meteorological Organisation, 2008. "The Potential Role of WMO in Space Weather." SAT-PUB WMO/TD-No. 1482; SP-No. 05. WMO. http://www.wmo.int/pages/prog/sat/documents/SAT-PUB_SP-5-TD-1482-Potential-Role-of-WMO-in-Space-Weather-2008.pdf.

——— 2016. "Four-Year Plan For WMO Activities Related to Space Weather 2016-2019." SAT-GEN Draft 2.1. WMO. http://www.wmo.int/pages/prog/sat/documents/SAT-GEN_Space-weather-Four-Year-Plan-2016-2019-en.pdf.

——— 2018a. "Who We Are." World Meteorological Organization. 2018. https://public.wmo.int/en/about-us/who-we-are.

——— 2018b. 'Report from Finland'. Inter-Programme Team on Space Weather Information, Systems and Services (IPT-SWeISS). http://www.wmo.int/pages/prog/sat/meetings/IPT-SWeISS-2/documents/4.2a_ReportfromFinland.pdf.

Zell, Holly, 2011. "Earth's Magnetosphere." National Aeronautical and Space Administration (NASA). 2018. http://www.nasa.gov/mission_pages/sunearth/multimedia/magnetosphere.html.

——— 2013. "Earth's Upper Atmosphere." National Aeronautical and Space Administration (NASA). January 2013. http://www.nasa.gov/mission_pages/sunearth/science/mos-upper-atmosphere.html.

Zheng, Y., and Evans, R., 2014. "Solar Energetic Particles (SEPs)." presented at the Space Weather Research, Education and Development Initiative (SW REDI), June 2014.

About ESPI

The European Space Policy Institute (ESPI) is an association ruled by Austrian Law, based in Vienna, funded at its inception (2003) by the Austrian Space Agency and ESA, and now supported by 17 members that include European national space agencies, the European Commission, and main European space services companies and manufacturers.

The Institute provides decision-makers with an informed view on mid-to-long-term issues relevant to Europe's space activities. In this context, ESPI acts as an independent platform for developing positions and strategies.

ESPI fulfils its objectives through various multidisciplinary research activities leading to the publication of books, reports, papers, articles, executive briefs, proceedings and position papers, and to the organisation of conferences and events including the annual ESPI Autumn Conference. Located in the heart of Vienna, the Institute has developed a privileged relationship with the United Nations Office for Outer Space Affairs and with a network of researchers and experts in Europe and across the globe.

ESPI More information on ESPI is available on our website: www.espi.or.at

About the Authors

This report was prepared with contributions from the following ESPI researchers:

- Marco Aliberti, Senior Research Fellow
- Leyton Wells, Research Intern

The study was conducted under the supervision of Jean-Jacques Tortora, Director of ESPI, and the coordination of Sebastien Moranta, Coordinator of ESPI Studies.

Publicly available data and information were completed with stakeholders and expert interviews. The list of interviewees is provided in Annex to this report.

ESPI is grateful to the many stakeholders that accepted to be interviewed and provided substantial contributions for this report.

Mission Statement of ESPI

The European Space Policy Institute (ESPI) provides decision-makers with an informed view on mid- to long-term issues relevant to Europe's space activities. In this context, ESPI acts as an independent platform for developing positions and strategies.